

PMATH 351 Real Analysis

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1 Set Theory

Lecture 1, 2026/01/05

1.1 Ordinals

$\emptyset = \{\} = 0 \neq \{\emptyset\} = 1$ and $\emptyset \neq \{\emptyset, \{\emptyset\}\} = \{0, 1\} = 2$.

$\{0, 1, 2\} = 3, \{0, 1, 2, 3\} = 4, \dots$

Example.

- $\mathbb{N} = \{1, 2, 3, \dots\}$.
- $\mathbb{N} \cup \{0\} = \{0, 1, 2, 3, \dots\} = \omega$, which does not contain ω itself.
- $\omega + 1 = \omega \cup \{\omega\} = \{0, 1, 2, 3, \dots, \omega\}$, which contains ω .
- $\omega + 2 = (\omega + 1) \cup \{\omega + 1\} = \{0, 1, 2, 3, \dots, \omega, \omega + 1\}$.
- $\omega + 3 = (\omega + 2) \cup \{\omega + 2\} = \{0, 1, 2, 3, \dots, \omega, \omega + 1, \omega + 2\}$.
- $\vdots$
- $2\omega = \omega + \omega = \{0, 1, 2, \dots\} \cup \{\omega, \omega + 1, \omega + 2, \dots\}$.
- $2\omega + 1, 2\omega + 2, \dots, 3\omega, 3\omega + 1, \dots, 4\omega, \dots, 5\omega, \dots$ can be defined similarly.
- $\omega^2 = \{0, 1, 2, \dots\} \cup \{\omega, \omega + 1, \dots\}$
 $\cup \{2\omega, 2\omega + 1, \dots\}$
 $\vdots$
 $\cup \{n\omega, n\omega + 1, \dots\}$.
- $\omega^2 + 1$.
- $\vdots$
- $\omega^\omega, \omega^\omega + 1, \omega^\omega + 2, \dots$
- $\omega^{\omega^\omega}, \dots, \omega^{\omega^{\omega^\omega}}, \dots$

In this course, $A \subset B$ could also imply the case of $A = B$.

Definition 1.1 (Union, Intersection).

Let X be a set. If $X_i \subset X$ are sets for $i \in I$, then

- the **union** of the sets X_i is $\bigcup_{i \in I} X_i = \{x \in X \mid x \in X_i \text{ for some } i \in I\}$,
- the **intersection** of the sets X_i is $\bigcap_{i \in I} X_i = \{x \in X \mid x \in X_i \text{ for all } i \in I\}$.

Note. $A \subset B$ if $a \in A \implies a \in B$. In this case, we say A is a **subset** of B or B is a **superset** of A .

Note. Let $X = \omega, I = \emptyset$. Then, $\bigcup_{i \in I} X_i = \emptyset$. That is, a union over an empty set is the empty set.

Example (Russell's Paradox).

Consider the following sets:

$$A = \{ \text{all sets} \}, \quad B = \{E \in A \mid E \notin E\}.$$

Question: Is $B \in B$?

- $B \in B \implies B \notin B$ by the definition of B .
- $B \notin B \implies B \in B$ by the definition of B .

This is a contradiction, so such a set A cannot exist. To fix this paradox, we restrict our universe to only sets that are subsets of some fixed set X .

Definition 1.2 (Power Set).

If X is a set, the **power set** of X is $\mathcal{P}(X) = \{E \mid E \subset X\}$.

Example.

- (1) $\mathcal{P}(\emptyset) = \{\emptyset\}$.
- (2) $\mathcal{P}(\{0\}) = \{\emptyset, \{0\}\} = 2$.
- (3) $\mathcal{P}(\{0, 1\}) = \{\emptyset, \{0, 1\}, \{0\}, \{1\}\}$. This is not an ordinal. Note that a power set is possible to not be an ordinal.
- (4) $\mathcal{P}(\mathbb{N}) = \{E \mid E \subset \mathbb{N}\}$. Then, $\emptyset, \{n\}, 2\mathbb{N}, \mathbb{N} \in \mathcal{P}(\mathbb{N})$.

Definition 1.3 (Ordered Pairs).

Let X, Y be subsets of Z (some set). For $x \in X$ and $y \in Y$, the **ordered pair** (x, y) is

$$(x, y) := \{\{x\}, \{x, y\}\}.$$

Note. If $x = y$, then $(x, y) = \{\{x\}, \{x\}\} = \{\{x\}\}$.

1.2 Relations

Definition 1.4 (Relation).

A **relation** on X is a subset $R \subseteq X \times X = \{(x, y) : x, y \in X\}$. We write xRy if $(x, y) \in R$.

Example.

- (1) Equality relation: $x = x$.
- (2) $R_{\leq}$: $(x, y) \in R_{\leq} \iff x \leq y$.
- (3) $R_{\subset}$ on 2^X : $(A, B) \in R_{\subset} \iff A \subset B$.

Definition 1.5 (Properties of Relations).

A relation $R \subset X \times X$ is

- (1) **reflexive** if xRx for all $x \in X$,
- (2) **symmetric** if $xRy \implies yRx$ for all $x, y \in X$,
- (3) **transitive** if xRy and $yRz \implies xRz$ for all $x, y, z \in X$.

Definition 1.6 (Equivalence Relation).

A relation $R \subset X \times X$ is an **equivalence relation** if it is reflexive, symmetric, and transitive.

Example.

- (1) Let X be a set. R is equality relation on X . Then, R is an equivalence relation.
- (2) Let $X = \mathbb{Z}$ and xRy if $y - x = 12m$ for some $m \in \mathbb{Z}$. Then, R is an equivalence relation.

Proof.

- (1) Reflexive: $x - x = 0 = 12 \cdot 0$, so xRx .
- (2) Symmetric: If $xRy \iff x - y = 12m$, then $y - x = 12 \cdot (-m)$, so yRx .
- (3) Transitive: If xRy and yRz , or equivalently $x - y = 12m$ and $y - z = 12n$, then

$$x - z = x - y + y - z = 12m + 12n = 12(m + n), \text{ so } xRz. \quad \square$$
- (3) Let X be number of students in class today. Let $x \sim y$ if x and y are in the same row. Then, $\sim$ is an equivalence relation.
- (4) Let $X = \mathbb{R}$ and $xRy \iff x < y$. Then, R is not an equivalence relation.

Proof. Note that $0 < 0$ is false, so R is not reflexive. □

(5) Let $X =$ all finite dimensional vector subspaces of $\mathbb{R}^n$ for some n and $V R W$ if $V \cong W$.

Then, R is an equivalence relation.

(6) Let X be any set and $x R y$ if $x, y \in X$. Then, R is an equivalence relation.

Note. A way to define the set of integers: $Z := \mathbb{N} \cup (\{1\} \times \mathbb{N}) \cup \{\emptyset\}$.

Note. We often use $\sim$, $\approx$, $\cong$, or $\simeq$ to denote an equivalence relation.

Definition 1.7 (Partition).

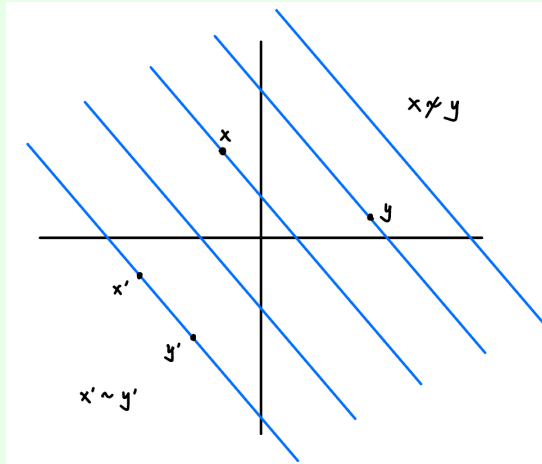
A **partition** of a set X is a family $\mathcal{P} \subset \mathcal{P}(X) \setminus \{\emptyset\}$ such that

(1) if $A, B \in \mathcal{P}$ and $A \neq B$, then $A \cap B = \emptyset$,

(2) $\bigcup \mathcal{P} = X$. Equivalently, $\bigcup_{A \in \mathcal{P}} A = X$.

Example. If $\mathcal{P}$ is a partition of X . For $x \in X$, define $[x]_{\mathcal{P}}$ to be the unique set in $\mathcal{P}$ such that $x \in [x]_{\mathcal{P}}$. Define $x \sim_{\mathcal{P}} y$ if $[x]_{\mathcal{P}} = [y]_{\mathcal{P}}$. Then, $\sim_{\mathcal{P}}$ is an equivalence relation on X .

Example. Consider the parallel lines with slope m in $\mathbb{R}^2$. Then, an equivalence relation can be defined as two points are related if they are on the same line. The corresponding partition is the set of all such lines.



Example. If $\sim$ is an equivalence relation on X . For $x \in X$, define $[x]_{\sim} = \{y \in X : y \sim x\}$.

Proposition 1.1. $\mathcal{P}_{\sim} = \{[x]_{\sim} : x \in X\}$ is a partition of X .

Proof. Note that $x \in [x]_{\sim}$ since $x \sim x$ by reflexivity, so $\emptyset \notin \mathcal{P}_{\sim}$. Also, $\bigcup \mathcal{P}_{\sim} = X$ since $\forall x \in X$, $x \in [x]_{\sim} \subset \bigcup \mathcal{P}_{\sim}$. Now, if $[x]_{\sim} \cap [y]_{\sim} \neq \emptyset$, then $z \in [x]_{\sim} \cap [y]_{\sim}$ for some $z \in X$. Thus, $z \sim x$ and $z \sim y$. Take $w \in [x]_{\sim}$. Then, $w \sim x$, and $z \sim x \implies x \sim z$ by symmetry. By transitivity, $w \sim z$. So, $w \in [y]_{\sim}$. Thus, $[x]_{\sim} \subset [y]_{\sim}$. Similarly, $[y]_{\sim} \subset [x]_{\sim}$ by exchanging the roles of x and y . Hence, $[x]_{\sim} = [y]_{\sim}$. Therefore, $\mathcal{P}_{\sim}$ is a partition of X . $\square$

Proposition 1.2. If $\sim$ is an equivalence relation. Then, $\sim$ is the equivalence relation coming from the partition $\mathcal{P}_{\sim}$ and vice versa: if $\mathcal{P}$ is a partition, then $\mathcal{P} = \mathcal{P}_{\sim_{\mathcal{P}}}$ where $\sim_{\mathcal{P}}$ is the associated equivalence relation.

1.3 Functions

Definition 1.8 (Function).

Let X, Y be sets. A function $f : X \rightarrow Y$ is a collection of ordered pairs $G_f = \{(x, y) \mid x \in X, y \in Y\}$ such that for each $x \in X$, $\exists! y \in Y$ (denoted by $f(x)$) such that $(x, y) \in G_f$.

Lecture 3, 2026/01/09

Another important equivalence relation below.

Example. Let $X = \mathbb{Z} \times \mathbb{N}$. Define an equivalence relation $(a, b) \sim (c, d)$ if $da = cb$. Note that

- (1) Reflexive: $(a, b) \sim (a, b)$ since $ba = ab$.
- (2) Symmetric: If $(a, b) \sim (c, d)$, then $da = cb \implies cb = da$, so $(c, d) \sim (a, b)$.
- (3) Transitive: If $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$, then $da = cb$ and $fc = ed$. Thus, $fa = f \frac{da}{d} = f \frac{cb}{d} = \frac{edb}{d} = eb$. So, $(a, b) \sim (e, f)$.

Then, $\mathcal{P}_{\sim} = \mathbb{Z} \times \mathbb{N} / \sim :=$ the set of equivalence classes. Define $\mathbb{Q} := \mathbb{Z} \times \mathbb{N} / \sim$.

Recall that a function denoted $f : X \rightarrow Y$ takes inputs x from X and outputs $f(x)$ in Y , where X is the **domain** and Y is the **codomain**.

Definition 1.9 (Image, Inverse Image, Range).

If $A \subset X$, the **image** of A under f is defined by

$$f(A) = \{f(x) \mid x \in A\} \subset Y.$$

The **inverse image** of $B \subset Y$ under f is defined by

$$f^{-1}(B) = \{x \in X \mid f(x) \in B\} \subset X.$$

The **range** of f is $f(X) \subset Y$.

Definition 1.10 (Composition).

If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, the **composition** $g \circ f : X \rightarrow Z$ is defined by

$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in X.$$

Definition 1.11 (Injective, Surjective).

A function $f : X \rightarrow Y$ is

- (1) **injective** (one-to-one) if whenever $f(a) = f(b)$, then $a = b$,
- (2) **surjective** (onto) if $f(X) = Y$, i.e. $\forall y \in Y, \exists x \in X$ such that $f(x) = y$.

Example.

- (1) Define $f : \mathbb{N} \rightarrow \mathbb{N}$ by $f(x) = 2x$. The range of f is the set of even natural numbers. Thus, f is injective but not surjective.
- (2) If $X \subset Y$, define $f : X \rightarrow Y$ by $f(x) = x$. Then, f is injective, and f is surjective $\Leftrightarrow X = Y$.
- (3) Define $f : \mathbb{Z} \rightarrow \mathbb{N}$ by

$$f(n) = \begin{cases} 2n, & n \geq 1 \\ -2n + 1, & n \leq -1 \\ 0, & n = 0. \end{cases}$$

Then, f is both injective and surjective, i.e. bijective.

Definition 1.12 (Bijective).

A function $f : X \rightarrow Y$ is **bijective** if it is both injective and surjective.

Lemma 1.3. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, then $g \circ f$ is injective if both f and g are injective, and $g \circ f$ is surjective if both f and g are surjective.

1.4 Cardinality**Definition 1.13 (Cardinality).**

Let X, Y be sets.

- (1) We write $|X| \leq |Y|$ if $\exists$ an injective function $f : X \rightarrow Y$.
- (2) We write $|X| = |Y|$ (same **cardinality**) if $\exists$ a bijective function $f : X \rightarrow Y$.

Definition 1.14 (Inverse).

Let $f : X \rightarrow Y$ be a bijection. The **inverse** of f is $f^{-1} : Y \rightarrow X$ defined by letting $f^{-1}(y)$ be the unique $x \in X$ such that $f(x) = y$ for each $y \in Y$.

Note. $(f^{-1} \circ f)(x) = f^{-1}(f(x)) = f^{-1}(y) = x$ for all $x \in X$.

Remark. In fact, a function is invertible $\iff$ it is bijective.

Claim. f^{-1} is bijective.

Proof. If $x \in X$, then $f^{-1}(f(x)) = x$, hence $x \in \text{Range}(f^{-1})$. Thus, f^{-1} is surjective. If $f^{-1}(y_1) = f^{-1}(y_2)$ for some $y_1, y_2 \in Y$, since f is surjective, $\exists x_1, x_2 \in X$ such that $f(x_1) = y_1$ and $f(x_2) = y_2$. Thus, $f^{-1}(f(x_1)) = f^{-1}(f(x_2)) \implies x_1 = x_2$ since f is injective. Hence, $y_1 = f(x_1) = f(x_2) = y_2$, so f^{-1} is injective. Therefore, f^{-1} is bijective. $\square$

Proposition 1.4.

- (1) If $|X| = |Y|$ and $|Y| = |Z|$, then $|X| = |Z|$. Similarly, if $|X| \leq |Y|$ and $|Y| \leq |Z|$, then $|X| \leq |Z|$.
- (2) $|X| = |X|$.
- (3) If $|X| = |Y|$, then $|Y| = |X|$.

Proof.

- (1) Follows from composition of functions.
- (2) Define the function $X \ni x \mapsto x \in X$, which is bijective.
- (3) Suppose $|X| = |Y|$. Then, $\exists$ a bijection $f : X \rightarrow Y$. The inverse $f^{-1} : Y \rightarrow X$ is also a bijection, so $|Y| = |X|$. $\square$

Example. If X and Y are finite sets, then $|X| = |Y| \iff X$ and Y have the same number of elements.

Exercise: Prove the following example.

Example. $|\mathbb{N}| = |\mathbb{Z}|$.

Theorem 1.5 (Cantor's Theorem).

If X is any set, then $|X| \not\leq |\mathcal{P}(X)|$.

Proof. Since $f : X \rightarrow \mathcal{P}(X)$ defined by $f(x) = \{x\}$ is injective, $|X| \leq |\mathcal{P}(X)|$.

Claim. f is not surjective, i.e. $|X| \not\leq |\mathcal{P}(X)|$.

Proof of Claim. Define $A = \{x \in X : x \notin f(x)\}$. If $x \in X$, then either $x \in A$ so $x \notin f(x) \implies f(x) \neq A$, or $x \notin A$ so $x \in f(x) \implies f(x) \neq A$. Hence, $A \notin f(X)$ so f is not surjective. $\square$

Therefore, $|X| \not\leq |\mathcal{P}(X)|$, i.e. $|X| < |\mathcal{P}(X)|$. $\square$

Lecture 4, 2026/01/12

Theorem 1.6 (Cantor's Theorem 2).

$|\mathbb{N}| < |\mathbb{R}|$.

Proof. Since $\mathbb{N} \subset \mathbb{R}$, the identity function $f : \mathbb{N} \rightarrow \mathbb{R}$ defined by $f(n) = n$ is injective, so $|\mathbb{N}| \leq |\mathbb{R}|$.

Suppose $f : \mathbb{N} \rightarrow (0, 1)$ is defined by

$$f(1) = 0.a_1^1 a_2^1 a_3^1 \dots$$

$$f(2) = 0.a_1^2 a_2^2 a_3^2 \dots$$

$$f(3) = 0.a_1^3 a_2^3 a_3^3 \dots$$

$\vdots$

such that $a_i \in \{0, 1, \dots, 9\}$ not ending in all 9s. Define $x = 0.b_1 b_2 b_3 \dots$ where

$$b_i = \begin{cases} a_i^i + 1, & a_i^i \neq 9, \\ a_i^i - 1, & a_i^i = 9. \end{cases}$$

Then, $x \neq f(1)$ since $b_1 \neq a_1^1$, $x \neq f(2)$ since $b_2 \neq a_2^2$, ..., $x \neq f(n)$ since $b_n \neq a_n^n$ for all $n \in \mathbb{N}$. Thus, $x \notin f(\mathbb{N})$, so f is not surjective. Hence, $|\mathbb{N}| \not\geq |\mathbb{R}|$. Therefore, $|\mathbb{N}| < |\mathbb{R}|$. $\square$

Remark. $|\mathbb{R}| = |\mathcal{P}(\mathbb{N})|$.

Theorem 1.7. $|\mathbb{N}| = |\mathbb{Q}|$.

Proof. We first prove that $|\mathbb{N}| = |\mathbb{Q}_{>0}|$. Note that

	1	2	3	4	...
1	1/1	2/1	3/1	4/1	...
2	1/2	2/2	3/2	4/2	...
3	1/3	2/3	3/3	4/3	...
4	1/4	2/4	3/4	4/4	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Exercise: Finish the rest. $\square$

Definition 1.15 (Countable Set).

- (1) A set X is **countable** if $|X| \leq |\mathbb{N}|$.
- (2) A set X is **denumerable** or **countably infinite** if it is countable but not finite, i.e. $|X| = |\mathbb{N}|$.

Definition 1.16 (Complement).

If $Y \subset X$, then $X \setminus Y = \{x \in X : x \notin Y\}$ is the **complement** of Y in X .

Theorem 1.8 (Cantor-Schroeder-Bernstein Theorem).

Let X and Y be sets. If $|X| \leq |Y|$ and $|Y| \leq |X|$, then $|X| = |Y|$.

Proof. Let $f : X \rightarrow Y$ and $g : Y \rightarrow X$ be injections. Set

$$B = (Y \setminus f(X)) \cup \bigcup_{n \geq 1} (f \circ g)^n (Y \setminus f(X)) \subset Y, \quad A = X \setminus g(B).$$

Note that

$$B = (f \circ g)(B) \cup (Y \setminus f(X)) \quad \text{and} \quad g(B) = X \setminus A.$$

Then, $f(A) = f(X) \setminus f(g(B))$ because f is injective. Then,

$$f(A) = Y \setminus ((Y \setminus f(X)) \cup (f \circ g)(B)) = Y \setminus B.$$

Define $h : X \rightarrow Y$ by

$$h(x) = \begin{cases} f(x), & x \in A, \\ g^{-1}(x), & x \in X \setminus A = g(B). \end{cases}$$

Exercise: Show that h is a bijection, so $|X| = |Y|$. □

1.5 Axiom of Choice and Zorn's Lemma

Note that if X, Y are non-empty sets, then $X \times Y \neq \emptyset$.

Question: Why?

Answer: Take $x \in X$ and $y \in Y$. Then, $(x, y) \in X \times Y$, so $X \times Y \neq \emptyset$.

Using induction, we can show that $X_1 \times X_2 \times \cdots \times X_n \neq \emptyset$ if $|X_i| \neq 0$ for each i .

If I is a set and for each $i \in I$ we have a set X_i , we define

$$\prod_{i \in I} X_i = \left\{ f : I \rightarrow \bigcup_{i \in I} X_i \mid f(i) \in X_i \text{ for all } i \in I \right\}.$$

Note that $X \times Y = \{f : \{1, 2\} \rightarrow X \cup Y \mid f(1) \in X, f(2) \in Y\}$.

Question: If $I \neq \emptyset$ and $X_i \neq \emptyset$ for each $i \in I$, then is $\prod_{i \in I} X_i \neq \emptyset$?

Answer: This is true if I is finite. For I infinite, this is still true but we cannot prove it. We take it as an axiom, i.e. the **Axiom of Choice**.

Definition 1.17 (Partial Ordering).

A **partial ordering** on a set X is a relation $<$ such that $<$ is

- (1) Reflexive: $x < x$ for all $x \in X$,
- (2) Transitive: $x < y$ and $y < z \implies x < z$ for all $x, y, z \in X$,
- (3) Antisymmetric: $x < y$ and $y < x \implies x = y$ for all $x, y \in X$.

Example. $(\mathbb{N}, \leq)$ is a partial ordering.

Example. $(\mathcal{P}(X), \subset)$ is a partial ordering.

Definition 1.18 (Chain, Upper Bound).

If $(X, <)$ is a partial ordering, a **chain** is a subset $\mathcal{C} \subset X$ such that $\forall x, y \in \mathcal{C}$, either $x < y$ or $y < x$. If $\mathcal{C} \subset X$, a point $x \in X$ is an **upper bound** in $\mathcal{C}$ if $\forall a \in \mathcal{C}$, we have $a < x$.

Definition 1.19 (Maximal).

An element $x \in X$ is **maximal** if $x < y \implies x = y$ for all $y \in X$.

Lemma 1.9 (Zorn's Lemma).

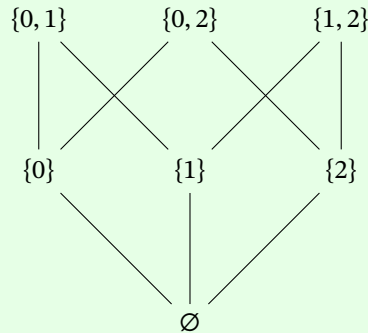
If $(X, <)$ is a partial ordering such that every chain has an upper bound, then X has a maximal element.

Note. Axiom of choice $\implies$ Zorn's lemma $\implies$ Axiom of choice. Thus, they are equivalent.

Example. Let $I = \mathcal{P}(\{0, 1, 2\}) \setminus \{\{0, 1, 2\}\}$. Then, $(I, \subset)$ is a partial ordering.

Let $\mathcal{C}_1 = \{\emptyset, \{0\}, \{0, 1\}\}$ and $\mathcal{C}_2 = \{\emptyset, \{1\}, \{0, 1\}\}$. Then, $\mathcal{C}_1$ and $\mathcal{C}_2$ are maximal chains in I and $\{0, 1\}$ is an upper bound for $\mathcal{C}_1$ and $\mathcal{C}_2$.

$\{0, 1\}$, $\{0, 2\}$, and $\{1, 2\}$ are maximal elements for this partial ordering.



Question: What about $\{0, 1, 2\}$? It is not in I !

Question: For Zorn's Lemma, what about $X = \emptyset$ and $< = \emptyset$?

Answer: Note that $(\emptyset, \emptyset)$ does not satisfy the hypothesis of Zorn's Lemma since the chain $\emptyset$ does not have an upper bound. Thus, Zorn's Lemma does not apply here.

Example. For $[0, \infty)$ with the usual ordering $\leq$, the chain $[0, \infty)$ does not have an upper bound.

Theorem 1.10. If X is infinite, then X contains a denumerable subset.

Proof. Since X is infinite, $\exists x_1 \in X$. We inductively define $x_n \in X$ by choosing x_n a point in $X \setminus \{x_1, x_2, \dots, x_{n-1}\}$ for $n \geq 2$. We have $f : \mathbb{N} \rightarrow X$ given by $f(n) = x_n$. If $n < m$, then $x_m \in X \setminus \{x_1, x_2, \dots, x_{m-1}\}$ so $f(n) = x_n \neq x_m = f(m)$. Thus, f is injective, so $|\mathbb{N}| \leq |X|$, i.e. $f(\mathbb{N}) \subset X$ and $f(\mathbb{N})$ is denumerable. $\square$

Theorem 1.11. Zorn's Lemma $\implies$ Axiom of Choice.

Proof. Axiom of Choice: If X is a set, then $\exists f : \mathcal{P}(X) \setminus \{\emptyset\} \rightarrow X$ such that $f(A) \in A$ for all $A \in \mathcal{P}(X) \setminus \{\emptyset\}$. Let $\mathcal{F} = \{f : \mathcal{E} \rightarrow X \mid \mathcal{E} \subset \mathcal{P}(X) \setminus \{\emptyset\} \text{ and } f(A) \in A \ \forall A \in \mathcal{E}\}$.

We will come back and finish the proof... □

Theorem 1.12 (Comparison Lemma).

If X and Y are sets, then either $|X| \leq |Y|$ or $|Y| \leq |X|$.

Proof. Let $\mathcal{F} = \{f : A \rightarrow Y \mid A \subset X \text{ and } f \text{ is injective}\}$. Define $f < g$ if $\text{Dom}(f) \subset \text{Dom}(g)$ and $g|_{\text{Dom}(f)} = f$. To check that $<$ is a partial ordering on $\mathcal{F}$:

- $f < f$ since $\text{Dom}(f) \subset \text{Dom}(f)$ and $f|_{\text{Dom}(f)} = f$.
- If $f < g$ and $g < h$, then $\text{Dom}(f) \subset \text{Dom}(g) \subset \text{Dom}(h)$, $h|_{\text{Dom}(g)} = g$ and $g|_{\text{Dom}(f)} = f$, so $h|_{\text{Dom}(f)} = f$. Thus, $f < h$.
- If $f < g$ and $g < f$, then $\text{Dom}(f) \subset \text{Dom}(g) \subset \text{Dom}(f)$, so $\text{Dom}(f) = \text{Dom}(g)$. Also, $g|_{\text{Dom}(f)} = f$ and $f|_{\text{Dom}(g)} = g$, so $f = g$.

If $\mathcal{C} \subset \mathcal{F}$ is a chain, set $D = \bigcup_{f \in \mathcal{C}} \text{Dom}(f) \subset X$. Define $\tilde{f} : D \rightarrow Y$ by $\tilde{f}(x) = f(x)$ if $x \in \text{Dom}(f)$ for some $f \in \mathcal{C}$. Note that if $x \in \text{Dom}(f)$ and $x \in \text{Dom}(g)$ for some $f, g \in \mathcal{C}$, then either $f < g$ or $g < f$. In both cases, $f(x) = g(x)$. So, $\tilde{f}$ is well-defined. If $f \in \mathcal{C}$, then $\forall x \in \text{Dom}(f)$, we have $\tilde{f}(x) = f(x)$. Therefore $\text{Dom}(f) \subset \text{Dom}(\tilde{f})$ and $\tilde{f}|_{\text{Dom}(f)} = f$. Note that if $x, y \in D$ such that $\tilde{f}(x) = \tilde{f}(y)$, then $x \in \text{Dom}(f), y \in \text{Dom}(g)$ for some $f, g \in \mathcal{C}$. WLOG, say $f < g$. Then, $x \in \text{Dom}(g)$ and g is injective, so $g(x) = \tilde{f}(x) = \tilde{f}(y) = g(y) \implies x = y$. Thus, $\tilde{f}$ is injective, so $\tilde{f} \in \mathcal{F}$ and hence an upper bound for $\mathcal{C}$, i.e. $f < \tilde{f}$ for all $f \in \mathcal{C}$.

By Zorn's Lemma, $\exists$ a maximal element $f \in \mathcal{F}$.

- Case 1: $\text{Dom}(f) = X$. Then, $|X| \leq |Y|$.
- Case 2: $\text{Range}(f) = Y$, then $f^{-1} : Y \rightarrow \text{Dom}(f) \subset X$ is injective, so $|Y| \leq |X|$.
- Case 3: $X \neq \text{Dom}(f)$ and $Y \neq \text{Range}(f)$. Take $x_0 \in X \setminus \text{Dom}(f)$ and $y_0 \in Y \setminus \text{Range}(f)$.

Define $g : \text{Dom}(f) \cup \{x_0\} \rightarrow Y$ by

$$g(x) = \begin{cases} y_0, & x = x_0, \\ f(x), & x \in \text{Dom}(f). \end{cases}$$

Then, g is injective, so $g \in \mathcal{F}$ and $f < g$ and $f \neq g$. Since f is maximal, we have a contradiction. Thus, this case cannot happen. □

Theorem 1.13. If X is infinite, then $|X| = |\{0, 1\} \times X|$.

Proof. Let $\mathcal{F} = \{f : A \rightarrow \{0, 1\} \times A \mid f \text{ is bijective, } A \subset X, \text{ and } A \neq \emptyset\}$. Note that if $D \subset X$ is denumerable, then $|D| = |\{0, 1\} \times D|$ (check?). So $\mathcal{F} \neq \emptyset$.

Exercise: Finish the rest. Once get a maximal element, do 2 cases similar to the previous proof. □

Lecture 6, 2026/01/16

Definition 1.20 (Linearly Independent, Span, Basis).

Let V be a vector space over a field F . A set $\beta \subset V$ is **linearly independent** if whenever $v_1, \dots, v_n \in \beta$ are distinct and $a_1, \dots, a_n \in F$ such that

$$\sum_{i=1}^n a_i v_i = 0,$$

then, $a_i = 0$ for all $i = 1, \dots, n$.

A set β **spans** V if

$$V = \left\{ \sum_{i=1}^n a_i v_i \mid v_i \in \beta \right\}.$$

A set β is a **basis** of V if β is linearly independent and spans V .

Theorem 1.14. If $\beta \subset V$ is linearly independent and no strictly larger set is linearly independent, then β spans V .

Proof. Suppose not. Then, $\exists v \in V \setminus \text{Span}(\beta)$. We will show that $\beta \cup \{v\}$ is linearly independent, contradicting the maximality of β .

Exercise: Finish the proof. □

Application of Zorn's Lemma:

Theorem 1.15. Every vector space V over a field F has a basis.

Proof. Let $\mathcal{F} = \{\beta \subset V \mid \beta \text{ is linearly independent}\}$ with partial ordering given by set inclusion. Note that $\emptyset \in \mathcal{F}$, so $\mathcal{F} \neq \emptyset$. Let $\mathcal{C} \in \mathcal{F}$ be a chain. Set $\beta = \bigcup \mathcal{C} \subset V$. Note that if $\alpha \in \mathcal{C}$, then $\alpha \subset \beta$. Suppose $v_1, \dots, v_n \in \beta$ are distinct and $a_1, \dots, a_n \in F$ such that

$$\sum_{i=1}^n a_i v_i = 0.$$

Since $v_i \in \beta$, $\exists \beta_i \in \mathcal{C}$ such that $v_i \in \beta_i$ for each $1 \leq i \leq n$. Note that $\{\beta_1, \dots, \beta_n\}$ is a finite chain and hence has a maximal element $\beta_0 \in \mathcal{C}$ and $v_1, \dots, v_n \in \beta_0$. Since β_0 is linearly independent, we have $a_1 = \dots = a_n = 0$. Thus, β is linearly independent, so $\beta \in \mathcal{F}$ and is an upper bound for $\mathcal{C}$. By Zorn's Lemma, $\exists$ a maximal element $\beta \in \mathcal{F}$. By the previous theorem, β spans V . Thus, β is a basis. □

1.6 Construction of the Real Numbers

Definition 1.21 (Sequence).

A **sequence** in a set X is a function from $\mathbb{N}$ to X , denoted by $\{a_n\}_{n \in \mathbb{N}}$ or $(a_n)_n$.

Definition 1.22 (Cauchy Sequence).

A sequence $\{a_n\}_{n \in \mathbb{N}}$ in $\mathbb{Q}$ is **Cauchy** if $\forall \epsilon > 0, \epsilon \in \mathbb{Q}, \exists N \in \mathbb{N}$ such that $\forall n, m \geq N$, we have $|a_n - a_m| < \epsilon$.

Definition 1.23 (Convergence).

A sequence $\{a_n\}_{n \in \mathbb{N}}$ in $\mathbb{Q}$ **converges** to $r \in \mathbb{Q}$ if $\forall \epsilon > 0, \epsilon \in \mathbb{Q}, \exists N \in \mathbb{N}$ such that $\forall n \geq N$, we have $|r - a_n| < \epsilon$.

Proposition 1.16. Every convergent sequence is Cauchy.

Proof. Suppose $(a_n)_n$ converges to r (we will write $a_n \rightarrow r$ or $\lim_{n \rightarrow \infty} a_n = r$). For $\epsilon > 0$, since $a_n \rightarrow r$, $\exists N \in \mathbb{N}$ s.t. for $n \geq N$, $|a_n - r| < \frac{\epsilon}{2}$. Then, for $n, m \geq N$,

$$|a_n - a_m| = |a_n - r + r - a_m| \leq |a_n - r| + |r - a_m| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

□

Definition 1.24 (Bounded Sequence).

A sequence $\{a_n\}_{n \in \mathbb{N}}$ is **bounded** if $\exists M \in \mathbb{N}$ such that $|a_n| \leq M$ for all $n \in \mathbb{N}$.

Lemma 1.17. Every Cauchy sequence is bounded.

Proof. Since $(a_n)_n$ is Cauchy, $\exists N \in \mathbb{N}$ such that $|a_n - a_m| < 1 \forall n, m \geq N$. Set M to be any natural number bigger than $|a_1|, |a_2|, \dots, |a_{N-1}|, |a_N| + 1$. If $n \geq N$, then

$$|a_n| = |a_n - a_N + a_N| \leq |a_n - a_N| + |a_N| < 1 + |a_N| \leq M.$$

□

Definition 1.25 (Equivalence).

Given $(a_n)_n$ and $(b_n)_n$ Cauchy sequences in $\mathbb{Q}$, we say that $(a_n)_n$ is **equivalent** to $(b_n)_n$, denoted by $(a_n)_n \sim (b_n)_n$, if $\lim_{n \rightarrow \infty} (a_n - b_n) = 0$.

Note. $\sim$ is an equivalence relation.

Proof.

- $(a_n)_n \sim (a_n)_n$ since $\lim_{n \rightarrow \infty} (a_n - a_n) = 0$.
- If $(a_n)_n \sim (b_n)_n$, then $\lim_{n \rightarrow \infty} (b_n - a_n) = -0 = 0$, so $(b_n)_n \sim (a_n)_n$.
- If $(a_n)_n \sim (b_n)_n$ and $(b_n)_n \sim (c_n)_n$, then $\lim_{n \rightarrow \infty} (a_n - b_n) = 0 = \lim_{n \rightarrow \infty} (b_n - c_n)$, so $\lim_{n \rightarrow \infty} (a_n - c_n) = 0$ (since a sum of convergent sequences converges to the sum of the limits), hence $(a_n)_n \sim (c_n)_n$.

So, $\sim$ is an equivalence relation on $\mathcal{C} = \{(a_n)_n \mid (a_n)_n \text{ is Cauchy}\}$.

□

Definition 1.26 (Real Numbers).

$\mathbb{R} = \mathcal{C} / \sim = \{[(a_n)_n]_{\sim} : (a_n)_n \in \mathcal{C}\}$ is the set of **real numbers**, where $[(a_n)_n]_{\sim} = \{(b_n)_n \in \mathcal{C} : (b_n)_n \sim (a_n)_n\}$ is the equivalence class of $(a_n)_n$.

Example. $1 = [(1)_n]$. If $q \in \mathbb{Q}$, then we represent q in $\mathbb{R}$ as $[(q)_n]$ so we consider $\mathbb{Q} \subset \mathbb{R}$.

Example. Consider π . Consider $a_1 = 3, a_2 = 3.1, a_3 = 3.14, \dots$. Then,

$$|a_n - a_{n+1}| \leq 10^{-n+1} \implies |a_n - a_{n+k}| < 10^{-n+1} + 10^{-n} + \dots + 10^{-n+k-1} \leq 10^{-n+2}.$$

So if $\epsilon > 0$ is given, take N such that $10^{-N+2} < \epsilon$. Then $\forall n, m \geq N$, we have $|a_n - a_m| < \epsilon$.

Thus, $(a_n)_n$ is Cauchy and we define $\pi = [(a_n)_n]$.

Definition 1.27 (Addition, Multiplication, Division).

If $x = [(x_n)_n], y = [(y_n)_n] \in \mathbb{R}$, then

- $x + y = [(x_n + y_n)_n]$.
- $x \cdot y = [(x_n \cdot y_n)_n]$.
- If $y \neq 0$, then $\frac{x}{y} = \left[\left(\frac{x_n}{y_n} \right)_n \right]$.

Exercise: Check that $x + y, x \cdot y$, and $\frac{x}{y}$ are well-defined.

Theorem 1.18. $(\mathbb{R}, +, \cdot, 0, 1)$ is a field.

Definition 1.28 (Ordering).

If $x = [(x_n)_n], y = [(y_n)_n] \in \mathbb{R}$, we say that $x \leq y$ if either $x = y$ or $\exists N \in \mathbb{N}$ s.t. $x_n \leq y_n$ $\forall n \geq N$.

Exercise: Check that $\leq$ is well-defined.

Lemma 1.19. If $x, y \in \mathbb{R}$, then either $x \leq y$ or $y \leq x$.

Theorem 1.20 (Least Upper Bound Principle).

If $A \subset \mathbb{R}$ with $A \neq \emptyset$ such that A has an upper bound (i.e. $\exists n \in \mathbb{R}$ s.t. $a \leq n \forall a \in A$), then $\exists r \in \mathbb{R}$ such that r is an upper bound for A and if s is any other upper bound for A , then $r \leq s$.

Note. We write $r = \sup A$. We may also define a greatest lower bound for a non-empty set with a lower bound, denoted by $\inf A$.

Proof. Assume that $A \cap [0, \infty) \neq \emptyset$ by replacing A with $\{a + m : a \in A\}$ for some $m \in \mathbb{R}$. Let $n \in \mathbb{N}$ be the smallest natural number that is an upper bound for A . Construct a real number as follows.

Let $a_1 = n - 1$. Note that a_1 is not an upper bound for A but $a_1 + 1$ is. Choose the smallest number $K \in \{1, \dots, 9, 10\}$ such that $a_1 + \frac{K}{10}$ is an upper bound for A . Then $a_1 + \frac{K}{10}$ is an upper bound, but $a_1 + \frac{K-1}{10}$ is not. Set $a_2 = a_1 + \frac{K-1}{10}$. We construct $a_k \in \mathbb{Q}$ such that a_k is not an upper bound but $a_k + \frac{1}{10^{k-1}}$ is. Having constructed $a_1, \dots, a_k$, let m be the smallest number in $\{1, \dots, 10\}$ such that $a_k + \frac{m}{10^{k+1}}$ is an upper bound. Set $a_{k+1} = a_k + \frac{m-1}{10^k}$. Note that $|a_k - a_{k+1}| \leq 10^{-k}$. Thus,

$$\begin{aligned} |a_k - a_{k+j}| &\leq |a_k - a_{k+1}| + |a_{k+1} - a_{k+2}| + \dots + |a_{k+j-1} - a_{k+j}| \\ &\leq 10^{-k} + 10^{-k-1} + \dots + 10^{-k+j} \leq 10^{-k+1}. \end{aligned}$$

Thus, $\{a_k\}_k$ is a Cauchy sequence of rational numbers. Set $r = [\{a_k\}_k] = \left[\left\{ a_k + \frac{1}{10^{k-1}} \right\}_k \right]$. If $a \in A$ s.t. $r < a$, then we may choose $b \in \mathbb{Q}$ with $r < b < a$. Then, $\exists N \in \mathbb{N}$ such that $a_k + \frac{1}{10^{k-1}} < b < a$ for all $k \geq N$. However $a_k + \frac{1}{10^{k-1}}$ is an upper bound, giving a contradiction. So, r is an upper bound. Suppose s is an upper bound with $s < r$. We may assume $s \in \mathbb{Q}$, then $\exists N \in \mathbb{N}$ s.t. for $k \geq N$, $s < a_k$. However, a_k is not an upper bound, giving a contradiction. Thus, $r \leq s$ for any upper bound s . $\square$

2 Metric Spaces

2.1 Metric Space and Examples

Definition 2.1 (Metric, Metric Space).

Let X be a set. A **metric** on X is a function $d : X \times X \rightarrow [0, \infty)$ such that

- (1) $d(x, y) = 0 \iff x = y$.
- (2) $d(x, y) = d(y, x)$ for all $x, y \in X$ (symmetry).
- (3) $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$ (triangle inequality).

The pair (X, d) is called a **metric space**.

Example. Let X be any set. Then,

$$d(x, y) = \begin{cases} 0, & \text{if } x = y, \\ 1, & \text{if } x \neq y. \end{cases}$$

is a metric on X . This is called the **discrete metric**.

Example. Let X be any set. Let $n > 0$ and define

$$d'(x, y) = \begin{cases} 0, & \text{if } x = y, \\ n, & \text{if } x \neq y. \end{cases}$$

Then, d' is also a metric on X .

Example. Let X be any finite set. Define $d_H : \mathcal{P}(X) \times \mathcal{P}(X) \rightarrow [0, \infty)$ by

$$d_H(A, B) = |A \Delta B|,$$

where $A \Delta B = (A \setminus B) \cup (B \setminus A)$. Then, d_H is a metric on $\mathcal{P}(X)$, called the **Hamming metric**.

Proof. Note that

- (1) $A \Delta A = \emptyset$, so $d_H(A, A) = 0$. If $A \neq B$, then $A \Delta B \neq \emptyset$, so $d_H(A, B) > 0$.
- (2) d_H is symmetric since $A \Delta B = B \Delta A$.
- (3) **Exercise:** check $d_H(A, C) \leq d_H(A, B) + d_H(B, C)$. □

Example. Let $X = \mathbb{R}^n$. Then,

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}$$

is the **Euclidean metric** on $\mathbb{R}^n$ and $(\mathbb{R}^n, d)$ is the **Euclidean space**.

Example. Let $X = \mathbb{N}^2$. Then,

$$d(x, y) = |x_1 - y_1| + |x_2 - y_2|$$

is the **Manhattan metric** on $\mathbb{N}^2$.

Example. Let $X = \mathbb{R}^n$. Then,

$$d_1(x, y) = \sum_{i=1}^n |x_i - y_i|$$

is the ℓ^1 **metric** on $\mathbb{R}^n$.

Example. Let $X = \mathbb{R}^n$. Then,

$$d_\infty(x, y) = \max_{1 \leq i \leq n} \{|x_i - y_i|\}$$

is the ℓ^∞ **metric** on $\mathbb{R}^n$.

Lecture 8, 2026/01/21

Example. Let $X = \mathbb{R}^2$. Then,

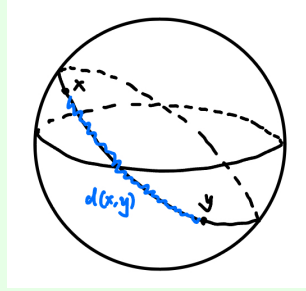
$$d_F(x, y) = \begin{cases} d_2(x, y), & \text{if } x, y, \text{ and } 0 \text{ are collinear,} \\ d_2(x, 0) + d_2(0, y), & \text{otherwise.} \end{cases}$$

is the **French train system metric**.

Example. Let X be a manifold (e.g. $S^d \subset \mathbb{R}^{d+1}$ the unit sphere). Define

$$d(x, y) = \text{length of the shortest path from } x \text{ to } y \text{ on the manifold.}$$

Then, d is a metric on X , called the **geodesic metric**.



Example. Let $X = \{0\}$. Define $d : X \times X \rightarrow [0, \infty)$ by $d(0, 0) = 0$. Then, (X, d) is a metric space.

Example. Let S = sequences of real numbers. Define

$$d((x_n)_n, (y_n)_n) = \begin{cases} n, & \text{if } n \text{ is the smallest natural number s.t. } x_n \neq y_n, \\ 0, & \text{if } (x_n)_n = (y_n)_n. \end{cases}$$

This is NOT a metric on S since the triangle inequality fails.

Note. Try replacing n with $\frac{1}{2}$ or $\frac{1}{2^n}$ and check if it becomes a metric.

Example. Let $X \subset \mathbb{R}^d$ be closed, bounded, and non-empty. Let $\mathcal{H}_d$ denote all closed non-empty subsets of X . If $x \in X$ and $A \in \mathcal{H}_d$, define

$$d_2(x, A) = \inf_{y \in A} d_2(x, y).$$

The **Hausdorff distance** (metric) between $A, B \in \mathcal{H}_d$ is defined by

$$d_H(A, B) = \max \left\{ \sup_{x \in A} d_2(x, B), \sup_{y \in B} d_2(y, A) \right\}.$$

Example. A graph is a pair (V, E) where V is a set (vertices) and $E \subset V^2$ (edges) such that $(a, b) \in E \implies (b, a) \in E$ and such that $(a, a) \notin E \forall a \in V$. Assume also that the graph is connected, i.e. for any $v_1, v_2 \in V$, there exists a path from v_1 to v_2 .

Define $d : V \times V \rightarrow [0, \infty)$ by

$$d(v_1, v_2) = \text{length of the shortest path from } v_1 \text{ to } v_2.$$

A path is a sequence of vertices $w_1, w_2, \dots, w_k$ such that $(w_i, w_{i+1}) \in E$ for all $1 \leq i < k$, and the length of the path is $k - 1$. Then, d is a metric on V .

2.2 Normed Space and Examples

Definition 2.2 (Norm).

Let V be a vector space over $\mathbb{R}$. A **norm** on V is a function $\| \cdot \| : V \rightarrow [0, \infty)$ such that

- (1) $\|v\| = 0 \iff v = 0$.
- (2) $\|av\| = |a|\|v\|$ for all $a \in \mathbb{R}$ and $v \in V$.
- (3) $\|v + w\| \leq \|v\| + \|w\|$ for all $v, w \in V$ (triangle inequality).

Example. Let $V = \mathbb{R}^d$. Then,

$$\|v\|_2 = \sqrt{\sum_{i=1}^d |v_i|^2} = \sqrt{v_1^2 + v_2^2 + \dots + v_d^2}$$

is the **Euclidean norm** on $\mathbb{R}^d$.

Definition 2.3 (Normed Space).

A **normed space** is a pair $(V, \| \cdot \|)$ where V is a vector space over $\mathbb{R}$ and $\| \cdot \|$ is a norm on V .

Example. If $(V, \| \cdot \|)$ is a normed space, then define

$$d_{\| \cdot \|}(v, w) = \|v - w\| \quad \forall v, w \in V.$$

Proposition 2.1. $d_{\|\cdot\|}$ is a metric on V .

Proof. Note that

$$(1) \quad d(v, w) = 0 \iff \|v - w\| = 0 \iff v - w = 0 \iff v = w.$$

$$(2) \quad d(v, w) = \|v - w\| = \|w - v\| = d(w, v).$$

$$(3) \quad d(v_1, v_2) + d(v_2, v_3) = \|v_1 - v_2\| + \|v_2 - v_3\| \\ \geq \|(v_1 - v_2) + (v_2 - v_3)\| = \|v_1 - v_3\| = d(v_1, v_3)$$

Thus, $d_{\|\cdot\|}$ is a metric on V . □

Lecture 9, 2026/01/23

Example. Let $V = \mathbb{R}^d$. Then,

$$\|v\|_1 = \sum_{i=1}^d |v_i|$$

is the 1-**norm** on $\mathbb{R}^d$.

Proof. Note that

$$(1) \quad \|v\|_1 = 0 \iff \sum_{i=1}^d |v_i| = 0 \iff v = (0, 0, \dots, 0).$$

$$(2) \quad \|\alpha v\|_1 = \sum_{i=1}^d |\alpha v_i| = |\alpha| \sum_{i=1}^d |v_i| = |\alpha| \|v\|_1.$$

$$(3) \quad \|v + w\|_1 = \sum_{i=1}^d |v_i + w_i| \leq \sum_{i=1}^d (|v_i| + |w_i|) = \sum_{i=1}^d |v_i| + \sum_{i=1}^d |w_i| = \|v\|_1 + \|w\|_1.$$

□

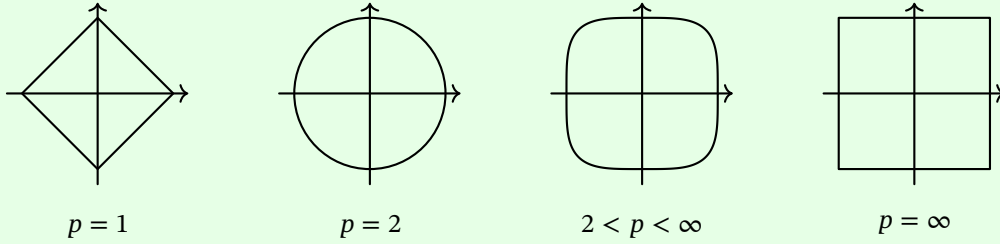
Example. Let $V = \mathbb{R}^d$. Then,

$$\|v\|_\infty = \max_{1 \leq i \leq d} \{|v_i|\}$$

is the ∞ -**norm** on $\mathbb{R}^d$.

Example. Let $V = \mathbb{R}^2$. The unit ball is $B_1 = \{v \in V : \|v\| \leq 1\}$. Then, the unit balls for $\|\cdot\|_2$,

$\|\cdot\|_1$, $\|\cdot\|_\infty$, and $\|\cdot\|_p$ (see the next example) look like:



Example. Fix $1 \leq p < \infty$. Let $V = \mathbb{R}^d$. Then,

$$\|v\|_p = \left(\sum_{i=1}^d |v_i|^p \right)^{1/p}$$

is the **p -norm** on $\mathbb{R}^d$.

Proof. Note that

(1) If $\|v\|_p = 0$, then $0 = \|v\|_p^p = \sum_{i=1}^d |v_i|^p \implies v_i = 0$ for all i . Conversely, if $v = 0$, then $\|v\|_p = 0$.

(2) Fix $v \in \mathbb{R}^d$ and $\alpha \in \mathbb{R}$. Then,

$$\|\alpha v\|_p^p = \sum_{i=1}^d |\alpha v_i|^p = |\alpha|^p \sum_{i=1}^d |v_i|^p = |\alpha|^p \|v\|_p^p \implies \|\alpha v\|_p = |\alpha| \|v\|_p.$$

(3) See below (Minkowski's inequality). □

Question: What about $0 < p < 1$?

Answer: It fails the triangle inequality, so it is NOT a norm.

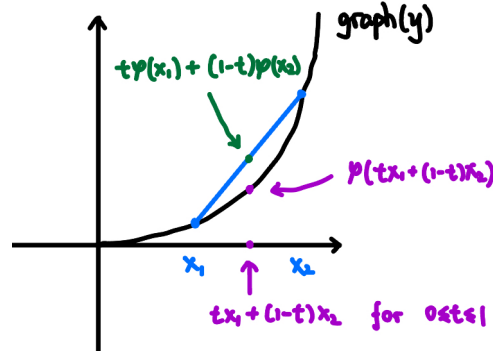
Theorem 2.2 (Minkowski's Inequality).

If $v, w \in \mathbb{R}^d$, then $\|v + w\|_p \leq \|v\|_p + \|w\|_p$ for all $1 \leq p < \infty$.

Proof. Fix $1 \leq p < \infty$. Consider the function $\varphi : [0, \infty) \rightarrow [0, \infty)$ given by $\varphi(x) = x^p$. Then,

$$\varphi'(x) = px^{p-1}, \quad \varphi''(x) = p(p-1)x^{p-2} > 0 \quad \forall x \in (0, \infty).$$

So, φ is a convex function.



So,

$$\varphi(tx_1 + (1-t)x_2) \leq t\varphi(x_1) + (1-t)\varphi(x_2) \quad \forall 0 \leq t \leq 1, 0 \leq x_1 \leq x_2 < \infty.$$

Fix $v, w \in \mathbb{R}^d$. Set $A = \|v\|_p$ and $B = \|w\|_p$. If $v = 0$ or $w = 0$, then the result holds. Assume $v \neq 0$ and $w \neq 0$. Set $v_0 = \frac{1}{A}v$ and $w_0 = \frac{1}{B}w$. Then, $\|v_0\|_p = 1 = \|w_0\|_p$. Consider the i -th coordinate of v_0 and w_0 , denoted by $(v_0)_i$ and $(w_0)_i$. Note that

$$\left(\frac{A}{A+B} |(v_0)_i| + \frac{B}{A+B} |(w_0)_i| \right)^p \leq \frac{A}{A+B} |(v_0)_i|^p + \frac{B}{A+B} |(w_0)_i|^p \quad \text{by convexity of } \varphi.$$

Then,

$$\begin{aligned} \|v + w\|_p^p &= \sum_{i=1}^d |v_i + w_i|^p \leq \sum_{i=1}^d (|v_i| + |w_i|)^p \\ &= \sum_{i=1}^d (A|(v_0)_i| + B|(w_0)_i|)^p \\ &= (A+B)^p \sum_{i=1}^d \left(\frac{A}{A+B} |(v_0)_i| + \frac{B}{A+B} |(w_0)_i| \right)^p \\ &\leq (A+B)^p \sum_{i=1}^d \left(\frac{A}{A+B} |(v_0)_i|^p + \frac{B}{A+B} |(w_0)_i|^p \right) \\ &= \frac{(A+B)^p}{A+B} (A\|v_0\|_p^p + B\|w_0\|_p^p) \\ &= (A+B)^p = (\|v\|_p + \|w\|_p)^p. \end{aligned}$$

Taking the p -th root gives the inequality. □

Definition 2.4 (ℓ_p^d Space).

We denote the normed space $(\mathbb{R}^d, \|\cdot\|_p)$ by ℓ_p^d .

Note. Other notations include $\ell_p(\{1, \dots, d\})$ or $\ell^p(d)$.

Definition 2.5 ($\ell_p(\mathbb{N})$).

For $1 \leq p < \infty$, define $\ell_p(\mathbb{N}) := \{(a_n)_n \subset \mathbb{R} : \sum_{n=1}^{\infty} |a_n|^p < \infty\}$.

Proposition 2.3. $\ell_p(\mathbb{N})$ is a vector space with the potential structure and we have a norm on $\ell_p(\mathbb{N})$ given by

$$\|(a_n)_n\|_p = \left(\sum_{n=1}^{\infty} |a_n|^p \right)^{1/p}.$$

Proof. For the triangle inequality, note that

$$\sum_{n=1}^{\infty} |a_n|^p := \lim_{N \rightarrow \infty} \sum_{n=1}^N |a_n|^p.$$

If $v = (a_n)_n$ and $w = (b_n)_n$ are in $\ell_p(\mathbb{N})$, then

$$\begin{aligned} \|v + w\|_p &= \lim_{N \rightarrow \infty} \left(\sum_{n=1}^N |a_n + b_n|^p \right)^{1/p} \leq \lim_{N \rightarrow \infty} \left(\sum_{n=1}^N |a_n|^p \right)^{1/p} + \lim_{N \rightarrow \infty} \left(\sum_{n=1}^N |b_n|^p \right)^{1/p} \\ &= \|v\|_p + \|w\|_p < \infty. \end{aligned}$$

Exercise: check the other properties of a norm. □

Remark. If X is any set, let $\ell_p(X) = \{f : X \rightarrow \mathbb{R} \mid \sum_{x \in X} |f(x)|^p < \infty\}$. Then, $\ell_p(X)$ is a vector space with the norm

$$\|f\|_p = \left(\sum_{x \in X} |f(x)|^p \right)^{1/p}$$

where $\sum_{x \in X} |f(x)| := \sup_{F \subset X \text{ finite}} \{\sum_{x \in F} |f(x)|\}$.

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Recall that $\ell_p(X) = \{f : X \rightarrow \mathbb{R} \mid \sum_{x \in X} |f(x)|^p < \infty\}$ with norm $\|f\|_p = (\sum_{x \in X} |f(x)|^p)^{1/p}$, for $1 \leq p < \infty$.

Definition 2.6 ($\ell_\infty(X)$).

Define $\ell_\infty(X) = \{f : X \rightarrow \mathbb{R} \mid \sup_{x \in X} |f(x)| < \infty\}$ with the ∞ -norm

$$\|f\|_\infty = \sup_{x \in X} |f(x)|.$$

Proposition 2.4. $\ell_\infty(X)$ is a normed space with the potential structure and the ∞ -norm.

Proof. Recall that $(\alpha f)(x) := \alpha(f(x))$ and $(f + g)(x) := f(x) + g(x)$. Note that

(1) If $\|f\|_\infty = 0$, then $|f(x)| = 0$ for all $x \in X \implies f = 0$.

Note. The first 0 is the number 0, the second 0 is the zero function, i.e. $0(x) = 0$ for all $x \in X$.

(2) Let $\alpha \in \mathbb{R}$. We have

$$\|\alpha f\|_\infty = \sup_{x \in X} |(\alpha f)(x)| = \sup_{x \in X} |\alpha(f(x))| = |\alpha| \sup_{x \in X} |f(x)| = |\alpha| \|f\|_\infty.$$

(3) If $f, g \in \ell_\infty(X)$, then

$$\begin{aligned} \|f + g\|_\infty &= \sup_{x \in X} |(f + g)(x)| = \sup_{x \in X} |f(x) + g(x)| \\ &\leq \sup_{x \in X} \{|f(x)| + |g(x)|\} \leq \underbrace{\sup_{x \in X} |f(x)| + \sup_{y \in X} |g(y)|}_{= \sup_{x, y \in X} \{|f(x)| + |g(y)|\}} = \|f\|_\infty + \|g\|_\infty. \end{aligned}$$

Thus, $\ell_\infty(X)$ is a normed space. □

Example. Let $C([a, b])$ denote the space of continuous functions on $[a, b]$. Note that $C([a, b]) \subset \ell_\infty([a, b])$. We may restrict the ∞ -norm $\|\cdot\|_\infty$ to $C([a, b])$.

If $f \in C([a, b])$, set

$$\|f\|_p = \left(\int_a^b |f(x)|^p dx \right)^{1/p} \quad \text{for } 1 \leq p < \infty.$$

Theorem 2.5 (Minkowski's Inequality for Integrals).

If $f, g \in C([a, b])$, then

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p \quad \forall 1 \leq p < \infty.$$

Proof. Exercise. □

Example. Suppose $(V_1, \|\cdot\|_1)$ and $(V_2, \|\cdot\|_2)$ are normed spaces.

Definition 2.7 (Bounded Linear Map, Operator Norm).

A linear map $T : V_1 \rightarrow V_2$ is **bounded** if $\exists C > 0$ s.t.

$$\|T(v)\|_2 \leq C\|v\|_1 \quad \forall v \in V_1.$$

Let $B(V_1, V_2) :=$ the set of all bounded linear maps from V_1 to V_2 . If $T \in B(V_1, V_2)$, define

$$\|T\| = \inf\{C \geq 0 : \|T(v)\|_2 \leq C\|v\|_1 \quad \forall v \in V_1\}.$$

Theorem 2.6. $(B(V_1, V_2), \|\cdot\|)$ is a normed space where

$$(\alpha T)(v) := \alpha(T(v)) \quad \alpha \in \mathbb{R}, T \in B(V_1, V_2), v \in V_1,$$

and

$$(T + S)(v) := T(v) + S(v) \quad T, S \in B(V_1, V_2), v \in V_1.$$

Proof. We check the properties of a norm.

(1) Suppose $\|T\| = 0$. Then, $\forall \epsilon > 0$, if $v \in V_1$, we have

$$\|T(v)\|_2 \leq \epsilon \|v\|_1.$$

Then, $\|T(v)\|_2 = 0 \implies T(v) = 0$. Since this is true for all $v \in V_1$, we have $T = 0$.

(2) Exercise.

(3) Suppose $T, S \in B(V_1, V_2)$. Then, for $v \in V_1$, $\epsilon > 0$, we have

$$\|T(v)\|_2 \leq (\|T\| + \epsilon)\|v\|_1, \quad \|S(v)\|_2 \leq (\|S\| + \epsilon)\|v\|_1.$$

So,

$$\begin{aligned} \|(T + S)(v)\|_2 &= \|T(v) + S(v)\|_2 \leq \|T(v)\|_2 + \|S(v)\|_2 \\ &\leq (\|T\| + \epsilon)\|v\|_1 + (\|S\| + \epsilon)\|v\|_1 = (\|T\| + \|S\| + 2\epsilon)\|v\|_1. \end{aligned}$$

Thus,

$$\|T + S\| \leq \|T\| + \|S\| + 2\epsilon \quad \forall \epsilon.$$

Since this is true for all $\epsilon > 0$, we have $\|T + S\| \leq \|T\| + \|S\|$. □

Example. Let $V_1 = \mathbb{R}^n$ and $V_2 = \mathbb{R}^m$ with the Euclidean norms $\|\cdot\|_2$. Let $B(V_1, V_2) = M_{m \times n}(\mathbb{R})$.

3 Topology

3.1 Topology on Metric Spaces

Definition 3.1 (Open Ball, Closed Ball).

Fix a metric space (X, d) . If $x \in X$ and $r > 0$, an **open ball** containing x is

$$B_r(x) := \{y \in X : d(x, y) < r\}.$$

A **closed ball** containing x is

$$\overline{B}_r(x) := \{y \in X : d(x, y) \leq r\}.$$

Definition 3.2 (Neighborhood).

If $x \in N \subset X$, then N is a **neighborhood** of x if $B_r(x) \subset N$ for some $r > 0$.

Definition 3.3 (Open Set, Closed Set).

- (1) A set $U \subset X$ is **open** if U is a neighborhood of each of its points, i.e. $\forall x \in U, \exists r > 0$ such that $B_r(x) \subset U$.
- (2) A set $F \subset X$ is **closed** if $F^c := X \setminus F$ is open.

Example. $U = \emptyset$ is open.

Example. $F = \emptyset$ is closed.

Note. $\emptyset$ is both open and closed.

Example. $\mathbb{R}$ is a metric space with the usual metric $d(x, y) = |x - y|$. Then,

- $U = (a, b)$ is open.
- $(-\infty, a) \cup (b, \infty)$ is open.
- $[a, b]$ is closed.
- $(0, 1]$ is neither open nor closed.

Proposition 3.1. If (X, d) is a metric space, then $\forall x \in X$ and $r > 0$, $B_r(x)$ is open and $\overline{B_r(x)}$ is closed.

Proof. Fix $y \in B_r(x)$ so $d(x, y) < r$. Consider $r' = r - d(y, x)$. If $z \in B_{r'}(y)$ so $d(y, z) < r' = r - d(y, x)$, then by the triangle inequality,

$$d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + (r - d(y, x)) = r.$$

Thus, $B_{r'}(y) \subset B_r(x)$, so $B_r(x)$ is a neighborhood of y and since y was arbitrary, $B_r(x)$ is open.

Exercise: Show that $\overline{B_r(x)}$ is closed. □

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Example. Consider (X, d) where d is the discrete metric, i.e.

$$d(x, y) = \begin{cases} 0, & \text{if } x = y, \\ 1, & \text{if } x \neq y. \end{cases}$$

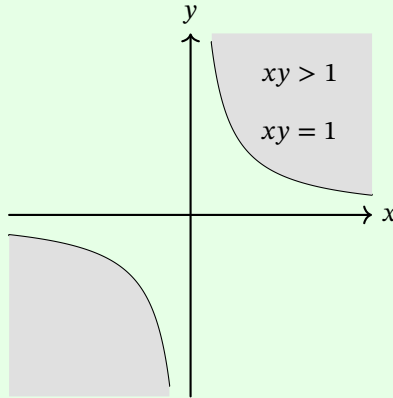
If $x \in X$ and $r > 0$, then

$$B_r(x) = \begin{cases} \{x\}, & \text{if } 0 < r \leq 1, \\ X, & \text{if } r > 1. \end{cases}$$

If $x \in N \subset X$, then N is a neighborhood of x since $B_{1/2}(x) = \{x\} \subset N$. Thus, every subset $U \subset X$ is open. Also, every subset $F \subset X$ is closed since F^c is also a subset of X and thus open.

Example. Let $X = \mathbb{R}^2$ with the Euclidean metric d_2 . Consider the set

$$A = \{(x, y) \in \mathbb{R}^2 : xy > 1\}.$$



Claim. A is open.

Proof. Fix $(x_0, y_0) \in A$ and assume that $x_0 > 0 \implies y_0 > 0$. Take

$$r < \min \left\{ x_0, y_0, \frac{x_0 y_0 - 1}{2x_0}, \frac{x_0 y_0 - 1}{2y_0} \right\}.$$

If $(u, v) \in B_r((x_0, y_0))$, then

$$d_2((x_0, y_0), (u, v)) = \sqrt{(u - x_0)^2 + (v - y_0)^2} < r.$$

So, $|u - x_0| < r$ and $|v - y_0| < r$, which implies

$$u > x_0 - r, \quad v > y_0 - r.$$

Thus,

$$\begin{aligned} uv &> (x_0 - r)(y_0 - r) = x_0 y_0 - r x_0 - r y_0 + r^2 \\ &> 1 + (x_0 y_0 - 1) - \frac{x_0 y_0 - 1}{2x_0} x_0 - \frac{x_0 y_0 - 1}{2y_0} y_0 + 0 = 1. \end{aligned}$$

So, $(u, v) \in A$. Thus, $B_r((x_0, y_0)) \subset A$, so A is a neighborhood of (x_0, y_0) . Since (x_0, y_0) was arbitrary, A is open. $\square$

Exercise: Prove that A is open using Proposition 3.6. Note that $\mathbb{R}^2 \ni (x, y) \mapsto xy \in \mathbb{R}$ is continuous and A is the inverse image of this continuous function applied to the open set $(1, \infty) \subset \mathbb{R}$.

Remark. Let $B = \{(x, y) \in \mathbb{R}^2 : xy \geq 1\}$. Then, B is closed since $B^c = \{(x, y) \in \mathbb{R}^2 : xy < 1\}$ is open (similar proof as above).

Proposition 3.2. Suppose (X, d) is a metric space. Then,

- (1) If $\{U_i\}_{i \in I}$ is a family of open sets, then $\bigcup_{i \in I} U_i$ is open.
- (2) If $\{F_i\}_{i \in I}$ is a family of closed sets, then $\bigcap_{i \in I} F_i$ is closed.
- (3) If $U_1, U_2, \dots, U_n$ are open, then $\bigcap_{k=1}^n U_k$ is open.
- (4) If $F_1, F_2, \dots, F_n$ are closed, then $\bigcup_{k=1}^n F_k$ is closed.

Proof.

- (1) Suppose $\{U_i\}_{i \in I}$ is a family of open sets. Fix $x \in \bigcup_{i \in I} U_i$. Then, $\exists i_0 \in I$ such that $x \in U_{i_0}$. Since U_{i_0} is open, $\exists r > 0$ such that $B_r(x) \subset U_{i_0} \subset \bigcup_{i \in I} U_i$. Thus, $\bigcup_{i \in I} U_i$ is open.

- (2) Suppose $\{F_i\}_{i \in I}$ is a family of closed sets, then

$$\left(\bigcap_{i \in I} F_i \right)^c = \bigcup_{i \in I} F_i^c$$

is open by (1). Thus, $\bigcap_{i \in I} F_i$ is closed.

- (3) Suppose $U_1, U_2, \dots, U_n$ are open. If $x \in U_1 \cap U_2 \cap \dots \cap U_n$, since U_i is open, $\exists r_i > 0$ such that $B_{r_i}(x) \subset U_i$ for each $i = 1, 2, \dots, n$. Set $r = \min\{r_1, r_2, \dots, r_n\}$. Then, $B_r(x) \subset B_{r_i}(x) \subset U_i$ for all i . Thus, $B_r(x) \subset \bigcap_{k=1}^n U_k$, so $\bigcap_{k=1}^n U_k$ is open.

- (4) Exercise (use De Morgan's laws and (3)). □

Definition 3.4 (Sequence).

A **sequence** in X is a function from $\mathbb{N}$ to X , usually denoted by $(x_n)_n$.

Definition 3.5 (Convergence).

A sequence $(x_n)_n$ **converges** to x_0 , i.e. $\lim_{n \rightarrow \infty} x_n = x_0$, if $\forall \epsilon > 0, \exists N \in \mathbb{N}$ such that $\forall n \geq N$, $d(x_n, x_0) < \epsilon$.

Definition 3.6 (Limit Point).

If $A \subset X$, a **limit point** of A is a point $x \in X$ such that $x = \lim_{n \rightarrow \infty} x_n$ for some sequence $(x_n)_n$ with $x_n \in A$ for all n .

Note. Every point in A is a limit point of A . If $a \in A$, set $x_n = a$ for all n .

Example. Let $A = \left\{\frac{1}{n}\right\}_{n \in \mathbb{N}} \subset \mathbb{R}$. Then, 0 is a limit point of A since $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Definition 3.7 (Accumulation Point).

Let $A \subset X$. A point $x \in X$ is an **accumulation point** of A if $x = \lim_{n \rightarrow \infty} x_n$ for some sequence $(x_n)_n$ with $x_n \in A$ and x_n distinct.

Note. An accumulation point is a limit point, but the converse is not necessarily true.

Example. Let $A = \left\{\frac{1}{n}\right\}_{n \in \mathbb{N}} \subset \mathbb{R}$. Then, 0 is an accumulation point of A since $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ and $\frac{1}{n}$ are distinct for all n .

Definition 3.8 (Closure).

If $A \subset X$ is a set. We let $\overline{A}$ denote the **closure** of A , the smallest closed set that contains A . Note that

$$\overline{A} = \bigcap \{E \subset X : E \text{ is closed and } A \subset E\}.$$

Theorem 3.3. If $A \subset X$ and (X, d) is a metric space, then

$$\begin{aligned} \overline{A} &= \{x \in X : x \text{ is a limit point of } A\} \\ &= A \cup \{x \in X : x \text{ is an accumulation point of } A\}. \end{aligned}$$

Lemma 3.4. If $B \subset X$ is closed, then B contains all its limit points.

Proof. If $x_n \in B$ and $\lim_{n \rightarrow \infty} x_n = x_0 \notin B$, then $x_0 \in B^c$ which is open. So, $\exists r > 0$ such that $B_r(x_0) \subset B^c$. But, $\exists N \in \mathbb{N}$ such that $\forall n \geq N$, $d(x_n, x_0) < r$, so $x_n \in B_r(x_0) \subset B^c$ for all $n \geq N$. Thus, x_0 is not a limit point of B . $\square$

Note. Prove by contraposition is preferred than proof by contradiction.

Lemma 3.5. A set $A \subset X$ is closed $\iff A$ contains all its limit points.

Proof.

($\Rightarrow$): We prove by contrapositive. Suppose A does not contain a limit point x . Say $x = \lim_{n \rightarrow \infty} a_n$ for some sequence $\{a_n\}_{n=1}^{\infty} \subset A$. Fix $r > 0$. Then, $\exists N \in \mathbb{N}$ s.t. $\forall n \geq N$, $d(a_n, x) < r$, so $a_n \in B_r(x) \cap A$. Thus, $B_r(x) \cap A \neq \emptyset$ for all $r > 0$, so $B_r(x) \not\subset A^c$, i.e. A^c is not a neighborhood of $x \in A^c$. Thus, A^c is not open, so A is not closed.

($\Leftarrow$): We prove by contrapositive again. Suppose A is not closed, i.e. A^c is not open. Then, $\exists x \in A^c$ s.t. A^c is not a neighborhood of x . For each $n \in \mathbb{N}$, $A \cap B_{1/n}(x) \neq \emptyset$. Let x_n be some point in $A \cap B_{1/n}(x)$. So, $\{x_n\}_{n=1}^{\infty} \subset A$. If $\epsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $\frac{1}{N} < \epsilon$. Then, $\forall n \geq N$,

$$x_n \in B_{1/n}(x) \cap A \subset B_{1/N}(x) \subset B_{\epsilon}(x) \implies d(x_n, x) < \epsilon.$$

Thus, $\lim_{n \rightarrow \infty} x_n = x$ and so x is a limit point of A . □

Remark. Note that $\lim_{n \rightarrow \infty} x_n = x$ but $x_n \neq x$. Then, we can choose a subsequence $\{x_{n_k}\}_{k=1}^{\infty}$ such that $\lim_{k \rightarrow \infty} x_{n_k} = x$ but $x_{n_k} \neq x_{n_j}$ for all $k \neq j$.

Note. Lemma 3.5 is also true if we replace “limit points” with “accumulation points”.

Recall Theorem 3.3.

Theorem 3.3. If $A \subset X$ and (X, d) is a metric space, then

$$\begin{aligned} \overline{A} &= \{x \in X : x \text{ is a limit point of } A\} \\ &= A \cup \{x \in X : x \text{ is an accumulation point of } A\}. \end{aligned}$$

Proof. By the remark above, any limit point is either in A or is also an accumulation point. This gives the last equality. To finish the proof, we just need to show that the set of limit points of A is closed.

Set $B = \{\text{limit points of } A\}$. Fix $x \in B^c$, $x \notin A$ and for some $r > 0$, $B_r(x) \subset A^c$ (by the proof of Lemma 3.5 that $B_r(x) \cap A = \emptyset$).

Claim. $B_{r/2}(x) \subset B^c$.

Proof of Claim. Suppose this is not true. Then, $\exists y \in B \cap B_{r/2}(x)$. Then, $\exists (x_n)_n \subset A$ s.t. $y = \lim_{n \rightarrow \infty} x_n$. Since $y \in B_{r/2}(x)$, $d(x, y) < r/2$. Then, $\exists N \in \mathbb{N}$ s.t.

$$d(x_n, x) \leq d(x_n, y) + d(y, x) < r/2 + r/2 = r \quad \forall n \geq N.$$

Thus, $x_n \in B_r(x) \cap A$, which is a contradiction since $B_r(x) \cap A = \emptyset$. Hence, $B_{r/2}(x) \subset B^c$. $\square$

Since x was arbitrary, B^c is open, so B is closed. $\square$

Definition 3.9 (Interior).

If $A \subset X$, the **interior** of A is

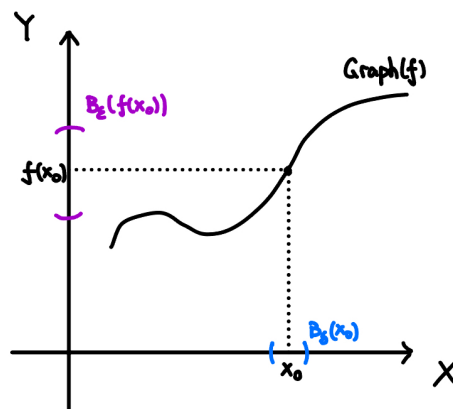
$$\text{Int } A := \bigcup \{G \subset \mathcal{P}(X) : G \text{ is open and } G \subset A\}.$$

Remark. We have $\text{Int } A = (\overline{A^c})^c$.

3.2 Continuity

Definition 3.10 (Continuity).

Let (X, d) and (Y, ρ) be metric spaces. A function $f : X \rightarrow Y$ is **continuous** at $x_0 \in X$ if $\forall \epsilon > 0, \exists \delta > 0$ s.t. if $x_1 \in X$ s.t. $d(x_0, x_1) < \delta$, then $\rho(f(x_0), f(x_1)) < \epsilon$.



Remark. f is continuous if it is continuous at each point $x \in X$.

Definition 3.11 (Uniform Continuity).

A function $f : X \rightarrow Y$ is **uniformly continuous** if $\forall \epsilon > 0, \exists \delta > 0$ s.t. if $x_0 \in X, x_1 \in X$ s.t. $d(x_0, x_1) < \delta$, then $\rho(f(x_0), f(x_1)) < \epsilon$.

Example. Let $X = (0, \infty)$ and $Y = (0, \infty)$. Consider $f(x) = \frac{1}{x}$. Then, f is continuous but not uniformly continuous.

Proposition 3.6. Let $f : X \rightarrow Y$ be a function. The following are equivalent:

- (1) f is continuous.
- (2) $f^{-1}(V)$ is open $\forall V \subset Y$ open.
- (3) $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$ whenever $\lim_{n \rightarrow \infty} x_n = x_0$.

Proof.

(1) $\implies$ (2): Suppose f is continuous. Take $V \subset Y$ open and take $x_0 \in f^{-1}(V)$. Then, $f(x_0) \in V$. Since V is open, $\exists \epsilon > 0$ such that $B_\epsilon(f(x_0)) \subset V$. Since f is continuous at x_0 , $\exists \delta > 0$ such that if $d(x_0, x_1) < \delta$, then $\rho(f(x_0), f(x_1)) < \epsilon$, so $f(x_1) \in B_\epsilon(f(x_0))$ if $x_1 \in B_\delta(x_0)$. Thus, $B_\delta(x_0) \subset f^{-1}(B_\epsilon(f(x_0))) \subset f^{-1}(V)$, so $f^{-1}(V)$ is a neighborhood of x_0 . Since x_0 was arbitrary, $f^{-1}(V)$ is open.

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(2) $\implies$ (3): Suppose $f^{-1}(V)$ is open $\forall V \subset Y$ open and $\lim_{n \rightarrow \infty} x_n = x_0$. Fix $\epsilon > 0$, let $V = B_\epsilon(f(x_0))$. Then, V is open, so $f^{-1}(V)$ is also open. Since $x_0 \in f^{-1}(V)$, $\exists \delta > 0$ s.t. $B_\delta(x_0) \subset f^{-1}(V)$. Hence, $\exists N \in \mathbb{N}$ s.t. $\forall n \geq N, d(x_n, x_0) < \delta$. Thus, $\forall n \geq N$,

$$d(x_n, x_0) < \delta \implies x_n \in B_\delta(x_0) \subset f^{-1}(V) \implies f(x_n) \in V = B_\epsilon(f(x_0)).$$

Thus, $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$.

(3) $\implies$ (1): We prove by contrapositive. Assume f is not continuous. Then, $\exists x_0 \in X$ s.t. f is not continuous at x_0 . Hence, $\exists \epsilon_0 > 0$ s.t. for any $\delta > 0, \exists x \in X$ s.t. $d(x, x_0) < \delta$ but $\rho(f(x), f(x_0)) \geq \epsilon_0$. Take $\delta = \frac{1}{n}$, we get $x_n \in X$ s.t. $d(x_n, x_0) < \frac{1}{n}$, but $\rho(f(x_n), f(x_0)) \geq \epsilon_0$ for $n = 1, 2, \dots$. This gives a sequence $(x_n)_n$ in X with $\lim_{n \rightarrow \infty} x_n = x_0$ but $\lim_{n \rightarrow \infty} f(x_n) \neq f(x_0)$. $\square$

Example. Suppose (X, d) is discrete. Then, any function $f : X \rightarrow Y$ is continuous.

Question: What about $f : Y \rightarrow X$?

Answer: We need $f^{-1}(\{x\}) = \{y \in Y : f(y) = x\}$ to be open for f to be continuous.

Definition 3.12 (Lipschitz, Lipschitz Constant, Contraction).

Let $f : X \rightarrow Y$ be a function. We say f is **Lipschitz** if $\exists C \geq 0$ such that

$$\rho(f(x), f(y)) \leq C d(x, y) \quad \forall x, y \in X$$

where C is called a **Lipschitz constant** for f . We say f is a **contraction** if it is Lipschitz with a Lipschitz constant $C < 1$.

Remark. **Exercise:** Lipschitz $\implies$ uniformly continuous.

Example. Let (Y, ρ) and (Z, σ) be metric spaces. On $Y \times Z$, take the metric

$$\max\{\rho(y_1, y_2), \sigma(z_1, z_2)\}.$$

Then, $\pi : Y \times Z \rightarrow Y$ is Lipschitz with Lipschitz constant $C = 1$ where $\pi(y, z) = y$.

Example. Let (X, d) be any metric space. Fix $x_0 \in X$ and let $f : X \rightarrow \mathbb{R}$ be the function $f(x) = d(x, x_0)$. Then, f is Lipschitz with Lipschitz constant $C = 1$ since

$$|f(x) - f(y)| = |d(x, x_0) - d(y, x_0)| \leq d(x, y) \quad \forall x, y \in X$$

by the reverse triangle inequality.

Example. If $f : (a, b) \rightarrow \mathbb{R}$ where a could be $-\infty$ and b could be ∞ , is differentiable with $|f'(s)| \leq C$ for all $s \in (a, b)$. If $s, t \in (a, b)$ with $s < t$, then by the Mean Value Theorem, $\exists c \in (s, t)$ such that

$$\left| \frac{f(t) - f(s)}{t - s} \right| = |f'(c)| \leq C \implies |f(t) - f(s)| \leq C|t - s|.$$

Thus, f is Lipschitz with Lipschitz constant C .

Example. Let $f : [0, \infty) \rightarrow [0, \infty)$ be given by $f(x) = \sqrt{x}$. Then, f is NOT Lipschitz.

To see this, note that

$$\frac{f\left(\frac{1}{n}\right) - f(0)}{\frac{1}{n} - 0} = \sqrt{n} \rightarrow \infty \quad \text{as } n \rightarrow \infty.$$

However, f is uniformly continuous. WLOG suppose $x > y$, then

$$\begin{aligned} |\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|} &\iff (\sqrt{x} - \sqrt{y})^2 \leq x - y \\ &\iff x + y - 2\sqrt{xy} \leq x - y \\ &\iff 2y \leq 2\sqrt{xy} \iff y^2 \leq xy. \end{aligned}$$

If $\epsilon > 0$, let $\delta = \epsilon^2$, then if $|x - y| < \delta$, we have

$$|f(x) - f(y)| = |\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|} < \sqrt{\delta} = \epsilon.$$

Proposition 3.7. Let (X, d) , (Y, ρ) , (Z, σ) be metric spaces. Then,

- (1) If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then $g \circ f : X \rightarrow Z$ is continuous.
- (2) If $f : X \rightarrow Y$ and $g : X \rightarrow Z$ are continuous, then $h = (f, g)$ is continuous where $h : X \rightarrow Y \times Z$ is defined by $h(x) = (f(x), g(x))$.

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Proof.

- (1) If $W \subset Z$ is open, then

$$(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W)).$$

Note that $g^{-1}(W)$ is open since g is continuous, so $f^{-1}(g^{-1}(W))$ is also open since f is continuous. Thus, $g \circ f$ is continuous by Proposition 3.6.

- (2) Suppose (x_n) is a sequence in X (which converges to x_0). Since f and g are continuous, $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$ and $\lim_{n \rightarrow \infty} g(x_n) = g(x_0)$. Thus,

$$\lim_{n \rightarrow \infty} (f, g)(x_n) = \left(\lim_{n \rightarrow \infty} f(x_n), \lim_{n \rightarrow \infty} g(x_n) \right) = (f(x_0), g(x_0)).$$

Hence, $h = (f, g)$ is continuous. □

Definition 3.13 ($C_b(X)$).

We let $C_b(X)$ denote the set of all real-valued and bounded continuous functions on X .

Note. $C_b(X) \subset \ell^\infty(X)$.

Proposition 3.8. $C_b(X)$ is a subalgebra of $\ell^\infty(X)$, i.e. $C_b(X)$ is a vector subspace and is also closed under multiplication.

Proof. Suppose $f, g \in C_b(X)$ and $\alpha, \beta \in \mathbb{R}$. Then, $(f, g) : X \rightarrow \mathbb{R}^2$ is continuous by the previous proposition. Then, $(\alpha f, \beta g) : X \rightarrow \mathbb{R}^2$ is also continuous since $(\alpha, \beta) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is continuous. This means $\alpha f + \beta g$ is continuous as it is the composition with $+$: $\mathbb{R}^2 \rightarrow \mathbb{R}$ given by $+(x, y) = x + y$ which is continuous. Similarly, $f \cdot g$ is continuous. Since f and g are bounded, $\alpha f + \beta g$ and $f \cdot g$ are also bounded. Thus, $\alpha f + \beta g, f \cdot g \in C_b(X)$. $\square$

Definition 3.14 (Homeomorphism).

We say $f : X \rightarrow Y$ is a **homeomorphism** if f is bijective, continuous, and f^{-1} is also continuous.

Note. f preserves open sets both ways.

Example. Let (X, d) be a metric space and set $\rho(x, y) = \frac{d(x, y)}{1 + d(x, y)}$. Then, ρ is a metric on X and the identity map $\text{id}_X : X \rightarrow X$ is a homeomorphism between (X, d) and (X, ρ) .

Proof. Note that if $\lim_{n \rightarrow \infty} x_n = x_0$ in (X, d) , then

$$\rho(x_n, x_0) = \frac{d(x_n, x_0)}{1 + d(x_n, x_0)} \leq d(x_n, x_0).$$

This shows that $\lim_{n \rightarrow \infty} x_n = x_0$ in (X, ρ) , i.e. $\text{id}_X : (X, d) \rightarrow (X, \rho)$ is continuous.

Note if $\rho(x_n, x_0) \leq \frac{1}{2}$, then

$$2d(x_n, x_0) \leq 1 + d(x_n, x_0) \implies 1 + d(x_n, x_0) \leq 2.$$

Hence,

$$d(x_n, x_0) = \rho(x_n, x_0)(1 + d(x_n, x_0)) \leq 2\rho(x_n, x_0).$$

So if $\lim_{n \rightarrow \infty} x_n = x_0$ in (X, ρ) , then $\lim_{n \rightarrow \infty} x_n = x_0$ in (X, d) , i.e. $\text{id}_X^{-1} : (X, \rho) \rightarrow (X, d)$ is continuous. Since id_X is bijective, $\text{id}_X : (X, d) \rightarrow (X, \rho)$ is a homeomorphism. $\square$

Example. Consider $f : [0, 2\pi) \rightarrow S^1 = \{z \in \mathbb{C} : |z| = 1\}$ given by $f(t) = e^{it}$. Then, f is continuous and bijective. However, f^{-1} is NOT continuous.

To see this, if $t_n \in [0, 2\pi)$ with $\lim_{n \rightarrow \infty} t_n = 2\pi$, then

$$\lim_{n \rightarrow \infty} f(t_n) = \lim_{n \rightarrow \infty} e^{it_n} = e^{i \cdot 2\pi} = 1.$$

However,

$$\lim_{n \rightarrow \infty} f^{-1}(f(t_n)) = \lim_{n \rightarrow \infty} t_n = 2\pi \neq 0 = f^{-1}(1).$$

Thus, f^{-1} is not continuous by Proposition 3.6.

3.3 Finite Dimensional Normed Spaces

Theorem 3.9. Suppose V is a finite dimensional real vector space, then any two norms on V are equivalent.

Note. That is, if $\|\cdot\|$ and $\|\cdot\|$ are two norms on V , then, $\exists C_1, C_2 > 0$ such that

$$C_1 \|v\| \leq \|v\| \leq C_2 \|v\| \quad \forall v \in V.$$

Proof. Fix a basis $\{V_1, V_2, \dots, V_n\}$ for V . Given $v \in V$, we can uniquely write v as

$$v = \sum_{i=1}^n a_i V_i, \quad a_1, \dots, a_n \in \mathbb{R}.$$

Consider the induced ℓ^1 -norm on V defined by $\|v\|_1 = \sum_{i=1}^n |a_i|$. If $\|\cdot\|$ is any norm on V , then

$$\begin{aligned} \|v\| &= \left\| \sum_{i=1}^n a_i V_i \right\| \leq \sum_{i=1}^n |a_i| \|V_i\| \quad \text{let } C = \max_{1 \leq i \leq n} \|V_i\| \\ &\leq C \sum_{i=1}^n |a_i| = C \|v\|_1. \end{aligned}$$

Note that this shows that $\|\cdot\|$ is a continuous function with respect to the norm $\|\cdot\|_1$. Let $S = \{v \in V : \|v\|_1 = 1\}$. If we identify $(V, \|\cdot\|_1)$ with $(\mathbb{R}^n, \|\cdot\|_1)$, we see that S is compact. This means $\|\cdot\|$ attains its minimum on S , say at $c = \|v_0\| \neq 0$ for some $v_0 \in S$. If $v \in V$ with $v \neq 0$, then $\frac{v}{\|v\|_1} \in S$, so

$$\frac{\|v\|}{\|v\|_1} = \left\| \frac{v}{\|v\|_1} \right\| \geq c \implies \|v\| \geq c\|v\|_1.$$

Thus, $\|v\| \leq C\|v\|_1$ and $\|v\| \geq c\|v\|_1$ for all $v \in V$. Hence, $\|\cdot\|$ and $\|\cdot\|_1$ are equivalent norms on V . Since $\|\cdot\|$ is also a norm on V , it is also equivalent to $\|\cdot\|_1$. Thus, $\|\cdot\|$ and $\|\cdot\|$ are equivalent. $\square$

4 Completeness

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Definition 4.1 (Complete).

A metric space (X, d) is **complete** if every Cauchy sequence in X converges to a point in X .

Example. We defined the real numbers $\mathbb{R}$ as equivalence classes of Cauchy sequences. Note that $\mathbb{R}$ is complete.

Proof. If $\{r_n\}_{n=1}^{\infty}$ is a Cauchy sequence in $\mathbb{R}$, then take $q_n \in \mathbb{Q}$ s.t. $|q_n - r_n| < 2^{-n}$. Consider $\{q_n\}_{n=1}^{\infty}$.

Claim. $\{q_n\}_{n=1}^{\infty}$ is a Cauchy sequence in $\mathbb{Q}$.

Proof of Claim. Fix $\epsilon > 0$. Since $\{r_n\}_{n=1}^{\infty}$ is Cauchy, $\exists N \in \mathbb{N}$ such that $\forall m, n \geq N$,

$$|r_n - r_m| < \frac{\epsilon}{2}.$$

Take $N' \in \mathbb{N}$ such that $2^{-N'} < \frac{\epsilon}{4}$. If $m, n \geq \max\{N, N'\}$, then

$$\begin{aligned} |q_n - q_m| &= |q_n - r_n + r_n - r_m + r_m - q_m| \\ &\leq |q_n - r_n| + |r_n - r_m| + |r_m - q_m| \\ &< 2^{-n} + \frac{\epsilon}{2} + 2^{-m} \\ &< \frac{\epsilon}{4} + \frac{\epsilon}{2} + \frac{\epsilon}{4} = \epsilon. \end{aligned}$$

Thus, $\lim_{n \rightarrow \infty} q_n = r$ for some $r \in \mathbb{R}$. If $\epsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $|r - q_n| < \epsilon$ for all $n \geq N$. Then, for $n \geq N$,

$$|r - r_n| = |r - q_n + q_n - r_n| \leq |r - q_n| + |q_n - r_n| < \epsilon + 2^{-n}. \quad \square$$

Thus, $\{r_n\}_{n=1}^{\infty}$ converges to $r \in \mathbb{R}$. Hence, $\mathbb{R}$ is complete. $\square$

Example. Fix p a prime. If $x \in \mathbb{Q}$, write $x = p^a \frac{s}{t}$ where $a \in \mathbb{Z}$ and $s, t \in \mathbb{Z}$ with $\gcd(p, s) = 1$, $\gcd(p, t) = 1$, and $\gcd(s, t) = 1$. Define

$$|x|_p = \begin{cases} p^{-a} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Lemma 4.1.

- (1) $|x|_p = 0 \iff x = 0$.
- (2) $|xy|_p = |x|_p |y|_p$.
- (3) $|x - y|_p \leq \max\{|x|_p, |y|_p\} \leq |x|_p + |y|_p$.

Proof. We will show (3). Let $x, y \in \mathbb{Q}$ and write $x = p^a \frac{s}{t}$ and $y = p^{a'} \frac{s'}{t'}$. Both are in reduced forms, and we assume $a \leq a'$. Then,

$$x - y = p^a \left(\frac{s}{t} - p^{a' - a} \frac{s'}{t'} \right) = p^a \left(\frac{st' - p^{a' - a} s' t}{tt'} \right).$$

So,

$$|x - y|_p = \begin{cases} p^{-a} & \text{if } a < a', \\ p^{-a-n} & \text{if } a = a', \text{ where } n \geq 0 \end{cases} \leq |x|_p.$$

□

Exercise: Finish the proof.

We can define a metric on $\mathbb{Q}$ by $d_p(x, y) = |x - y|_p \geq 0$. Note that

$$d_p(x, z) = |x - y + y - z|_p \leq |x - y|_p + |y - z|_p = d_p(x, y) + d_p(y, z).$$

Example. Consider $\{p^n\}_{n=1}^\infty$. Note that $|p^n|_p = d_p(p^n, 0) = p^{-n}$, so

$$\lim_{n \rightarrow \infty} p^n = 0.$$

Example. Consider $1, 1 + p, 1 + p + p^2, 1 + p + p^2 + p^3, \dots$. Note that $x_n = \sum_{i=1}^n p^{i-1}$. Then, $|x_n|_p = 1$, so the sequence is bounded. In fact, this sequence is Cauchy.

To see this, note that for $n \geq m$,

$$x_n - x_m = \sum_{i=m}^{n-1} p^i = p^m \sum_{i=0}^{n-m-1} p^i.$$

So, $|x_n - x_m|_p = p^{-m}$. If $\epsilon > 0$, take $N \in \mathbb{N}$ s.t. $p^{-N} < \epsilon$. Then, for $n, m \geq N$, $|x_n - x_m|_p < \epsilon$.

Thus, $\{x_n\}_{n=1}^\infty$ is Cauchy.

Note. $(\mathbb{Q}, |\cdot|_p)$ is not complete.

Example. Any set X with the discrete metric d is complete.

Example. Consider the set $(-1, 1)$ with the usual metric. Note that $x_n = \frac{n}{n+1}$ gives a Cauchy sequence that does not converge in $(-1, 1)$. Thus, $(-1, 1)$ is not complete.

Also if we take $f : (-1, 1) \rightarrow \mathbb{R}$ defined by $f(x) = \tan\left(\frac{2}{\pi}x\right)$, then f is a homeomorphism.

Note that $\mathbb{R}$ is complete, but $(-1, 1)$ is not complete. Thus, Cauchyness is not preserved under homeomorphism.

Note. Convergence is preserved under homeomorphism, but Cauchyness is not.

Proposition 4.2. Suppose (X, d) is complete and $Y \subset X$. Then, $(Y, d|_{Y \times Y})$ is complete $\iff$ Y is closed in X .

Proof.

($\Rightarrow$): Fix $y \in X$ a limit point of Y . Say $y = \lim_{n \rightarrow \infty} y_n$ for some sequence $\{y_n\}_{n=1}^\infty$ in Y . Since $\{y_n\}_{n=1}^\infty$ converges in X , it is Cauchy. Since Y is complete, $\{y_n\}_{n=1}^\infty$ converges in Y . Thus, $y = \lim_{n \rightarrow \infty} y_n \in Y$. Hence, Y is closed in X by Lemma 3.5.

($\Leftarrow$): Suppose $\{y_n\}_{n=1}^\infty$ is a Cauchy sequence in Y . Since X is complete, $y_n \rightarrow y \in X$. Since Y is closed, $y \in Y$. Thus, $\{y_n\}_{n=1}^\infty$ converges in Y . Hence, Y is complete. $\square$

Definition 4.2 (Banach Space).

A complete normed space is called a **Banach space**.

Definition 4.3 (Summable Sequence, Absolutely Summable Sequence).

If V is a normed space, a sequence $\{a_n\}_{n=1}^{\infty} \subset V$ is **summable** if the sequence of partial sums $S_n = \sum_{k=1}^n a_k$ converges in V .

The sequence $\{a_n\}_{n=1}^{\infty}$ is **absolutely summable** if $\sum_{n=1}^{\infty} \|a_n\| < \infty$.

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Theorem 4.3. A normed space $(V, \|\cdot\|)$ is a Banach space $\iff$ every absolutely summable sequence in V is summable.

Proof.

($\Rightarrow$): Suppose V is a Banach space and $(a_n)_n$ is an absolutely summable sequence in V . Set $S_N = \sum_{n=1}^N a_n$. For all $\epsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $\sum_{k=N}^{\infty} \|a_k\| < \epsilon$. If $n, m \geq N$, say $m \geq n$, then

$$\|S_m - S_n\| = \left\| \sum_{k=n+1}^m a_k \right\| \leq \sum_{k=n+1}^m \|a_k\| < \sum_{k=N}^{\infty} \|a_k\| < \epsilon.$$

Thus, $(S_n)_n$ is Cauchy. Since V is a Banach space, $(S_n)_n$ converges in V . Hence, $(a_n)_n$ is summable.

($\Leftarrow$): Suppose every absolutely summable sequence in V is summable. Take $(a_n)_n$ a Cauchy sequence, then $\exists$ an increasing sequence $n_k \in \mathbb{N}$ s.t. for $n, m \geq n_k$,

$$\|a_n - a_m\| < 2^{-k}.$$

Define $b_1 = a_{n_1}$ and $b_k = a_{n_k} - a_{n_{k-1}}$ for $k > 1$ so that $a_{n_k} = \sum_{n=1}^k b_n$. Note that

$$\begin{aligned} \sum_{k=1}^N \|b_k\| &\leq \|a_{n_1}\| + \sum_{k=2}^N \|a_{n_k} - a_{n_{k-1}}\| \\ &< \|a_{n_1}\| + \sum_{k=2}^N 2^{-k} < \|a_{n_1}\| + 2. \end{aligned}$$

Thus, $\sum_{k=1}^{\infty} \|b_k\| \leq \|a_{n_1}\| + 2$. By hypothesis, $(a_{n_k})_k$ converges to some $a \in V$. If $\epsilon > 0$, $\exists K \in \mathbb{N}$ s.t.

for $k \geq K$, $\|a - a_{n_k}\| < \epsilon$. If $m \geq n_K$, then

$$\|a - a_m\| = \|a - a_{n_k} + a_{n_k} - a_m\| \leq \|a - a_{n_k}\| + \|a_{n_k} - a_m\| < \epsilon + 2^{-k}.$$

Note that we can take k arbitrarily large. □

Theorem 4.4. $\ell_p(\mathbb{N})$ is a Banach space for $1 \leq p \leq \infty$.

Proof. Case $p = \infty$: Exercise.

Fix $1 \leq p \leq \infty$. Suppose $f_n \in \ell_p(\mathbb{N})$ is absolutely summable. Set $M = \sum_{n=1}^{\infty} \|f_n\|_p < \infty$. Define $g_n : \mathbb{N} \rightarrow \mathbb{R}$ by $g_n(k) = \sum_{j=1}^n |f_j(k)|$. For each $k \in \mathbb{N}$,

$$\|g_n\|_p \leq \sum_{j=1}^n \|f_j\|_p \leq M$$

by Minkowski's inequality. Thus for each k , $(g_n(k))_n$ converges. Set $g(k) := \lim_{n \rightarrow \infty} g_n(k)$. Also, for each k , $\sum_{n=1}^{\infty} |f_n(k)| \leq M$. Then, $f(k) := \sum_{n=1}^{\infty} f_n(k)$ is well-defined. Note that

$$\left| \sum_{n=1}^N f_n(k) \right| \leq g(k).$$

Then,

$$\sum_{k=1}^{\infty} \left| \sum_{n=1}^N f_n(k) \right|^p \leq \sum_{k=1}^{\infty} g(k)^p = \|g\|_p^p = \lim_{K \rightarrow \infty} \sum_{k=1}^K g(k)^p \leq M^p.$$

Thus,

$$\|f\|_p^p \leq \|g\|_p^p \leq M^p = \left(\sum_{n=1}^{\infty} \|f_n\|_p \right)^p.$$

Hence, $f \in \ell_p(\mathbb{N})$. Also,

$$\left\| f - \sum_{k=1}^n f_k \right\|_p \leq \sum_{k=n}^{\infty} \|f_k\|_p$$

by the same argument as above. Since $(f_k)_k$ is absolutely summable, we can choose $N \in \mathbb{N}$ s.t. for $n \geq N$, $\sum_{k=n}^{\infty} \|f_k\|_p < \epsilon$. Thus, $\left\| f - \sum_{k=1}^n f_k \right\|_p < \epsilon$ for $n \geq N$. Hence, $\sum_{k=1}^n f_k \rightarrow f$ in $\ell_p(\mathbb{N})$. □

Recall that if (X, d) is a metric space, then

$$C_b(X) = \{f : X \rightarrow \mathbb{R} \mid f \text{ is continuous and bounded}\}, \quad \|f\|_\infty = \sup_{x \in X} |f(x)|.$$

Theorem 4.5. $(C_b(X), \|\cdot\|_\infty)$ is a Banach space.

Definition 4.4 (Pointwise Convergence, Uniform Convergence).

A sequence of functions $(f_n)_n$ with $f_n : X \rightarrow Y$ is said to **converge pointwise** to a function $f : X \rightarrow Y$ if $\lim_{n \rightarrow \infty} f_n(x) = f(x) \forall x \in X$.

The sequence $(f_n)_n$ is said to **converge uniformly** to f if $\forall \epsilon > 0, \exists N \in \mathbb{N}$ such that for $n \geq N$, $d_Y(f_n(x), f(x)) < \epsilon \forall x \in X$.

Lemma 4.6. Let (X, d_X) and (Y, d_Y) be metric spaces. If $f_n : X \rightarrow Y$ are continuous and $(f_n)_n$ converges uniformly to $f : X \rightarrow Y$, then f is continuous.

Example. Let $f_n : [0, 1] \rightarrow [0, 1]$ be defined by $f_n(x) = x^n$. Note that

$$f_n(x) \rightarrow f(x) := \begin{cases} 0 & \text{if } x < 1, \\ 1 & \text{if } x = 1. \end{cases}$$

Thus, f_n converges pointwise to f and f is not continuous. However, f_n does not converge uniformly to f .

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Proof of Lemma 4.6.

This proof is called the 3ϵ argument. Fix $x_0 \in X$ and $\epsilon > 0$. Since $(f_n)_n$ converges uniformly to f , $\exists N \in \mathbb{N}$ such that

$$\sup_{x \in X} d_Y(f(x), f_N(x)) < \epsilon.$$

Since f_N is continuous at x_0 , $\exists \delta > 0$ such that if $d_X(y, x_0) < \delta$, then $d_Y(f_N(x_0), f_N(y)) < \epsilon$. If $y \in X$

and $d_X(x_0, y) < \delta$, then

$$\begin{aligned} d_Y(f(x_0), f(y)) &\leq d_Y(f(x_0), f_N(x_0)) + d_Y(f_N(x_0), f_N(y)) + d_Y(f_N(y), f(y)) \\ &< \epsilon + \epsilon + \epsilon = 3\epsilon. \end{aligned}$$

□

Since $\epsilon > 0$ was arbitrary and $x_0 \in X$ was arbitrary, f is continuous.

Example. Let $X = Y = [0, 1]$ and $f_n(x) = x^n$. Then,

$$\lim_{n \rightarrow \infty} f_n(x) = f(x) = \begin{cases} 1 & \text{if } x = 1, \\ 0 & \text{if } x < 1. \end{cases}$$

Thus, f_n converges pointwise to f . However, f_n does not converge uniformly to f . To see this, note that for each n , $\sup_{x \in [0, 1]} |f_n(x) - f(x)| = 1$. Hence, $\lim_{n \rightarrow \infty} \sup_{x \in [0, 1]} |f_n(x) - f(x)| = 1 \neq 0$. Thus, f_n does not converge uniformly to f . Then, f is not continuous.

Theorem 4.7 (Weierstrass M-Test).

Suppose (X, d) is a metric space and $f_n : X \rightarrow \mathbb{R}$ are continuous, bounded such that

$$\sum_{n=1}^{\infty} \|f_n\|_{\infty} \leq M < \infty.$$

Then, $\sum_{n=1}^{\infty} f_n$ is continuous and bounded and we have

$$\lim_{N \rightarrow \infty} \left\| f - \sum_{n=1}^N f_n \right\|_{\infty} = 0,$$

where $f : X \rightarrow \mathbb{R}$ is defined by $f(x) = \sum_{n=1}^{\infty} f_n(x)$.

Remark. This theorem is equivalent to saying $C_b(X)$ is a Banach space.

Proof. If $x \in X$, then

$$\sum_{n=1}^{\infty} |f_n(x)| \leq \sum_{n=1}^{\infty} \|f_n\|_{\infty} \leq M.$$

So we can define the function

$$f(x) := \sum_{n=1}^{\infty} f_n(x).$$

Then,

$$|f(x)| \leq \sum_{n=1}^{\infty} |f_n(x)| \leq M.$$

So $f \in \ell_{\infty}(X)$. If $N \in \mathbb{N}$,

$$\left| f(x) - \sum_{n=1}^N f_n(x) \right| = \left| \sum_{n=N+1}^{\infty} f_n(x) \right| \leq \sum_{n=N+1}^{\infty} \|f_n\|_{\infty}.$$

Since $\sum_{n=1}^{\infty} \|f_n\|_{\infty} < \infty$, $\forall \epsilon > 0$, $\exists N \in \mathbb{N}$ s.t. for $N' \geq N$,

$$\left\| f - \sum_{n=1}^{N'} f_n \right\|_{\infty} \leq \sum_{n=N'+1}^{\infty} \|f_n\|_{\infty} < \epsilon.$$

Thus, $\sum_{n=1}^N f_n$ converges uniformly to f . Since each f_n is continuous, f is continuous by Lemma 4.6.

□

5 Compactness

5.1 Compactness

Definition 5.1 (Cover, Subcover, Open Cover).

Suppose (X, d) is a metric space and $A \subset X$.

- (1) A **cover** of A is a family of sets $\{G_\lambda : \lambda \in \Lambda\}$ such that $G_\lambda \subset X$ such that $A \subset \bigcup_{\lambda \in \Lambda} G_\lambda$.
- (2) A **subcover** consists of a subfamily $\{G_{\lambda_i} : \lambda \in \Lambda'\}$ where $\Lambda' \subset \Lambda$ such that this is also a cover of A .

A cover $\{G_\lambda : \lambda \in \Lambda\}$ of A is **open** if each G_λ is open for $\lambda \in \Lambda$.

Definition 5.2 (Compactness).

A is **compact** if every open cover has a finite subcover.

Definition 5.3 (Sequentially Compact).

A is **sequentially compact** if every sequence in A has a convergent subsequence in A .

Example. Let $A \subset X$ be finite. Then, A is compact and sequentially compact.

Example (Heine-Borel Theorem).

Let $A \subset \mathbb{R}^n$. Then, A is compact $\iff A$ is closed and bounded.

Example. Let (X, d) be a metric space with the discrete metric d . Then, X is compact $\iff X$ is finite.

Suppose X is infinite. Consider the open cover $\{B_{1/2}(x) : x \in X\}$. Note that

$$\bigcup_{x \in X} B_{1/2}(x) = X.$$

However, $B_{1/2}(x) = \{x\}$ for all $x \in X$. Thus, any finite subcover of $\{B_{1/2}(x) : x \in X\}$ cannot cover X . Hence, X is not compact.

Note. In this example, X is closed (even complete) and bounded, but not compact. Thus, the Heine-Borel Theorem does not hold in general metric spaces.

Example (Comb Space).

Let $X = [0, 1] \times [0, 1]$ and define

$$d((x_1, y_1), (x_2, y_2)) = \begin{cases} |y_1 - y_2| & \text{if } x_1 = x_2 \\ y_1 + y_2 + |x_1 - x_2| & \text{if } x_1 \neq x_2 \end{cases}$$

Think of X as uncountably many vertical lines between $x = 0$, $x = 1$ and $y = 0$, $y = 1$.

Note that X is closed and bounded, but not compact. Consider

$$A = \{(x, 1) : 0 \leq x \leq 1\}.$$

Then,

$$d((x, 1), (y, 1)) \begin{cases} = 0 & \text{if } x = y \\ \geq 2 & \text{if } x \neq y \end{cases}$$

Note that $\left\{\left(\frac{1}{n}, 1\right)\right\}$ has no convergent subsequence in A . Thus, A is not sequentially compact.

Hence, X is not compact (exercise).

We can also construct an open cover of A with no finite subcover. For each $x \in [0, 1]$, let

$$G_x = \{(x, r) : r \in \left(\frac{1}{2}, 1\right]\}.$$

Let $U = \{(x, r) : r < \frac{3}{4}\}$. Then, $\{G_x : x \in [0, 1]\} \cup \{U\}$ is an open cover of A . However, it has no finite subcover.

Definition 5.4 (Totally Bounded).

A metric space (X, d) is **totally bounded** if $\forall \epsilon > 0$, $\exists$ a finite set $x_1, x_2, \dots, x_n \in X$ such that $X = \bigcup_{i=1}^n B_\epsilon(x_i)$.

Note. In the previous example, X is not totally bounded.

Remark. $A \subset X$ is **totally bounded** if $\forall \epsilon > 0$, $\exists a_1, a_2, \dots, a_n \in A$ such that $A \subset \bigcup_{i=1}^n B_\epsilon(a_i)$.

Definition 5.5 (Finite Intersection Property (FIC)).

Let $\{F_\lambda\}_{\lambda \in \Lambda}$ be a family of subsets of X . We say this family has the **finite intersection property** if for any $\lambda_1, \lambda_2, \dots, \lambda_n \in \Lambda$,

$$\bigcap_{i=1}^n F_{\lambda_i} \neq \emptyset.$$

Remark. $A \subset X$ is compact $\iff$ the metric space $(A, d|_{A \times A})$ is compact. Compactness does not depend on the metric and compactness is also the property of metric space (and subsets).

Note. Compactness is preserved under homeomorphisms.

Theorem 5.1 (Borel-Lebesgue Theorem).

Fix a metric space (X, d) . The following are equivalent:

- (1) X is compact.
- (2) If $\{F_\lambda\}_{\lambda \in \Lambda}$ is a family of closed sets with the finite intersection property, then $\bigcap_{\lambda \in \Lambda} F_\lambda \neq \emptyset$.
- (3) X is sequentially compact.
- (4) X is complete and totally bounded.

Proof.

(1) $\implies$ (2): Suppose X is compact and $\{F_\lambda\}_{\lambda \in \Lambda}$ is a family of closed sets with the finite intersection property. Set $O_\lambda = F_\lambda^c$ open. Then, FIC implies that if $\lambda_1, \dots, \lambda_n \in \Lambda$, then

$$\bigcup_{i=1}^n O_{\lambda_i} = \left(\bigcap_{i=1}^n F_{\lambda_i} \right)^c \neq X.$$

By compactness, $\{O_\lambda\}$ is not a cover of X . Thus, $\exists x \in \left(\bigcup_{\lambda \in \Lambda} O_\lambda \right)^c = \bigcap_{\lambda \in \Lambda} F_\lambda$.

(2) $\implies$ (3): Suppose (2) holds. Take $(x_n)_n$ a sequence in X . Define $F_n = \overline{\{x_k : k \geq n\}}$. If $n_1, n_2, \dots, n_l \in \mathbb{N}$, then $x_k \in \bigcap_{i=1}^l F_{n_i}$ whenever $k \geq \max\{n_1, n_2, \dots, n_l\}$. By (2), $\exists x \in \bigcap_{n=1}^\infty F_n$. For each n , $\exists$ some n_k such that $d(x_{n_k}, x) < \frac{1}{n}$. Then, $x = \lim_{k \rightarrow \infty} x_{n_k}$.

(3) $\implies$ (4): We prove the contrapositive. Suppose X is not complete. Let $\{x_n\}_{n=1}^\infty$ be a non-convergent Cauchy sequence. Then, no subsequence of $\{x_n\}_{n=1}^\infty$ can converge. Thus, X is not se-

quentially compact.

Suppose X is not totally bounded. Then, $\exists c > 0$ s.t. X cannot be covered by finitely many balls of radius c . Fix $x_1 \in X$, and define x_2 s.t. $d(x_2, x_1) > \frac{c}{2}$. Define x_3 s.t. $d(x_3, x_2), d(x_3, x_1) > \frac{c}{2}$. Inductively, define x_n such that $d(x_n, x_k) > \frac{c}{2}$ for all $1 \leq k < n$. Then, $(x_n)_n$ is a sequence in X with no convergent subsequence since $d(x_n, x_k) > \frac{c}{2}$ for all $n \neq k$. Hence, X is not sequentially compact. $\square$

Exercise: Suppose $(x_n)_n$ is Cauchy and a subsequence $(x_{n_k})_k$ converges to $x \in X$. Show that $\lim_{n \rightarrow \infty} x_n = x$.

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Lemma 5.2. If X is totally bounded and $A \subset X$, then A is totally bounded.

Proof. Fix $\epsilon > 0$. Find $x_1, \dots, x_n \in X$ such that $X = \bigcup_{i=1}^n B_{\epsilon/2}(x_i)$. Let $I_0 = \{i : B_{\epsilon/2}(x_i) \cap A \neq \emptyset\}$. For each $i \in I_0$ s.t. $B_{\epsilon/2}(x_i) \cap A \neq \emptyset$, choose $a_i \in B_{\epsilon/2}(x_i) \cap A$, then $B_\epsilon(a_i) \supset B_{\epsilon/2}(x_i)$. Hence,

$$\bigcup_{i \in I_0} B_\epsilon(a_i) \supset \bigcup_{i \in I_0} B_{\epsilon/2}(x_i) \supset A.$$

Thus, A is totally bounded. $\square$

Proposition 5.3. If X is compact and $A \subset X$ is closed, then A is compact.

Proof. Suppose X is compact. Then, X is complete and totally bounded by the Borel-Lebesgue Theorem. Since $A \subset X$ is closed and X is complete, then A is complete. Since X is totally bounded, then A is totally bounded by the previous lemma. Then, A is complete and totally bounded. Hence, A is compact by the Borel-Lebesgue Theorem. $\square$

Definition 5.6 (Separable).

A metric space X is **separable** if X has a countable dense subset.

Note. Let $A \subset X$. Recall that A is dense in X if $\overline{A} = X$.

Proposition 5.4. Every compact metric space X is separable.

Proof. Note that X is totally bounded. For any $n \in \mathbb{N}$, $\exists x_1^n, x_2^n, \dots, x_{k_n}^n \in X$ such that

$$X = \bigcup_{i=1}^{k_n} B_{1/n}(x_i^n).$$

Let $A := \bigcup_{n \in \mathbb{N}} \{x_1^n, x_2^n, \dots, x_{k_n}^n\}$. Then, A is countable. If $x \in X$ and $n \in \mathbb{N}$, then $\exists y \in A$ such that $d(x, y) < \frac{1}{n}$. Hence, $\overline{A} = X$. Thus, X is separable. $\square$

Theorem 5.5. Suppose $f : X \rightarrow Y$ is continuous and X is compact. Then, $f(X)$ is compact.

Remark. Memorize the proof!

Proof. Suppose X is compact and $f : X \rightarrow Y$ is continuous. Let $\{O_\lambda : \lambda \in \Lambda\}$ be an open cover of $f(X)$. Then, $\{f^{-1}(O_\lambda) : \lambda \in \Lambda\}$ is an open cover of X . Since X is compact, $\exists \lambda_1, \dots, \lambda_n \in \Lambda$ such that $\{f^{-1}(O_{\lambda_1}), \dots, f^{-1}(O_{\lambda_n})\}$ is a finite subcover of X . Then, $\{O_{\lambda_1}, \dots, O_{\lambda_n}\} = \{f(f^{-1}(O_{\lambda_1})), \dots, f(f^{-1}(O_{\lambda_n}))\}$ is a finite subcover of $f(X)$. Hence, $f(X)$ is compact. $\square$

Theorem 5.6 (Extreme Value Theorem).

If $f : X \rightarrow \mathbb{R}$ is continuous and $X \neq \emptyset$ is compact, then f attains its maximum and minimum.

Proof. Since f is continuous and X is compact, $f(X) \subset \mathbb{R}$ is compact by the previous theorem. By the Heine-Borel Theorem, $f(X)$ is closed and bounded. Since $f(X)$ is bounded, $\sup_{x \in X} \{f(x)\} < \infty$. Since $f(X)$ is closed, $\sup_{x \in X} \{f(x)\} \in f(X)$. Take $x_0 \in X$ such that $f(x_0) = \sup_{x \in X} \{f(x)\}$. Then, f attains its maximum at x_0 . Similarly, f attains its minimum. $\square$

Theorem 5.7. Suppose $f : X \rightarrow Y$ is a continuous bijection and X is compact, then f is a homeomorphism.

Proof. We need to show that the image of an open set is open. If $O \subset X$ is open, then O^c is closed. Since X is compact, O^c is compact by Proposition 5.3. Since f is a continuous bijection, $f(O^c)$ is compact by Theorem 5.5. Then, $f(O^c)$ is complete by the Borel-Lebesgue Theorem, hence, $f(O^c)$ is closed. So, $f(O^c)^c$ is open, but $f(O^c)^c = f(O)$ since f is a bijection. Hence, $f(O)$ is open. $\square$

Now, we will prove (4) $\implies$ (1) of the Borel-Lebesgue Theorem. Recall the theorem.

Theorem 5.1 (Borel-Lebesgue Theorem).

Fix a metric space (X, d) . The following are equivalent:

- (1) X is compact.
- (2) If $\{F_\lambda\}_{\lambda \in \Lambda}$ is a family of closed sets with the finite intersection property, then $\bigcap_{\lambda \in \Lambda} F_\lambda \neq \emptyset$.
- (3) X is sequentially compact.
- (4) X is complete and totally bounded.

Proof.

(4) $\implies$ (1): Strategy: assume X is not compact, but is totally bounded, then show that X is not complete.

Let $\{O_\lambda\}_{\lambda \in \Lambda}$ be an open cover with no finite subcover. Find $x_1^1, x_2^1, \dots, x_{k_1}^1 \in X$ such that

$$X = \bigcup_{i=1}^{k_1} B_1(x_i^1).$$

Claim. For some $1 \leq i \leq k_1$, the cover $\{O_\lambda\}_{\lambda \in \Lambda}$ has no finite subcover of $B_1(x_i^1)$.

Note that otherwise we could take the finite union of all these finite subcovers to get a finite subcover of X , which is a contradiction.

Proof of Claim. Set $y_1 = x_i^1$. Since $B_1(y_1)$ is totally bounded, we can find $x_1^2, x_2^2, \dots, x_{k_2}^2 \in B_1(y_1)$ such that

$$B_1(y_1) \subset \bigcup_{i=1}^{k_2} B_{1/2}(x_i^2).$$

As before, there is some $1 \leq i \leq k_2$ such that the cover $\{O_\lambda\}_{\lambda \in \Lambda}$ has no finite subcover of $B_{1/2}(x_i^2)$. Set $y_2 = x_i^2$. Repeat this process. By induction, we construct y_n such that $B_{1/n}(y_n)$ has no finite subcover of $\{O_\lambda\}_{\lambda \in \Lambda}$. Also,

$$d(y_n, y_m) \leq \frac{1}{\min(n, m)}.$$

So $\{y_n\}_{n=1}^\infty$ is Cauchy. If $y_0 \in X$ such that $\lim_{n \rightarrow \infty} y_n = y_0$. Then, $\exists \lambda_0 \in \Lambda$ such that $y_0 \in O_{\lambda_0}$. So,

$\exists m \in \mathbb{N}$ s.t.

$$B_{1/m}(y_0) \subset O_{\lambda_0}.$$

Thus, $\exists N \in \mathbb{N}$ s.t.

$$B_{1/N}(y_N) \subset B_{1/m}(y_0) \subset O_{\lambda_0},$$

so O_{λ_0} is a finite subcover of $B_{1/N}(y_N)$, which is a contradiction. $\square$

So, X is not complete. $\square$

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Theorem 5.8. Suppose (X, d) is a compact metric space, and $f : X \rightarrow Y$ is continuous where (Y, ρ) is another metric space. Then, f is uniformly continuous.

Proof. Fix $\epsilon > 0$. Since f is continuous, for each $x \in X$, $\exists \delta_x > 0$ s.t. if $d(x, x') < \delta_x$, then $\rho(f(x), f(x')) < \frac{\epsilon}{2}$. Note that $\{B_{\delta_x/2}(x)\}_{x \in X}$ is an open cover of X . By compactness, $\exists x_1, \dots, x_n \in X$ s.t.

$$X = \bigcup_{i=1}^n B_{\delta_{x_i}/2}(x_i).$$

Set $\delta = \min_{1 \leq i \leq n} \left\{ \frac{\delta_{x_i}}{2} \right\}$. If $x, x' \in X$ s.t. $d(x, x') < \delta$, then take $1 \leq i \leq n$ s.t. $x \in B_{\delta_{x_i}/2}(x_i)$. Then,

$$d(x, x_i) < \frac{\delta_{x_i}}{2},$$

also

$$d(x', x_i) \leq d(x', x) + d(x, x_i) < \frac{\delta_{x_i}}{2} + \frac{\delta_{x_i}}{2} = \delta_{x_i}.$$

So, $\rho(f(x), f(x_i)), \rho(f(x'), f(x_i)) < \frac{\epsilon}{2}$. By the triangle inequality,

$$\rho(f(x), f(x')) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \quad \square$$

Example. Let $X = [-1, 1] \setminus \{0\}$ and $Y = \mathbb{R}$. Consider $f : X \rightarrow Y$ given by $f(x) = \frac{1}{x}$. Note that f is continuous, X is not compact, and f is not uniformly continuous.

Example. Consider $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$.

Claim. f is not uniformly continuous.

Theorem 5.9. There is a continuous surjective function $f : [0, 1] \rightarrow [0, 1]^2$.

Note. f maps a point to some paths.

Example (Cantor Set).

Let $C_0 = [0, 1]$ and $J_1 = \left(\frac{1}{3}, \frac{2}{3}\right)$. Then,

$$\begin{aligned} C_1 &= C_0 \setminus J_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right], & J_2 &= \left(\frac{1}{9}, \frac{2}{9}\right) \cup \left(\frac{7}{9}, \frac{8}{9}\right), \\ C_2 &= C_1 \setminus J_2 = \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right] \\ &\vdots \end{aligned}$$

If $x \in [0, 1]$, then x can be written as

$$x = 0.x_1x_2x_3 \dots = \sum_{i=1}^{\infty} \frac{x_i}{3^i}$$

in ternary where $x_i \in \{0, 1, 2\}$ for all $i \in \mathbb{N}$. Note that $J_1 = \{x \in [0, 1] : x_1 = 1\}$, so

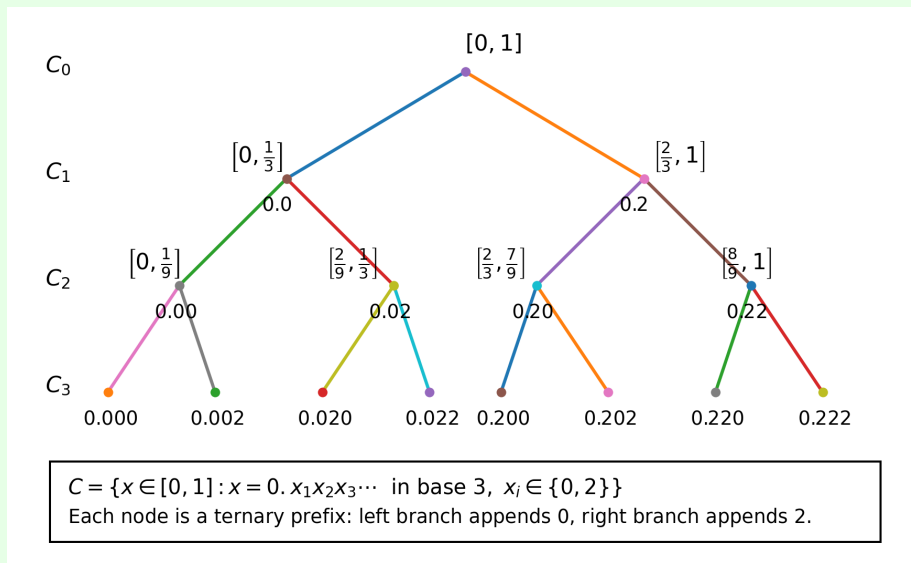
$$C_1 = \{x \in [0, 1] : x_1 \in \{0, 2\}\}, \quad J_2 = \{x \in [0, 1] : x_2 = 1\}$$

and so on. Note that

$$C := \{x \in [0, 1] : x \text{ has a ternary representation } x = 0.x_1x_2x_3 \dots \text{ with all } x_i \in \{0, 2\}\}.$$

From A5, C is closed (hence compact) and is in fact perfect. Also, $C \cap \left[0, \frac{1}{3}\right] = C \cap \left(-1, \frac{1}{2}\right)$ is both closed and open subset of C .

Another perspective is to view C as a tree:



Theorem 5.10. If (X, d) is any non-empty compact metric space, then there is a continuous surjection $f : C \rightarrow X$ where C is the Cantor set.

Proof. Suppose X is compact, hence totally bounded. Take $x_1, x_2, \dots, x_{2^n}$ such that

$$X = \bigcup_{i=1}^{2^n} B_1(x_i).$$

Relabel the x_i 's as x_a where $a = (a_1, a_2, \dots, a_n)$ and $a_i \in \{0, 1\}$. Define $f_1 : C \rightarrow X$ by $f_1(x) = x_a$ for x of the form $x = 0.(a_1a_2a_3 \dots a_n)x_{n+1}x_{n+2} \dots$

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Note that f_1 is “locally constant”, i.e. $\forall x \in C, \exists$ an open neighborhood O of x such that $f(x') = f(x)$ for all $x' \in O$. Hence, f_1 is continuous. Also, $f_1(C) = \{x_1, x_2, \dots, x_{2^n}\}$ is 1-dense.

Repeat this process by choosing a finite $\frac{1}{2}$ -dense subset of $\overline{B_1(x_i)}$. We may assume these are labeled by x_1 for $a = (a_1, a_2, \dots, a_k)$ and $a_i \in \{0, 1\}$. Define $f_2 : C \rightarrow X$ by $f_2(x) = x_a^i$ if $x = 0.x_1x_2 \dots x_{n_1}a_1a_2 \dots a_kx_{n_k+1} \dots$ where $x_1x_2 \dots x_{n_1}$ corresponds to x_i . Note that f_2 is locally constant,

so f_2 is continuous. Also, $f_2(C)$ is $\frac{1}{2}$ -dense. Note that

$$d(f_2(x), f_1(x)) < 1 \quad \forall x \in C.$$

By induction, we construct a sequence $f_n : C \rightarrow X$ such that

- (1) f_n is locally constant, hence continuous.
- (2) $f_n(C)$ is $\frac{1}{2^n}$ -dense in X .
- (3) If $x \in C$, then $d(f_k(x), f_{k+1}(x)) \leq \frac{1}{2^k}$ for all $k \in \mathbb{N}$.

By (3), $\forall x \in C$, we have

$$\sum_{k=1}^{\infty} d(f_k(x), f_{k+1}(x)) < \infty.$$

Thus $\{f_k(x)\}_{k=1}^{\infty}$ is Cauchy. So we may set $f(x) = \lim_{k \rightarrow \infty} f_k(x)$. We have

$$\sup_{x \in X} d(f(x), f_k(x)) \leq \frac{1}{2^{k-1}}.$$

So, $\{f_k\}_{k=1}^{\infty}$ converges uniformly to f . Thus, f is continuous. If $y \in X$, for $\epsilon > 0$, take k such that $\frac{1}{2^k} < \frac{\epsilon}{2}$. Then, $\exists x \in C$ such that $d(f_k(x), y) < \frac{\epsilon}{2}$. Thus,

$$d(f(x), y) < d(f_{k+1}(x), y) + d(f_{k+1}(x), f(x)) < \frac{\epsilon}{2} + \frac{1}{2^k} < \epsilon.$$

Hence, f is surjective since $f(C)$ is compact and hence closed.

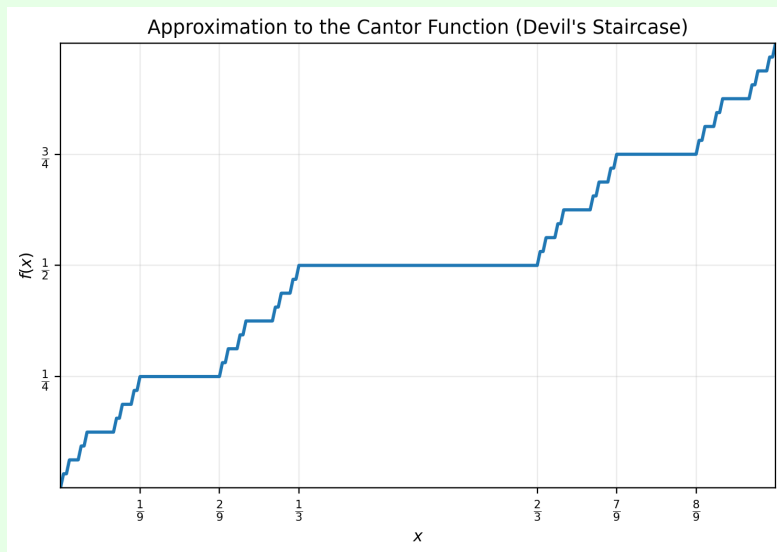
□

Remark. If X' is a compact metric space, then $|X'| \leq |\mathbb{R}|$.

Example (Cantor's Function).

The Devil's Staircase.

Define $f : [0, 1] \rightarrow [0, 1]$ such that f is continuous non-decreasing locally constant on $C^c = [0, 1] \setminus C$ and $f(0) = 0, f(1) = 1$. The function looks like the following:



If $x = 0.x_1x_2x_3 \dots \in C$ where $x_i \in \{0, 2\}$. Then,

$$f(x) = 0.\frac{x_1}{2}\frac{x_2}{2}\frac{x_3}{2} \dots \text{ in base 2.}$$

Note that

$$x = \sum_{i=1}^{\infty} \frac{x_i}{3^i} \xrightarrow{f} \sum_{i=1}^{\infty} \frac{x_i}{2^{i+1}}.$$

Off the Cantor set, we just set $f(x) = f(c)$ where $c \in C$ is the largest point in C such that $c \leq x$.

Recall the following theorem.

Theorem 5.9. There is a continuous surjective function $f : [0, 1] \rightarrow [0, 1]^2$.

Proof. Since $[0, 1]^2$ is compact, $\exists$ a continuous surjection $g : C \rightarrow [0, 1]^2$ by Theorem 5.10. We want to find $f : [0, 1] \rightarrow [0, 1]^2$ continuous and $f|_C = g$. Write

$$C = [0, 1] \setminus \left(\bigcup_{i=1}^{\infty} (a_i, b_i) \right)$$

where (a_i, b_i) are disjoint open intervals. $f(x)$ is defined by setting $f(x) = g(x)$ for $x \in C$ and f is linear on each interval (a_i, b_i) , i.e. $f(ta_i + (1-t)b_i) = tg(a_i) + (1-t)g(b_i)$. $\square$

Note that a continuous surjection means space filling curve. Below theorem is a generalization of the previous theorem.

Theorem 5.11. Suppose X is a non-empty compact convex subset of a normed vector space. If $f : C \rightarrow X$ is continuous, then f has a continuous extension to $g : [0, 1] \rightarrow X$, i.e. $g(x) = f(x) \forall x \in C$.

Note. Convex: if $x_1, x_2 \in X$ and $t \in [0, 1]$, then $tx_1 + (1 - t)x_2 \in X$.

Proof. Write

$$C = [0, 1] \setminus \left(\bigcup_{n=1}^{\infty} (a_n, b_n) \right)$$

where the (a_n, b_n) are pairwise disjoint. Given $f : C \rightarrow X$ continuous, define $g : [0, 1] \rightarrow X$ by

$$g(x) = \begin{cases} f(x) & \text{if } x \in C \\ tf(a_n) + (1 - t)f(b_n) & \text{if } x = ta_n + (1 - t)b_n \text{ for } t \in (0, 1). \end{cases}$$

Fix $\epsilon > 0$. Since f is continuous and C is compact, f is uniformly continuous, so $\exists \delta_1 > 0$ s.t. if $|x - y| < \delta_1$, then

$$\|f(x) - f(y)\| < \frac{\epsilon}{3}.$$

Note that there are only finitely many n such that $b_n - a_n \geq \delta_1$. Let F denote the set of such n . Define $D = \max \left\{ \|f(b_n) - f(a_n)\| \text{ for } n \in F, \frac{\epsilon}{3} \right\}$. Set $\delta = \frac{\delta_1 \epsilon}{3D} \leq \delta_1$.

Claim. If $x, y \in [0, 1]$ and $|x - y| < \delta$, then $\|g(x) - g(y)\| < \epsilon$.

Case 1: $x, y \in C$. Then, $|x - y| < \delta_1$, so $\|f(x) - f(y)\| < \frac{\epsilon}{3}$. Hence, $\|g(x) - g(y)\| = \|f(x) - f(y)\| < \epsilon$.

Case 2: $x, y \in (a_n, b_n)$ where $n \in F$. Then,

$$\|g(x) - g(y)\| = \frac{|x - y|}{b_n - a_n} \|f(b_n) - f(a_n)\| \leq \frac{|x - y|}{b_n - a_n} D < |x - y| \frac{D}{\delta_1} < \frac{\epsilon}{3}.$$

Case 3: $x, y \in (a_n, b_n)$ where $n \notin F$. Then, $b_n - a_n < \delta_1$. SO,

$$\|g(x) - g(y)\| \leq \|f(b_n) - f(a_n)\| < \frac{\epsilon}{3}.$$

Case 4: $x \in (a_n, b_n)$ and $y \in C$ where $n \in F$. Assume $y \geq b_n$ (the other case is similar). Then, $y - b_n < \delta$ and $b_n - x < \delta$. So,

$$\begin{aligned}\|g(x) - g(y)\| &\leq \|g(x) - g(b_n)\| + \|g(b_n) - g(y)\| \\ &\leq \frac{\epsilon}{3} + \frac{\epsilon}{3} < \epsilon\end{aligned}$$

where the first $\frac{\epsilon}{3}$ comes from Case 2 and the second $\frac{\epsilon}{3}$ comes from Case 1.

Case 5: $x \in (a_n, b_n)$ and $y \in (a_m, b_m)$ where $n \neq m$. Assume $b_n < a_m$. Then,

$$\begin{aligned}\|g(x) - g(y)\| &\leq \|g(x) - g(b_n)\| + \|g(b_n) - g(a_m)\| + \|g(a_m) - g(y)\| \\ &\leq \frac{\epsilon}{3} + D + \frac{\epsilon}{3} < \epsilon\end{aligned}$$

□

5.2 Arzela-Ascoli Theorem

Arzela-Ascoli Theorem describes compact subsets of $C(X)$, where X is a compact metric space. Note that $C(X) = C_b(X)$ since X is compact. Recall that

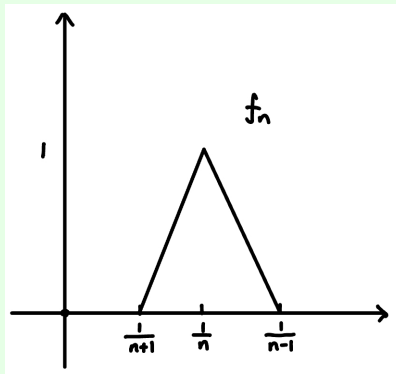
$$C(X) = \{f : X \rightarrow \mathbb{R} : f \text{ is continuous}\}$$

Example. Let $K = \{x^n\}_{n=1}^\infty \subset C([0, 1])$. Note that if $f_n(x) = x^n$, then $(f_n)_n$ converges pointwise to f where

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1 \end{cases}$$

Note that f is not continuous, so $(f_n)_n$ does not converge uniformly to any function in $C([0, 1])$. The sequence (f_n) has no convergent subsequence in K . Hence $K = \{x^n\}_{n=1}^\infty$ is not compact since in a compact metric space, every sequence has a convergent subsequence.

Example. Define $f_n \in C([0, 1])$ as follows:



where $f_n\left(\frac{1}{n}\right) = 1$ and $0 = f_n\left(\frac{1}{n+1}\right) = f_n\left(\frac{1}{n-1}\right) = f_n(0) = f_n(1)$. Note that f_n is linear between these points and $(f_n)_n$ converges pointwise to 0, which is continuous.

Question: Is $K = \{f_n\}_{n=1}^\infty$ compact?

Answer: This does not have a convergent subsequence. Also note that $\|f_n - f_m\|_\infty \geq 1$ with $n \neq m$. In fact, K is discrete with the induced metric. Hence, K is not compact.

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Definition 5.7 (Equicontinuity, Uniform Equicontinuity).

Let (X, d) be a metric space. A family $\mathcal{F} \subset C(X, \mathbb{R})$ is **equicontinuous** at $x \in X$ if $\forall \epsilon > 0$, $\exists \delta > 0$ s.t. if $f \in \mathcal{F}$, then $d(x, x') < \delta \implies |f(x) - f(x')| < \epsilon$. $\mathcal{F}$ is **equicontinuous** if it is equicontinuous at every point $x \in X$.

$\mathcal{F}$ is **uniformly equicontinuous** if $\forall \epsilon > 0$, $\exists \delta > 0$ s.t. $d(x, y) < \delta \implies |f(x) - f(y)| < \epsilon$ $\forall x, y \in X$ and $\forall f \in \mathcal{F}$.

Example. Consider $\mathcal{F} = \{x^n\}_{n=1}^\infty \subset C([0, 1])$.

Question: Is $\mathcal{F}$ equicontinuous at $\frac{1}{2}$?

Answer: Yes, we can take $\delta = \min\left\{\epsilon, \frac{1}{4}\right\}$.

Question: Is $\mathcal{F}$ equicontinuous?

Answer: No, $\mathcal{F}$ is not equicontinuous at 1.

Example. Define $f_n \in C([0, 1])$ the same as Example 5.2. Let $\mathcal{F} = \{f_n\}_{n=1}^\infty$.

Question: Is $\mathcal{F}$ equicontinuous at $\frac{1}{2}$?

Answer: Yes, f_n is equal to 0 in $\left(\frac{1}{3}, 1\right)$ for $n \geq 4$.

Question: Is $\mathcal{F}$ equicontinuous?

Answer: No. If $\delta > 0$, take n s.t. $\frac{1}{n} < \delta$. Then, $f_n\left(\frac{1}{n}\right) = 1$ and $f_n(0) = 0$, so the condition fails for $\epsilon = 1$.

Lemma 5.12. Suppose X is a compact metric space and $K \subset C(X)$ is compact, then K is uniformly equicontinuous.

Proof. Fix $\epsilon > 0$. Since K is compact, $\exists f_1, f_2, \dots, f_n \in K$ s.t. if $f \in K$, then $\exists i$ such that $\|f - f_i\|_\infty < \frac{\epsilon}{3}$. Each f_i is uniformly continuous, $\exists \delta_i > 0$ s.t. if $x, y \in X$ and $d(x, y) < \delta_i$, then $|f_i(x) - f_i(y)| < \frac{\epsilon}{3}$. Set $\delta = \min\{\delta_1, \dots, \delta_n\}$. If $x, y \in X$, $d(x, y) < \delta$ and $f \in K$, take $1 \leq i \leq n$ such that $\|f - f_i\|_\infty < \frac{\epsilon}{3}$. Since $d(x, y) < \delta \leq \delta_i$ and $|f_i(x) - f_i(y)| < \frac{\epsilon}{3}$, we have

$$\begin{aligned} |f(x) - f(y)| &\leq |f(x) - f_i(x)| + |f_i(x) - f_i(y)| + |f_i(y) - f(y)| \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon. \end{aligned} \quad \square$$

Lemma 5.13. If X is a compact metric space and $\mathcal{F} \subset C(X)$, then $\mathcal{F}$ is equicontinuous $\iff \mathcal{F}$ is uniformly equicontinuous.

Proof.

($\implies$): If $\mathcal{F}$ is equicontinuous, then $\forall \epsilon > 0, \forall x \in X, \exists \delta_x > 0$ s.t. $d(x, y) < \delta_x \implies |f(x) - f(y)| < \frac{\epsilon}{2}$ $\forall f \in \mathcal{F}$. Note that $\{B_{\delta_x/2}(x)\}_{x \in X}$ is an open cover of X . Since X is compact, $\exists x_1, \dots, x_n \in X$ s.t. $X = \bigcup_{i=1}^n B_{\delta_{x_i}/2}(x_i)$. Set $\delta = \min\left\{\frac{\delta_{x_1}}{2}, \dots, \frac{\delta_{x_n}}{2}\right\}$. If $d(x, y) < \delta$ and $f \in \mathcal{F}$, $\exists i$ s.t. $d(x, x_i) < \frac{\delta_{x_i}}{2} \implies d(y, x_i) < \delta_{x_i}$, so

$$\begin{aligned} |f(x) - f(y)| &\leq |f(x) - f(x_i)| + |f(x_i) - f(y)| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

($\impliedby$): this follows from definition. $\square$

Theorem 5.14 (Arzela-Ascoli Theorem).

Let (X, d) be a compact metric space. A subset $K \subset C(X)$ is compact $\iff$ K is closed, bounded and equicontinuous.

Proof.

($\Rightarrow$): this follows from the previous lemma and the fact that a compact metric space is closed and bounded.

($\Leftarrow$): Suppose $K \subset C(X)$ is closed, bounded and equicontinuous (hence uniformly equicontinuous). Since $C(X)$ is complete and K is closed, K is complete. Fix $\epsilon > 0$. Since K is uniformly equicontinuous, $\exists \delta > 0$ s.t. $f \in K, x, y \in X, d(x, y) < \delta \implies |f(x) - f(y)| < \frac{\epsilon}{4}$. Since X is totally bounded, $\exists x_1, \dots, x_n \in X$ s.t. $X = \bigcup_{i=1}^n B_\delta(x_i)$.

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Consider $T : C(X) \rightarrow (\mathbb{R}^n, \|\cdot\|_\infty)$ given by $T(f) = (f(x_1), \dots, f(x_n))$. Note that $\|T\|_\infty \leq \|f\|_\infty$. Since K is bounded, $T(K)$ is bounded, hence totally bounded. Take $f_1, f_2, \dots, f_m \in K$ s.t. if $f \in K$, then $\|T(f) - T(f_i)\|_\infty < \frac{\epsilon}{4}$ for some $1 \leq i \leq m$. If $f \in K$, take i with $1 \leq i \leq m$ s.t. $|f(x_j) - f_i(x_j)| < \frac{\epsilon}{4}$ for all $1 \leq j \leq n$. If $y \in X$, then $\exists 1 \leq j \leq n$ s.t. $d(y, x_j) < \delta$. So,

$$\begin{aligned} |f(y) - f_i(y)| &\leq |f(y) - f(x_j)| + |f(x_j) - f_i(x_j)| + |f_i(x_j) - f_i(y)| \\ &< \frac{\epsilon}{4} + \frac{\epsilon}{4} + \frac{\epsilon}{4} = \frac{3\epsilon}{4}. \end{aligned}$$

Hence, $\|f - f_i\|_\infty < \epsilon$ for some $1 \leq i \leq m$. So, K is totally bounded. Since K is complete and totally bounded, K is compact. □

Example. Consider $K = \{f_n\}_{n=1}^\infty \subset C([0, 1])$ where $f_n(t) = nt$. Since K is discrete, K is closed. Also K is equicontinuous. However, K is not compact since no subsequence converges.

6 Connectedness

Definition 6.1 (Disconnected, Connected).

A metric space (X, d) is **disconnected** if we can write $X = U_1 \cup U_2$ where each U_i is non-empty, open, and $U_1 \cap U_2 = \emptyset$. We say X is **connected** if it is not disconnected.

Remark. X is disconnected $\iff \exists U_1 \subset X$ such that U_1 is non-empty, proper (i.e. $U_1 \neq X$), and clopen.

Proof.

($\Rightarrow$): If X is disconnected. Then, $X = U_1 \cup U_2$ where each U_i is non-empty, open, and $U_1 \cap U_2 = \emptyset$. Since $U_2 = X \setminus U_1$, U_1 is also closed. Hence, U_1 is clopen. Also, U_1 is proper since U_2 is non-empty.

($\Leftarrow$): If $\exists U_1 \subset X$ such that U_1 is non-empty, proper and clopen. Then, let $U_2 = X \setminus U_1$. Since U_1 is clopen, U_2 is also clopen (hence open). Since U_1 is proper, U_2 is non-empty. Also, $U_1 \cap U_2 = \emptyset$ and $X = U_1 \cup U_2$. Hence, X is disconnected. $\square$

Example. Let $X = [0, 1] \cup [2, 3]$ with the usual metric. Then, X is disconnected.

Proof. Note that $X = \left([0, 1] \cap \left(-\frac{1}{2}, \frac{3}{2}\right)\right) \cup \left([2, 3] \cap \left(\frac{3}{2}, \frac{7}{2}\right)\right)$ where each set is non-empty, open in X since $\left(-\frac{1}{2}, \frac{3}{2}\right)$ and $\left(\frac{3}{2}, \frac{7}{2}\right)$ are open in $\mathbb{R}$, and the two sets are disjoint. Hence, X is disconnected. $\square$

Example. Let $X = \mathbb{Q}$. Then, X is disconnected.

Proof. Note that $X = \left((-\infty, \sqrt{2}) \cap \mathbb{Q}\right) \cup \left((\sqrt{2}, \infty) \cap \mathbb{Q}\right)$ where each set is non-empty, open in X since $(-\infty, \sqrt{2})$ and $(\sqrt{2}, \infty)$ are open in $\mathbb{R}$, and the two sets are disjoint. Hence, X is disconnected. $\square$

Proposition 6.1. $[0, 1]$ is connected.

Proof. We will prove by contradiction. Suppose $[0, 1]$ is disconnected. Then, $[0, 1] = U_1 \cup U_2$ where each U_i is non-empty, open in $[0, 1]$, and $U_1 \cap U_2 = \emptyset$. Suppose $1 \in U_2$. Set $x = \sup U_1$ where $x \in [0, 1]$. If $x \notin U_1$, then $x \in U_2$. Since U_2 is open in $[0, 1]$, $\exists \epsilon > 0$ s.t. $(x - \epsilon, x] \subset U_2$. Thus, $x - \epsilon$ is an upper bound for U_1 smaller than x , contradiction. If $x \in U_1$, then $x < 1$ since $1 \in U_2$. Since U_1 is

open in $[0, 1]$, $\exists \epsilon > 0$ s.t. $[x, x + \epsilon) \subset U_1$. Thus, x is not an upper bound for U_1 , contradiction. Hence, $[0, 1]$ is connected. $\square$

Remark. $[a, b]$ is connected for $a \leq b$. In fact, all intervals in $\mathbb{R}$ are connected.

Also, if $\emptyset \neq A \subset \mathbb{R}$ is not an interval, then A is disconnected.

Proof. If A is not an interval, then $\exists a, c \in A$ and $b \in \mathbb{R} \setminus A$ such that $a < b < c$. Then, $A = (A \cap (-\infty, b)) \cup (A \cap (b, \infty))$ where each set is non-empty, open in A since $(-\infty, b)$ and (b, ∞) are open in $\mathbb{R}$, and the two sets are disjoint. Hence, A is disconnected. $\square$

Example. $\mathbb{R}^2 \setminus \mathbb{Q}^2$ is connected.

Definition 6.2 (Path Connected).

A metric space (X, d) is **path connected** if $\forall x, y \in X$, $\exists$ a continuous function $f : [0, 1] \rightarrow X$ such that $f(0) = x$ and $f(1) = y$.

Theorem 6.2 (Intermediate Value Theorem).

If X is connected and $f : X \rightarrow Y$ is continuous, then $f(X)$ is connected.

Remark. Let $X = [a, b]$ and $Y = \mathbb{R}$. Then, $f(X)$ is an interval.

Proof. Suppose $f(X)$ is disconnected. Then, $\exists U_1, U_2$ open in Y such that

$$f(X) = (U_1 \cap f(X)) \sqcup (U_2 \cap f(X))$$

where $(U_1 \cap f(X))$ and $(U_2 \cap f(X))$ are non-empty, open in $f(X)$, and disjoint. Then,

$$X = f^{-1}(f(X)) = f^{-1}(U_1) \sqcup f^{-1}(U_2)$$

where $f^{-1}(U_1)$ and $f^{-1}(U_2)$ are non-empty, open in X since f is continuous, and disjoint. Since X is connected, $f^{-1}(U_1) = X$ or $f^{-1}(U_2) = X$, i.e. $f(X) \subset U_1$ or $f(X) \subset U_2$, a contradiction. Hence, $f(X)$ is connected. $\square$

Lemma 6.3. Suppose $X_\lambda \subset X$ with $\lambda \in \Lambda \neq \emptyset$ such that each X_λ is connected and $\bigcap_{\lambda \in \Lambda} X_\lambda \neq \emptyset$. Then, $\bigcup_{\lambda \in \Lambda} X_\lambda$ is connected.

Proof. We replace X with $\bigcup_{\lambda \in \Lambda} X_\lambda$ and assume $X = \bigcup_{\lambda \in \Lambda} X_\lambda$. Suppose $\bigcup_{\lambda \in \Lambda} X_\lambda = U_1 \cup U_2$ where each U_i is non-empty, open in $\bigcup_{\lambda \in \Lambda} X_\lambda$, and $U_1 \cap U_2 = \emptyset$. Fix $x \in \bigcap_{\lambda \in \Lambda} X_\lambda$. Assume $x \in U_1$. Fix $\lambda \in \Lambda$. Then,

$$(X_\lambda \cap U_1) \cup (X_\lambda \cap U_2) = X_\lambda$$

so $x \in X_\lambda \subset U_1$. Since X_λ is connected and $X_\lambda \cap U_1$. We have $\bigcup_{\lambda \in \Lambda} X_\lambda = U_1$, hence, $\bigcup_{\lambda \in \Lambda} X_\lambda$ is connected. $\square$

Corollary 6.4. If X is path connected, then X is connected.

Proof. Fix $x \in X$. Since X is path connected, then $\forall y \in X, \exists f_y : [0, 1] \rightarrow X$ continuous such that $f_y(0) = x$ and $f_y(1) = y$. Set $X_y = f_y([0, 1])$, which is connected. Note that $x \in \bigcap_{y \in X} X_y$ and $\bigcup_{y \in X} X_y = X$. Hence, X is connected by Lemma 6.3. $\square$

Example. $\mathbb{R}^2 \setminus \{(0, 0)\}$ is path connected, hence connected.

Definition 6.3 (Locally Connected).

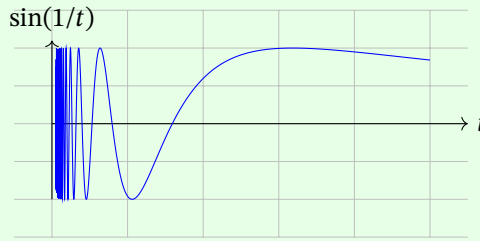
A metric space X is **locally connected** if $\forall x \in X, O$ is a neighborhood of $x, \exists$ a connected neighborhood U of x such that $x \in U \subset O$.

Remark. We can replace “connected ” with “path connected” to define **locally path connected**.

Example. Let $X = [0, 1] \cup [2, 3]$. Then, X is locally connected but not connected.

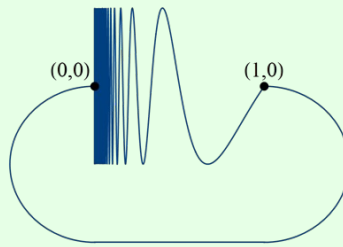
Example (Topologist's Sine Curve).

Let $X_0 = \left\{ \left(t, \sin\left(\frac{1}{t}\right) \right) : t \in (0, 1] \right\}$. Note that X_0 is path connected.



Claim. $X = X_0 \cup \{(0, 0)\}$ is connected, not path connected, and not locally connected.

Example. The Warsaw Circle is path connected but not locally connected.



Definition 6.4 (Connected Component).

If $x \in X$, the **connected component** of x is the largest connected set that contains x .

Note. Let $\mathcal{F}$ be all subsets of X that are connected and contain x . Note that $\{x\} \in \mathcal{F}$. By Lemma 6.3, $\bigcup_{A \in \mathcal{F}} A$ is connected.

Proposition 6.5. If $A \subset X$ is connected, then $\bar{A}$ is connected.

Proof. Assume that A is connected with $A \neq \emptyset$. Suppose $\bar{A} = O_1 \cup O_2$ where each O_i is non-empty, open in $\bar{A}$, and $O_1 \cap O_2 = \emptyset$. Then, $A = (A \cap O_1) \cup (A \cap O_2)$. Since A is connected, one of these, say $A \cap O_2$ is empty. Thus, $A \subset O_1^c$ is closed and $\bar{A} \subset O_1^c$. Hence, $O_2 = \emptyset$, which is a contradiction. Therefore, $\bar{A}$ is connected. $\square$

Remark. If A is the connected component of $x \in X$ and B is the connected component of $y \in X$ such that $A \cap B \neq \emptyset$, then $A = B$.

Proof. If $A \cap B \neq \emptyset$, then $A \cup B$ is connected by Lemma 6.3 and contains x and y . Thus, $A \cup B = A = B$ by the definition of connected component. $\square$

7 Approximations

Lecture 26, 2026/03/16

7.1 Polynomial Approximations

Recall $C([a, b]) = \{f : [a, b] \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$ with $a < b$.

Problem: Given $f : [a, b] \rightarrow \mathbb{R}$ continuous and $\epsilon > 0$, can we find a polynomial p with

$$\|f - p\|_{\infty} = \sup_{x \in [a, b]} |f(x) - p(x)| < \epsilon?$$

First attempt: Lagrange Interpolation.

Let $x_i = a + i \frac{b-a}{n}$ for $0 \leq i \leq n$. Then, $\exists!$ polynomial q of degree $\leq n$ with $q(x_i) = f(x_i)$ for $i = 0, 1, \dots, n$. Set

$$q_i(x) = \prod_{0 \leq j \leq n} \frac{x - x_j}{x_i - x_j} \quad \text{with } \deg(q_i) = n$$

and

$$q_i(x_j) = \begin{cases} 1 & i = j \\ 0 & i \neq j. \end{cases}$$

Let $q(x) = \sum_{i=0}^n f(x_i) q_i(x)$, then q is a polynomial with $\deg(q) \leq n$. If p is another polynomial with $\deg(p) \leq n$ and $q(x_i) = p(x_i) = f(x_i)$ for $i = 0, 1, \dots, n$, then $q = p$ ($\deg(q - p) \leq n$, $x_0, x_1, \dots, x_n$ are $n + 1$ distinct roots of $q - p$), so $q - p = 0$.

Unfortunately, for a given $f \in C([a, b])$, defining q_n to be the unique polynomial with $q_n(x_i) = f(x_i)$ for $i = 0, 1, \dots, n$ does not guarantee $\|q_n - f\|_{\infty} \rightarrow 0$. Find an example online.

Second attempt: Taylor series.

Let $f(x) = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$, which is valid for $|x| < 1$. This does not work if we want convergence on $[-2, 2]$.

Even worse: let

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0 \\ 0 & x = 0. \end{cases}$$

Note that f is infinitely differentiable (i.e. f is C^∞), but $f^{(k)}(0) = 0$ for all $k \geq 0$. So the Taylor series of f at 0 is 0, which does not converge to f on any interval containing 0.

Theorem 7.1 (Weierstrass Approximation Theorem).

The polynomials are dense in $C([a, b])$.

Remark (Binomial Theorem).

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}. \text{ In particular, } 1 = (x + (1 - x))^n = \sum_{k=0}^n \binom{n}{k} x^k (1 - x)^{n-k}.$$

Setup: We may assume $a = 0$ and $b = 1$. Set $p_k^n(x) = \binom{n}{k} x^k (1 - x)^{n-k}$ for $0 \leq k \leq n$. Note that $p_k^n(x) \geq 0$ for all $x \in [0, 1]$. Define Bernstein's map $B_n : C([0, 1]) \rightarrow \mathbb{R}[x]$ by

$$(B_n f)(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) p_k^n(x).$$

Note that $B_n(f)$ is a polynomial with $\deg(B_n(f)) \leq n$.

Lemma 7.2. B_n is a positive linear map. That is,

- (1) $B_n(af + bg) = aB_n(f) + bB_n(g)$ for all $f, g \in C([0, 1])$ and $a, b \in \mathbb{R}$.
- (2) If $f \geq 0$, then $B_n(f) \geq 0$.
- (3) If $f \geq g$, then $B_n(f) \geq B_n(g)$.
- (4) If $|f| \leq g$, then $|B_n(f)| \leq B_n(g)$.

Proof.

(1) Exercise.

(2) If $f \geq 0$, then

$$(B_n f)(x) = \sum_{k=0}^n \underbrace{f\left(\frac{k}{n}\right)}_{\geq 0} p_k^n(x) \geq 0$$

on $[0, 1]$.

(3) If $f \geq g$, then $f - g \geq 0$, so $B_n(f - g) \geq 0$ by (2). Thus, $B_n(f) - B_n(g) \geq 0 \implies B_n(f) \geq B_n(g)$.

(4) If $|f| \leq g$, then $-g \leq f \leq g$, so $-B_n(g) \leq B_n(f) \leq B_n(g)$ by (3). Thus, $|B_n(f)| \leq B_n(g)$.

□

Lemma 7.3.

(1) $B_n(1) = 1$ where the first 1 means $1(t) = 1 \forall t \in [0, 1]$.

(2) $B_n(x) = x$ where the first x means $x(t) = t \forall t \in [0, 1]$.

(3) $B_n(x^2) = \frac{n-1}{n}x^2 + \frac{1}{n}x = x^2 + \frac{1}{n}(x - x^2)$.

Hence $B_n(x^2)$ converges uniformly to x^2 as $n \rightarrow \infty$.

Differentiate the Binomial Theorem in x :

$$n(x+y)^{n-1} = \sum_{k=0}^n \binom{n}{k} k x^{k-1} y^{n-k}.$$

Differentiate again:

$$\begin{aligned} n(n-1)(x+y)^{n-2} &= \sum_{k=0}^n \binom{n}{k} k(k-1) x^{k-2} y^{n-k} \\ &= \sum_{k=0}^n \binom{n}{k} k^2 x^{k-2} y^{n-k} - \sum_{k=0}^n \binom{n}{k} k x^{k-2} y^{n-k} \\ &= \sum_{k=0}^n \binom{n}{k} k^2 x^{k-2} y^{n-k} - \frac{1}{x} n(x+y)^{n-1} \end{aligned}$$

Proof.

(1) Note that

$$\begin{aligned} (B_n 1)(t) &= \sum_{k=0}^n 1 \binom{n}{k} p_k^n(t) \\ &= \sum_{k=0}^n p_k^n(t) \\ &= \sum_{k=0}^n \binom{n}{k} t^k (1-t)^{n-k} \\ &= (t + (1-t))^n \\ &= 1. \end{aligned}$$

(2) Note that

$$\begin{aligned}
(B_n x)(t) &= \sum_{k=0}^n \frac{k}{n} p_k^n(t) \\
&= \sum_{k=0}^n \frac{k}{n} \binom{n}{k} t^k (1-t)^{n-k} \\
&= \frac{t}{n} \sum_{k=0}^n \binom{n}{k} k t^{k-1} (1-t)^{n-k} \\
&= \frac{t}{n} \cdot n(t + (1-t))^{n-1} \quad \text{by differentiating the Binomial Theorem} \\
&= t.
\end{aligned}$$

(3) Note that

$$\begin{aligned}
(B_n x^2)(t) &= \sum_{k=0}^n \frac{k^2}{n^2} p_k^n(t) \\
&= \frac{t^2}{n^2} \sum_{k=0}^n \binom{n}{k} k^2 t^{k-2} (1-t)^{n-k} \\
&= \frac{t^2}{n^2} \left(n(n-1)(t + (1-t))^{n-2} + \frac{1}{t} n(t + (1-t))^{n-1} \right) \\
&= \left(\frac{n-1}{n} \right) t^2 + \frac{1}{n} t
\end{aligned}$$

□

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Corollary 7.4. $|(B_n(x-a)^2)(a)| \leq \frac{1}{4n}$ for all $a \in [0, 1]$.

Proof. Note that

$$\begin{aligned}
|B_n(x-a)^2(a)| &= (B_n(x-a)^2)(a) \\
&= (B_n(x^2 - 2ax + a^2))(a) \\
&= (B_n(x^2))(a) - 2a(B_n x)(a) + a^2(B_n 1)(a) \\
&= a^2 + \frac{1}{n}(a - a^2) - 2a^2 + a^2 \\
&= \frac{1}{n}(a - a^2) \\
&\leq \frac{1}{4n}.
\end{aligned}$$

□

Now, we prove the Weierstrass Approximation Theorem.

Proof. Fix $f \in C([0, 1])$ and $\epsilon > 0$. By Theorem 5.8, f is uniformly continuous, $\exists \delta > 0$ s.t. $|x - a| < \delta \implies |f(x) - f(a)| < \epsilon$ for all $x, a \in [0, 1]$. If $|x - a| \geq \delta$, then

$$|f(x) - f(a)| \leq |f(x)| + |f(a)| \leq 2\|f\|_\infty \leq \frac{2\|f\|_\infty}{\delta^2}(x - a)^2.$$

So, for all $x, a \in [0, 1]$, we have

$$|f(x) - f(a)| \leq \epsilon + \frac{2\|f\|_\infty}{\delta^2}(x - a)^2.$$

Treat x as a variable and $a \in [0, 1]$ as a fixed constant. This is an inequality of functions which we can apply B_n to:

$$|(B_n f)(t) - f(a)| \leq \epsilon + \frac{2\|f\|_\infty}{\delta^2} (B_n(x - a)^2)(t)$$

which is valid $\forall t, a \in [0, 1]$. In particular, taking $t = a$ gives

$$\begin{aligned} |(B_n f)(a) - f(a)| &\leq \epsilon + \frac{2\|f\|_\infty}{\delta^2} (B_n(x - a)^2)(a) \\ &\leq \epsilon + \frac{2\|f\|_\infty}{\delta^2} \cdot \frac{1}{4n} \\ &= \epsilon + \frac{\|f\|_\infty}{2n\delta^2}. \end{aligned}$$

Then, $\|B_n f - f\|_\infty = \sup_{a \in [0, 1]} |(B_n f)(a) - f(a)| \leq \epsilon + \frac{\|f\|_\infty}{2n\delta^2}$. Take n large enough can make $\frac{\|f\|_\infty}{2n\delta^2} < \epsilon$. For this n , we have

$$\|B_n f - f\|_\infty < 2\epsilon.$$

This makes $p = B_n f \in \mathbb{R}[x]$ the desired polynomial. □

Remark. We assumed $a = 0$ and $b = 1$ in the setup. For general $a < b$, $f \in C([a, b])$, and $\epsilon > 0$, we can define $g \in C([0, 1])$ by $g(t) = f(a + t(b - a))$ for $t \in [0, 1]$. Then, find a polynomial $q \in \mathbb{R}[x]$ with $\|g - q\|_\infty < \epsilon$. Define $p(x) = q\left(\frac{x-a}{b-a}\right) \in \mathbb{R}[x]$. Then, $\|f - p\|_\infty = \|g - q\|_\infty < \epsilon$.

Note. If $f : [a, b] \rightarrow \mathbb{C}$ is continuous, by taking the real and imaginary parts of f separately, for any $\epsilon > 0$, we can find a polynomial $p \in \mathbb{C}[x]$ with $\|f - p\|_\infty < \epsilon$.

Exercise: Let $D = \{z \in \mathbb{C} : |z| < 1\}$. Are the complex polynomials dense in $C(\overline{D})$?

7.2 Stone-Weierstrass Theorem

Definition 7.1 (Algebra, Vector Lattice).

Let (X, d) be a compact metric space. A subset $A \subset C(X)$ is

- (1) an **algebra** if it is a linear subspace and $f, g \in A \implies f \cdot g \in A$ where $(f \cdot g)(x) = f(x)g(x)$ for all $x \in X$.
- (2) a **vector lattice** if it is a linear subspace and $f, g \in A \implies \max\{f, g\}, \min\{f, g\} \in A$.

Example. Algebras include polynomials, $C(X)$, differentiable functions, constant functions, $\{0\}$, etc. Vector lattices are much harder to come up with. Simple examples include $C(X)$, constant functions, $\{0\}$, etc.

Lemma 7.5. If $A \subset C(X)$ is an algebra, then $\overline{A}^{\|\cdot\|_\infty}$ is an algebra and a vector lattice.

Proof. Suppose $(f_n)_{n=1}^\infty$ and $(g_n)_{n=1}^\infty$ are sequences in A with $f_n \rightarrow f$ and $g_n \rightarrow g$ both uniformly. Then, $af + bg = \lim_{n \rightarrow \infty} (af_n + bg_n) \in \overline{A}^{\|\cdot\|_\infty}$ for all $a, b \in \mathbb{R}$, so $\overline{A}^{\|\cdot\|_\infty}$ is a linear subspace of $C(X)$. Also, $fg = \lim_{n \rightarrow \infty} f_n g_n \in \overline{A}^{\|\cdot\|_\infty}$, so $\overline{A}^{\|\cdot\|_\infty}$ is an algebra.

Note that $\max\{f, g\} = \frac{1}{2}(f + g + |f - g|)$ and $\min\{f, g\} = \frac{1}{2}(f + g - |f - g|)$. It suffices to prove that if $f \in \overline{A}^{\|\cdot\|_\infty}$, then $|f| \in \overline{A}^{\|\cdot\|_\infty}$. By Weierstrass Approximation Theorem, $\exists p_n \in \mathbb{R}[x]$ which converges uniformly on $[-\|f\|_\infty, \|f\|_\infty]$ to the absolute value function $|x|$. Write $p_n(x) = \sum_{k=0}^{M_n} a_{n,k} x^k$, where M_n is an integer depending on n .

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Let $q_n(x) = p_n(x) - a_{n,0}$. Note that $a_{n,0} = p_n(0) \rightarrow |0| = 0$. Hence, $q_n(x) \rightarrow |x|$ uniformly on $[-\|f\|_\infty, \|f\|_\infty]$. Now if $f \in \overline{A}^{\|\cdot\|_\infty}$, then $q_n \circ f = \sum_{k=1}^{M_n} a_{n,k} f^k \in \overline{A}^{\|\cdot\|_\infty}$. since $f^k \in \overline{A}^{\|\cdot\|_\infty}$ for all $k \geq 1$ and $\overline{A}^{\|\cdot\|_\infty}$ is an algebra. Then,

$$\begin{aligned} \|q_n \circ f - |f|\|_\infty &= \sup_{x \in X} |q_n(f(x)) - |f(x)|| \\ &\leq \sup_{t \in [-\|f\|_\infty, \|f\|_\infty]} |q_n(t) - |t|| \\ &= \|q_n - |\cdot|\|_\infty \rightarrow 0 \quad \text{on } [-\|f\|_\infty, \|f\|_\infty]. \end{aligned}$$

As $q_n \circ f \in \overline{A}^{\|\cdot\|_\infty}$, we have $|f| \in \overline{A}^{\|\cdot\|_\infty}$. Thus, $\overline{A}^{\|\cdot\|_\infty}$ is a vector lattice. □

Definition 7.2 (Separate Points, Vanish at).

A subset $A \subset C(X)$ is said to

- (1) **separate points** if $\forall x, y \in X$ with $x \neq y$, $\exists f \in A$ such that $f(x) \neq f(y)$.
- (2) **vanish at** $x_0 \in X$ if $\forall f \in A$, $f(x_0) = 0$.

Example. $C(X)$ does not vanish at any $x \in X$ since $1 \in C(X)$. If $x \neq y$, let $f(z) = d(x, z)$. Then, $f \in C(X)$ and $f(x) = 0 \neq d(x, y) = f(y)$, so $C(X)$ separates points.

Example. If $x_0 \in X$, then $A = \{f \in C(X) : f(x_0) = 0\}$ separates points (exercise) and vanishes at x_0 .

Lemma 7.6. Suppose $A \subset C(X)$ is an algebra which separates points and does not vanish at any point. Given $x, y \in X$ with $x \neq y$ and $r, s \in \mathbb{R}$, $\exists f \in A$ such that $f(x) = r$ and $f(y) = s$.

Proof. Since A separates points, $\exists f \in A$ such that $a = f(x) \neq f(y) = b$.

Claim. There exists an $f \in A$ with $a \neq 0$ and $b \neq 0$.

Proof of Claim. Note that at most one of a and b can be 0. Assume that $a = 0 = f(x)$. Since A does not vanish at x , $\exists g \in A$ such that $g(x) \neq 0$. Find an $\epsilon > 0$ such that $h = f + \epsilon g$ satisfies $h(y) = f(y) + \epsilon g(y) \neq 0$. Then, $h(x) = \epsilon g(x) \neq 0$ and $h(y) \neq h(x)$. So we may assume $a = f(x), b = f(y) \neq 0$. □

Define $T : \text{Span}\{f, f^2\} \rightarrow \mathbb{R}^2$ by $T(g) = (g(x), g(y))$. Then, $T(f) = (a, b)$ and $T(f^2) = (a^2, b^2)$. Note that

$$\det \begin{pmatrix} a & a^2 \\ b & b^2 \end{pmatrix} = ab(b - a) \neq 0.$$

Hence, T is surjective. So $\exists g \in \text{Span}\{f, f^2\} \subset A$ with $(g(x), g(y)) = T(g) = (r, s)$. □

Theorem 7.7 (Stone-Weierstrass Theorem).

Let (X, d) be a compact metric space. Suppose $A \subset C(X)$ is an algebra that separates points and does not vanish at any point in X . Then, $\overline{A}^{\|\cdot\|_\infty} = C(X)$.

Proof. Note that $\overline{A}^{\|\cdot\|_\infty}$ is a vector lattice, an algebra, separates points, and does not vanish any point. Fix $f \in C(X)$ and $\epsilon > 0$. Fix $a \in X$ and $\forall x \in X \setminus \{a\}$, choose $h_x \in \overline{A}^{\|\cdot\|_\infty}$ such that $h_x(x) = f(x)$ and $h_x(a) = f(a)$ by Lemma 7.6. Let $U_x = \{y \in X : h_x(y) > f(y) - \epsilon\}$ for $x \in X \setminus \{a\}$. Then, U_x is open. Since $x, a \in U_x$, it follows that $\{U_x : x \in X \setminus \{a\}\}$ is an open cover for X . By compactness, $\exists x_1, \dots, x_n \in X \setminus \{a\}$ such that $X = U_{x_1} \cup \dots \cup U_{x_n}$. Let $g_a = \max\{h_{x_1}, \dots, h_{x_n}\} \in \overline{A}^{\|\cdot\|_\infty}$. Note that if $x \in X$, then $x \in U_{x_j}$ for some j , so $g_a(x) \geq h_{x_j}(x) > f(x) - \epsilon$. Now, we have, for all $a \in X$, $g_a \in \overline{A}^{\|\cdot\|_\infty}$ with $g_a > f - \epsilon$ and $g_a(a) = f(a)$. Let $V_a = \{y \in X : g_a(y) < f(y) + \epsilon\}$. Since $a \in V_a$, $\{V_a : a \in X\}$ is an open cover for X . Let $X = V_{a_1} \cup \dots \cup V_{a_m}$ for some $a_1, \dots, a_m \in X$. Let $f_0 = \min\{g_{a_1}, \dots, g_{a_m}\}$. Then, $f_0 > f - \epsilon$ and if $x \in V_{a_j}$, then $f_0(x) \leq g_{a_j}(x) < f(x) + \epsilon$. Hence, $f_0 < f + \epsilon$. Thus, $\|f - f_0\|_\infty < \epsilon$. Since $f_0 \in \overline{A}^{\|\cdot\|_\infty}$, we have $\overline{A}^{\|\cdot\|_\infty} = C(X)$. $\square$

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7.3 Applications

Corollary 7.8. Suppose $K \subset \mathbb{R}^n$ is compact. Consider

$$\mathcal{A} = \{P(x_1, \dots, x_n) : P \text{ is a polynomial in } n \text{ variables}\} \subset C(K).$$

Then, $\overline{\mathcal{A}} = C(K)$. That is, every continuous function on a compact subset of $\mathbb{R}^n$ is uniformly closed to a polynomial in n variables.

Proof. Note that $\mathcal{A}$ is an algebra. Let $\vec{a} \neq \vec{b}$ be points in K . Then, $\exists i$ such that $a_i \neq b_i$. If $P(\vec{x}) = x_i$, then $P(\vec{a}) = a_i \neq b_i = P(\vec{b})$, so $\mathcal{A}$ separates points. Also, $\mathcal{A}$ does not vanish at any point since $1 \in \mathcal{A}$. By Stone-Weierstrass Theorem, $\overline{\mathcal{A}} = C(K)$. $\square$

Corollary 7.9. Suppose (X_1, d_1) and (X_2, d_2) are compact metric spaces and $(X_1 \times X_2, d)$ is a compact metric space where $d((a, b), (a', b')) = \max\{d_1(a, a'), d_2(b, b')\}$. Consider all continuous real-valued functions on $X_1 \times X_2$ of the form

$$h(x, y) = \sum_{i=1}^n f_i(x)g_i(y).$$

If $\mathcal{A}$ is the space of such functions, then $\overline{\mathcal{A}} = C(X_1 \times X_2)$.

Proof. Note that $\mathcal{A}$ is an algebra. Since $1 \in \mathcal{A}$, so $\mathcal{A}$ does not vanish at a point. If $(x_1, y_1) \neq (x_2, y_2)$, then either $x_1 \neq x_2$ or $y_1 \neq y_2$. If $x_1 \neq x_2$, take $f \in C(X_1)$ such that $f(x_1) \neq f(x_2)$, then $h(x, y) = f(x) \in \mathcal{A}$ separates (x_1, y_1) and (x_2, y_2) . If $y_1 \neq y_2$, take $g \in C(X_2)$ such that $g(y_1) \neq g(y_2)$, then $h(x, y) = g(y) \in \mathcal{A}$ separates (x_1, y_1) and (x_2, y_2) . By Stone-Weierstrass Theorem, $\overline{\mathcal{A}} = C(X_1 \times X_2)$. $\square$

Example. Let $\mathcal{A} = \{\sum_{k=0}^n a_k z^k : a_k \in \mathbb{C}\} \subset C(\overline{\mathbb{D}})$ where $\overline{\mathbb{D}} = \{z \in \mathbb{C} : |z| \leq 1\}$. Suppose $f(z) \in \mathcal{A}$, then $f(z) = \sum_{k=0}^n a_k z^k$. Note that

$$\int_0^{2\pi} f(e^{it}) dt = \sum_{k=0}^n a_k \int_0^{2\pi} e^{ikt} dt = 2\pi a_0 = 2\pi f(0).$$

Hence, if $f \in \overline{\mathcal{A}}$, then

$$\int_0^{2\pi} f(e^{it}) dt = 2\pi f(0).$$

For example, if $f(z) = 1 - |z|$. Then,

$$0 = \int_0^{2\pi} f(e^{it}) dt, \quad f(0) = 1.$$

So $f \notin \overline{\mathcal{A}}$. Hence, $\overline{\mathcal{A}} \neq C(\overline{\mathbb{D}})$.

Definition 7.3 (Self-Adjoint).

Let (X, d) be a metric space. A subset $\mathcal{A} \subset C(X; \mathbb{C})$ is **self-adjoint** if $f \in \mathcal{A} \implies \overline{f} \in \mathcal{A}$ where $\overline{f}(z) = \overline{f(z)}$.

Theorem 7.10 (Stone-Weierstrass Theorem: Complex Version).

If X is a compact metric space and $\mathcal{A} \subset C(X; \mathbb{C})$ is an algebra such that

- (1) $\mathcal{A}$ separates points.
- (2) $\mathcal{A}$ does not vanish at any point in X .
- (3) $\mathcal{A}$ is self-adjoint.

Then, $\overline{\mathcal{A}} = C(X; \mathbb{C})$.

Proof. If $f \in \mathcal{A}$, then $\operatorname{Re}(f) = \frac{f + \overline{f}}{2}$. Let $\mathcal{A}_0 = \{f \in C(X) \mid f \in \mathcal{A}\}$. Then, $\mathcal{A}_0$ is an algebra which separates points and does not vanish at any point in X . By Stone-Weierstrass Theorem, $\overline{\mathcal{A}_0} = C(X)$.

Exercise: Finish the proof. $\square$

8 Baire Category Theorem

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8.1 Baire Category Theorem

Motivation

Definition 8.1 (Algebraic Number).

A number $a \in \mathbb{R}$ is **algebraic** if $\exists$ some polynomial with integer coefficients such that a is a root of the polynomial. That is, $P(x) = \sum_{k=0}^n a_k x^k$ where $a_k \in \mathbb{Z}$ and $P(a) = 0$.

Example. $n \in \mathbb{Z}$ is algebraic. Take $P(x) = x - n$.

Example. $\sqrt{2}$ is algebraic. Take $P(x) = x^2 - 2$.

Question: Is every number algebraic?

Answer: No. Fact: each polynomial has only finitely many zeros. Note that there are only countably many polynomials with integer coefficients, so we have a surjection

$$\underbrace{\mathbb{Z}^n}_{\text{countable}} \rightarrow \text{polynomials of deg } n \text{ with } \mathbb{Z}\text{-coefficients.}$$

Thus, there are countably many polynomials with $\mathbb{Z}$ -coefficients. So, there are countably many algebraic numbers. Therefore, $\exists$ a non-algebraic number.

Motivation Problems

- (1) Is there a function that is continuous at every irrational number, but discontinuous at every rational number?
- (2) Is there a continuous function f on $[0, 1]$ such that f is not differentiable at any point?

Definition 8.2 (Nowhere Dense, First Category, Residual).

Let (X, d) be a metric space and $A \subset X$.

- (1) A is **nowhere dense** if $\text{Int}(\overline{A}) = \emptyset$.

- (2) A is of the **first category** (or **meager**) if it is a countable union of nowhere dense sets.
- (3) $B \subset X$ is **residual** if B^c is of the first category.

Note. First category sets are “small” and residual sets are “large”.

Example.

- (1) A singleton $\{x\} \subset \mathbb{R}$ is nowhere dense.
- (2) Every countable subset of $\mathbb{R}$ is of the first category.
- (3) $\mathbb{Q} \subset \mathbb{R}$ is of the first category. Note that $\overline{\mathbb{Q}} = \mathbb{R}$, so $\mathbb{Q}$ is dense, not nowhere dense.
- (4) The Cantor set C is closed and nowhere dense. Also, $|C| = |\mathbb{R}|$.

Note. The countable union of first category sets is of the first category.

Theorem 8.1 (Baire Category Theorem).

If (X, d) is a complete metric space, then X is not of the first category (as a subset of itself).

Proof. Suppose X is complete and we have $\bigcup_{n=1}^{\infty} A_n$ where each $A_n \subset X$ is nowhere dense. Then, $X = \bigcup_{n=1}^{\infty} \overline{A_n}$, so assume $A_n = \overline{A_n}$ for $n \geq 1$. We will show that $\bigcup_{n=1}^{\infty} A_n \neq X$. Fix $x \in X$. Note that $\text{Int}(A_n) = \emptyset$ and A_n^c is dense and open, so $x \in \overline{A_1^c}$. Choose $x_1 \in B_r(x) \cap A_1^c$ for some $r > 0$. Then, choose $r_1 < \frac{1}{2}r$ such that $\overline{B_{r_1}(x_1)} \subset B_r(x) \cap A_1^c$. Choose $x_2 \in B_{r_1}(x_1) \cap A_2^c$ and $0 < r_2 < \frac{1}{2^2}r$ such that $\overline{B_{r_2}(x_2)} \subset B_{r_1}(x_1) \cap A_2^c$. By induction, we choose $x_n \in B_{r_{n-1}}(x_{n-1}) \cap A_n^c$ and $0 < r_n < \frac{1}{2^n}r$ such that $\overline{B_{r_n}(x_n)} \subset B_{r_{n-1}}(x_{n-1}) \cap A_n^c$. Note if $n, m \geq N$, then $x_n, x_m \in B_{r_N}(x_N)$, so $d(x_n, x_m) < 2r_N < 2 \cdot \frac{1}{2^N}r$. Since X is complete, $\exists x_0 \in X$ such that $\lim_{n \rightarrow \infty} x_n = x_0$. Note that $d(x_0, x) \leq r$. Also, $x_0 \in \bigcap_{n=1}^{\infty} \overline{B_{r_n}(x_n)}$ and $\overline{B_{r_n}(x_n)} \cap A_n = \emptyset$, so $x_0 \notin \bigcup_{n=1}^{\infty} A_n$. In fact, this shows that $\left(\bigcup_{n=1}^{\infty} A_n\right)^c$ is dense in X . $\square$

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Corollary 8.2. If X is a non-empty complete metric space and if $\{U_n\}_{n=1}^{\infty}$ are open and dense, then $\bigcap_{n=1}^{\infty} U_n$ is dense in X .

Note. This is another version of the Baire Category Theorem.

Proof. If U_n is open and dense, then U_n^c is nowhere dense. Also, $\bigcup_{n=1}^{\infty} U_n^c \neq X$ by the Baire Category Theorem, so $\bigcap_{n=1}^{\infty} U_n = \left(\bigcup_{n=1}^{\infty} U_n^c\right)^c$ is dense in X . $\square$

Definition 8.3 (G_δ -set).

A subset $A \subset X$ is a G_δ -set if A is the countable intersection of open sets.

Note. That is, $A = \bigcap_{n=1}^{\infty} U_n$ where each U_n is open.

Example.

- $\mathbb{R} \setminus \mathbb{Q}$ is a G_δ -set.
- Every open set is a G_δ -set.

Definition 8.4 (F_σ -set).

A subset $B \subset X$ is an F_σ -set if B is the countable union of closed sets.

Note. That is, $B = \bigcup_{n=1}^{\infty} C_n$ where each C_n is closed.

Example. A set B is $F_\sigma \iff B^c$ is G_δ .

Corollary 8.3. If X is a non-empty complete metric space and if $X = \bigcup_{n=1}^{\infty} F_n$ where each F_n is closed, then $\exists n \geq 1$ such that $\text{Int}(F_n) \neq \emptyset$.

8.2 Applications

Theorem 8.4. If $f : [a, b] \rightarrow \mathbb{R}$ is a pointwise limit of continuous functions, then f is continuous on a residual set.

Definition 8.5 (Oscillation).

Let X be a metric space and $f : X \rightarrow \mathbb{R}$ be a function. Set

$$\omega_f(\delta)(x) = \sup_{y, z \in B_\delta(x)} \{|f(y) - f(z)|\}.$$

An **oscillation** of f at x is

$$\omega_f(x) = \inf_{\delta > 0} \omega_f(\delta)(x).$$

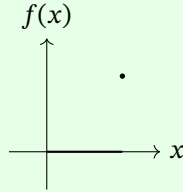
Note. $\omega_f(x)$ measures how much f oscillates at x .

Remark. f is continuous at $x \iff \omega_f(x) = 0$.

Example. Consider

$$f(x) = \begin{cases} 0 & x \in [0, 1), \\ 1 & x = 1. \end{cases}$$

The function looks like:



Then,

$$\omega_f(\delta)(x) = \omega_f(x) = \begin{cases} 0 & x \in [0, 1), \\ 1 & x = 1. \end{cases}$$

Example. Consider

$$f(x) = \begin{cases} 0 & x \in \mathbb{R} \setminus \mathbb{Q}, \\ \frac{1}{q} & x = \frac{p}{q} \text{ in reduced form.} \end{cases}$$

Then,

$$\omega_f(x) = \begin{cases} 0 & x \in \mathbb{R} \setminus \mathbb{Q}, \\ \frac{1}{q} & x = \frac{p}{q} \text{ in reduced form.} \end{cases}$$

Proof of Theorem 8.4. Suppose $f : [a, b] \rightarrow \mathbb{R}$ continuous and $f_i(x) \rightarrow f(x)$ pointwise. Note that f is continuous at $x \iff \omega_f(x) = 0$. That is, at the set

$$\{x \in [a, b] : \omega_f(x) = 0\} = \bigcap_{n=1}^{\infty} \left\{x \in [a, b] : \omega_f(x) < \frac{1}{n}\right\}.$$

Lemma 8.5. For any function $f : X \rightarrow \mathbb{R}$, $A = \{x \in X : \omega_f(x) < \frac{1}{n}\}$ is open.

Proof. If $x \in A$, then $\exists \delta > 0$ s.t. $\omega_f(\delta)(x) < \frac{1}{n}$. If $y \in B_{\frac{\delta}{2}}(x)$, then

$$\omega_f(y) \leq \omega_f\left(\frac{\delta}{2}\right)(y) = \sup_{z_1, z_2 \in B_{\frac{\delta}{2}}(y)} \{|f(z_1) - f(z_2)|\} \leq \sup_{z_1, z_2 \in B_{\delta}(x)} \{|f(z_1) - f(z_2)|\} = \omega_f(\delta)(x) < \frac{1}{n}.$$

So, $B_{\frac{\delta}{2}}(x) \subset A$, so A is open. □

So we just need to show that each set

$$\left\{x \in [a, b] : \omega_f(\delta)(x) < \frac{1}{n}\right\}$$

is dense. Let $I \subset [a, b]$ be a non-empty interval. Set $\epsilon < \frac{1}{3n}$. For $i, j \geq 1$, set

$$X_{i,j} = \{x \in \bar{I} : |f_i(x) - f_j(x)| \leq \epsilon\} = |f_i - f_j|^{-1}([0, \epsilon]) \cap \bar{I}.$$

For $N \in \mathbb{N}$, set

$$E_N = \bigcap_{i,j \geq N} X_{i,j}.$$

Note that $X_{i,j}$ is closed since f_i and f_j are continuous, so E_N is closed. Since $f_i \rightarrow f$ pointwise, $\bigcup_{N=1}^{\infty} E_N = \bar{I}$, which is complete. By the Baire Category Theorem, $\exists N \in \mathbb{N}$ s.t. E_N has non-empty interior, i.e. $\text{Int}(E_N) \neq \emptyset$. Say $J \subset E_N$ where J is a non-empty interval. Then,

$$J \subset \{x \in \bar{I} : |f_N(x) - f_j(x)| \leq \epsilon \forall j \geq N\}.$$

Thus,

$$I \subset \{x \in [a, b] : |f_N(x) - f(x)| \leq \epsilon\}.$$

Since f_N is uniformly continuous, $\exists \delta_0 > 0$ s.t. if $|x - y| < \delta_0$, then $|f_N(x) - f_N(y)| < \epsilon$. If $x \in J$, set $\delta_x = \min\{\delta_0, \text{dist}(x, J^c)\}$, so if $y, z \in B_{\delta_x}(x)$, then

$$\begin{aligned} |f(y) - f(z)| &\leq |f(y) - f_N(y)| + |f_N(y) - f_N(z)| + |f_N(z) - f(z)| \\ &< 3\epsilon < \frac{1}{n}. \end{aligned}$$

Thus,

$$x \in \{z \in [a, b] : \omega_f(z) < \frac{1}{n}\}, \quad \text{and} \quad J \subset \{z \in [a, b] : \omega_f(z) < \frac{1}{n}\}.$$

This set is dense. □

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Another Application: Nowhere Differentiable Functions

Definition 8.6 (Lipschitz at a Point).

A function $f : [a, b] \rightarrow \mathbb{R}$ is **Lipschitz** at x if $\exists L$ s.t. $|f(x) - f(y)| \leq L|x - y|$ for all $y \in [a, b]$.

Lemma 8.6. If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable at $x \in [a, b]$, then f is Lipschitz at x .

Proof. Since f is differentiable at x , $\exists \delta > 0$ such that if $|y - x| < \delta$, then

$$\left| \frac{f(x) - f(y)}{x - y} - f'(x) \right| < 1.$$

Thus,

$$\begin{aligned} \left| \frac{f(x) - f(y)}{x - y} \right| &< 1 + |f'(x)| \\ |f(x) - f(y)| &< (1 + |f'(x)|)|x - y|. \end{aligned}$$

Since f is continuous on $[a, b]$, which is compact, $\|f\|_\infty < \infty$. Set $L = \max\left\{1 + |f'(x)|, \frac{2\|f\|_\infty}{\delta}\right\}$. If $|x - y| \geq \delta$, then

$$|f(x) - f(y)| \leq 2\|f\|_\infty = \frac{2\|f\|_\infty}{\delta}\delta \leq \frac{2\|f\|_\infty}{\delta}|x - y| \leq L|x - y|.$$

Therefore, f is Lipschitz at x . □

Theorem 8.7. The set of continuous functions on $[a, b]$ that are nowhere differentiable is residual in $C([a, b])$.

Proof. Let $A_K = \{f \in C([a, b]) \mid f \text{ is } K\text{-Lipschitz at some point in } [a, b]\}$. We need to show that $\bigcup_{K=1}^{\infty} A_K$ is of the first category.

Claim (1). A_K is closed.

Proof. Suppose $f_n \in A_K$ and $f_n \rightarrow f$ in $C([a, b])$ (uniform convergence). Take $x_n \in [a, b]$ s.t. $|f_n(x_n) - f_n(y)| \leq K|x_n - y| \forall y \in [a, b]$. Take a subsequence $\{x_{n_j}\}$ s.t. $x = \lim_{j \rightarrow \infty} x_{n_j}$.

Note. When you see a sequence in a compact set, you should think about taking a subsequence that converges.

If $y \in [a, b]$, fix $\epsilon > 0$. Since $f_n \rightarrow f$ uniformly, $\exists N \in \mathbb{N}$ s.t. $\forall n \geq N, \|f - f_n\|_{\infty} < \frac{\epsilon}{3}$. Since f is continuous at x , $\exists \delta > 0$ and $\delta < \frac{\epsilon}{4K}$ s.t. if $|x' - x| < \delta$, then $|f(x') - f(x)| < \frac{\epsilon}{3}$. Also, since $x_n \rightarrow x$, $\exists N' \in \mathbb{N}$ s.t. for $n \geq N'$, we have $|x - x_n| < \delta$. Take $n \geq \max\{N, N'\}$. Then,

$$\begin{aligned} |f(x) - f(y)| &\leq |f(x) - f(x_n)| + |f(x_n) - f(y)| \\ &\leq \frac{\epsilon}{3} + |f(x_n) - f_n(x_n)| + |f_n(x_n) - f_n(y)| + |f_n(y) - f(y)| \\ &\leq \frac{\epsilon}{3} + \frac{\epsilon}{3} + K|x_n - y| + \frac{\epsilon}{3} \\ &\leq \epsilon + K(|x_n - x| + |x - y|) \\ &\leq \epsilon + K \cdot \delta + K|x - y| \\ &< 2\epsilon + K|x - y|. \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, $|f(x) - f(y)| \leq K|x - y|$. Thus, $f \in A_K$. □

Claim (2). Each A_k has empty interior, i.e. $\text{Int}(A_K) = \emptyset$.

Idea: Take $f \in A_k$ and add to f a very small but very wiggly function.

Proof. Fix $\epsilon > 0$. Fix $f \in A_K$. Since f is uniformly continuous, $\exists \delta > 0$ s.t. $|x - y| < \delta \implies |f(x) - f(y)| < \frac{\epsilon}{2}$. Choose $x_0 = a, x_1 = a + \frac{\delta}{2}, \dots, x_{n-1} < b$ s.t. $|b - x_{n-1}| < \delta$. Set $x_n = b$. Let h be the function such that $h(x_K) = f(x_K)$ and h is linear on each interval $[x_i, x_{i+1}]$. Then, $\|h - f\|_{\infty} < \frac{\epsilon}{2}$. Since h is piecewise linear, h is Lipschitz with Lipschitz constant L . Set $g(x) = h(x) + \frac{\epsilon}{2} \sin(Mx)$ where $M > \frac{4\pi K + L}{\epsilon}$. Then, $\|g - f\|_{\infty} \leq \epsilon$. Suppose $x_0 \in [a, b]$, $\sin(Mx)$ takes the values ± 1 on any

interval of length at least $\frac{2\pi}{M}$. Choose x s.t. $|x - x_0| < \frac{2\pi}{M}$ and $|\sin(Mx) - \sin(Mx_0)| \geq 1$. Then,

$$\begin{aligned}
|g(x) - g(x_0)| &= |h(x) - h(x_0) + \frac{\epsilon}{2}(\sin(Mx) - \sin(Mx_0))| \\
&\geq \frac{\epsilon}{2} - |h(x) - h(x_0)| \\
&\geq \frac{\epsilon}{2} - L|x - x_0| \\
&\geq \frac{\epsilon}{2} \cdot \frac{M}{2\pi}|x - x_0| - L|x - x_0| \\
&\geq \left(\frac{\epsilon}{2} \cdot \frac{M}{2\pi} - L\right)|x - x_0| \\
&\geq K|x - x_0|.
\end{aligned}$$

Thus, g is not K -Lipschitz at x_0 . Since x_0 was arbitrary, g is not K -Lipschitz at any point in $[a, b]$, so $g \notin A_K$. Thus, $\text{Int}(A_K) = \emptyset$. □

By Claim 1 and Claim 2, A_K is nowhere dense, so $\bigcup_{K=1}^{\infty} A_K$ is of the first category. Let

$$D = \{f \in C([a, b]) : f \text{ is differentiable at some point in } [a, b]\}.$$

By Lemma 8.6, $D \subset \bigcup_{K=1}^{\infty} A_K$, so D is of the first category. Thus,

$$D^c = C([a, b]) \setminus D = \{f \in C([a, b]) : f \text{ is nowhere differentiable}\}$$

is residual in $C([a, b])$. □

9 Contraction Mapping Principle

Lecture 33, 2026/04/01

9.1 Contraction Mapping Principle

Motivation

Example. Let $T : \ell^2(\mathbb{N}) \rightarrow \ell^2(\mathbb{N})$ be linear with $\|T\| < 1$. Then,

$$\|Tf\|_2 \leq \|T\| \|f\|_2.$$

Note that $T0 = 0$, so 0 is a fixed point of T . If $Tf = f$, then

$$\|f\|_2 = \|Tf\|_2 \leq \|T\| \|f\|_2.$$

Since $\|T\| < 1$, we must have $\|f\|_2 = 0$, so $f = 0$. Therefore, 0 is the unique fixed point of T .

Definition 9.1 (Contraction).

Let (X, d) be a metric space. A **contraction** is a function $T : X \rightarrow X$ that is Lipschitz with Lipschitz constant $L < 1$.

Theorem 9.1 (Contraction Mapping Principle).

Suppose (X, d) is non-empty, complete, and $T : X \rightarrow X$ is a contraction. Then, there is a unique point $x^* \in X$ such that $T(x^*) = x^*$.

Proof. Fix any point $x_0 \in X$. Set $x_n = T^n(x_0) = \underbrace{T(T(\cdots T(x_0)))}_n$. Note that

$$\begin{aligned} d(x_{n+1}, x_n) &= d(T^n(T(x_0)), T^n(x_0)) \\ &\leq Ld(T^{n-1}(T(x_0)), T^{n-1}(x_0)) \\ &\leq \cdots \leq L^n d(T(x_0), x_0). \end{aligned}$$

Then,

$$\begin{aligned}
d(x_{n+k}, x_n) &\leq \sum_{\ell=0}^{k-1} d(x_{n+\ell+1}, x_{n+\ell}) \\
&\leq \sum_{\ell=0}^{k-1} L^{n+\ell} d(T(x_0), x_0) \\
&\leq L^n \sum_{\ell=0}^{\infty} L^{\ell} d(T(x_0), x_0) \\
&= L^n \frac{1}{1-L} d(T(x_0), x_0) \rightarrow 0 \quad \text{as } n \rightarrow \infty.
\end{aligned}$$

Therefore, $\{x_n\}_n$ is a Cauchy sequence. Since $x^* = \lim_{n \rightarrow \infty} x_n$ exists, we have

$$T(x^*) = \lim_{n \rightarrow \infty} T(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = x^*.$$

If $y \in X$ s.t. $T(y) = y$, then

$$d(x^*, y) = d(T(x^*), T(y)) \leq Ld(x^*, y).$$

Since $L < 1$, we must have $d(x^*, y) = 0$, so $x^* = y$. This proves the uniqueness. $\square$

Example. Consider $T : \mathbb{R} \rightarrow \mathbb{R}$ given by $T(x) = x + \frac{1}{2}$. Then, T is Lipschitz with $L = 1$ since

$$|T(x) - T(y)| = |x - y| \quad \text{for all } x, y \in \mathbb{R}.$$

Example. Consider $T : [1, \infty) \rightarrow [1, \infty)$ given by $T(x) = x + \frac{1}{x}$. If $x, y \in [1, \infty)$, then

$$\begin{aligned}
|T(x) - T(y)| &= \left| x + \frac{1}{x} - y - \frac{1}{y} \right| \\
&= \left| \frac{(x^2 + 1)y - x(y^2 + 1)}{xy} \right| \\
&= |x - y| \left| 1 - \frac{1}{xy} \right| < |x - y|.
\end{aligned}$$

Note that

$$\sup_{x \neq y} \frac{|T(x) - T(y)|}{|x - y|} = \sup_{x \neq y} \left| 1 - \frac{1}{xy} \right| = 1.$$

So, T is not a contraction. Also, if $T(x) = x$, then $x + \frac{1}{x} = x \implies \frac{1}{x} = 0$, which has no solution. Therefore, T has no fixed point.

9.2 Applications

Example. Consider $\frac{dy}{dx} = y + x$ with $y(0) = 0$. A solution to this DE is a function φ that is defined in a neighborhood of 0 and satisfies $\varphi(0) = 0$ and $\varphi'(t) = \varphi(t) + t$. Note that φ is a solution $\iff$

$$\varphi(x) = 0 + \int_0^x (\varphi(t) + t) dt.$$

Define $T : C([-1, 1]) \rightarrow C([-1, 1])$ by

$$T(\varphi)(x) = \int_0^x (\varphi(t) + t) dt.$$

We want to find a fixed point of T . Set $\varphi_0 = 0$, $\varphi_1 = T(\varphi_0)$ so that

$$\varphi_1(x) = \int_0^x t dt = \frac{1}{2!}x^2.$$

Then, $\varphi_2 = T(\varphi_1)$ so that

$$\varphi_2(x) = \int_0^x \left(\frac{1}{2!}t^2 + t \right) dt = \frac{1}{3!}x^3 + \frac{1}{2!}x^2.$$

set $\varphi_n = T(\varphi_{n-1})$. By induction,

$$\varphi_n(x) = \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots + \frac{1}{(n+1)!}x^{n+1}.$$

Note that $\{\varphi_n\}_n$ converges uniformly on compact sets to $\varphi(x) = e^x - x - 1$. Note that $\varphi(0) = 0$ and

$$\varphi'(t) = e^t - 1 = e^t - t - 1 + t = \varphi(t) + t.$$

Theorem 9.2 (Picard).

Suppose f is a continuous function on a rectangle $D \subset \mathbb{R}^2$ containing the point (x_0, y_0) in its

interior and $\exists M > 0$ s.t.

$$|f(x, y_1) - f(x, y_2)| \leq M|y_1 - y_2| \quad \text{for all } (x, y_1), (x, y_2) \in D.$$

Then, $\exists \delta > 0$ s.t. the DE

$$\frac{dy}{dx} = f(x, y)$$

has a unique solution in the interval $(x_0 - \delta, x_0 + \delta)$ s.t. $y(x_0) = y_0$.

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Recall that a solution to the DE $\frac{dy}{dx} = f(x, y)$ with $y(x_0) = y_0$ is a function $\varphi : (x_0 - \delta, x_0 + \delta) \rightarrow \mathbb{R}$ with $\varphi(x_0) = y_0$ and φ is differentiable such that $\varphi'(x) = f(x, \varphi(x))$.

Proof of Picard's Theorem. We may assume $K = \|f\|_\infty < \infty$ since f is continuous on a compact set $\implies f$ is bounded (by EVT). Choose $\delta > 0$ s.t. if $|x - x_0| < \delta$ and $|y - y_0| < K\delta$, then $(x, y) \in D$. Also take $\delta > 0$ s.t. $\delta M < 1$. Set $C^* \subset C([x - \delta, x + \delta])$ by

$$C^* = \{g \in C([x - \delta, x + \delta]) : |g(x) - y_0| \leq K\delta\}.$$

Note that C^* is complete since $C([x - \delta, x + \delta])$ is complete and C^* is closed. Define $T : C^* \rightarrow C^*$ by $T(\varphi) = \psi$ where

$$\psi(x) = y_0 + \int_{x_0}^x f(t, \varphi(t)) dt.$$

Claim. $\exists! \psi \in C^*$ s.t. $T(\varphi) = \psi$.

Proof of Claim. Suppose $\varphi_1, \varphi_2 \in C^*$ s.t. $\psi_1 = T(\varphi_1)$ and $\psi_2 = T(\varphi_2)$. Then,

$$\begin{aligned} \|\psi_1 - \psi_2\|_\infty &= \sup_{x \in [x_0 - \delta, x_0 + \delta]} |\psi_1(x) - \psi_2(x)| \\ &= \sup_{x \in I} \left| \int_{x_0}^x f(t, \varphi_1(t)) - f(t, \varphi_2(t)) dt \right| \\ &\leq \sup_{x \in I} \left| \int_{x_0}^x M |\varphi_1(t) - \varphi_2(t)| dt \right| \\ &\leq \delta M \|\varphi_1 - \varphi_2\|_\infty. \end{aligned}$$

Since $\delta M < 1$, T is a contraction. By Contraction Mapping Principle, $\exists! \psi \in C^*$ s.t. $T(\varphi) = \psi$. □

This concludes the proof. □

Theorem 9.3. Let (X, d) be a non-empty complete metric space. Suppose $T_1, T_2, \dots, T_k : X \rightarrow X$ are such that each map is Lipschitz with Lipschitz constant $c_i < 1$ for $1 \leq i \leq k$. Let

$$\mathcal{H}(X) = \{A \subset X : A \text{ is non-empty and closed}\}.$$

Recall that the Hausdorff metric $d_{\mathcal{H}}$ on $\mathcal{H}(X)$ is defined by

$$d_{\mathcal{H}}(A, B) = \max \left\{ \sup_{a \in A} d(a, B), \sup_{b \in B} d(b, A) \right\}.$$

Define $T : \mathcal{H}(X) \rightarrow \mathcal{H}(X)$ by

$$T(A) = \overline{T_1(A)} \cup \overline{T_2(A)} \cup \dots \cup \overline{T_k(A)}.$$

Then, $\exists! A_{\infty} \in \mathcal{H}(X)$ s.t. $T(A_{\infty}) = A_{\infty}$ (i.e. A_{∞} is a fixed point of T).

Exercise: If (X, d) is non-empty and complete, then $(\mathcal{H}(X), d_H)$ is also complete.

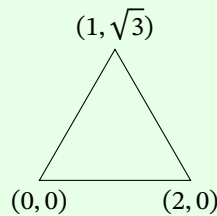
Hint: If $\{A_n\}_{n=1}^{\infty}$ is Cauchy. Show that $\lim_{n \rightarrow \infty} A_n = \bigcap_{n \geq 1} \overline{\bigcup_{m \geq n} A_m}$.

Example (Sierpinski Triangle).

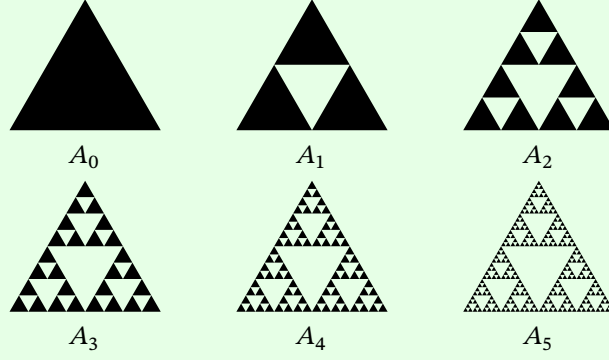
Consider $X = \mathbb{R}^2$. Define

$$T_1(x) = \frac{1}{2}x, \quad T_2(x) = (1, 0) + \frac{1}{2}x, \quad T_3(x) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) + \frac{1}{2}x.$$

Then, we have an equilateral triangle with vertices at $(0, 0)$, $(2, 0)$, $(1, \sqrt{3})$ and side length 2:



Define $A_n = T_1(A_{n-1}) \cup T_2(A_{n-1}) \cup T_3(A_{n-1})$ with $A_0 = \{(0, 0), (2, 0), (1, \sqrt{3})\}$. Then, A_1 is the union of three triangles and looks like A_0 was divided into four triangles with the middle triangle removed. Likewise, A_2 looks like each solid triangle in A_1 was divided into four triangles with the middle triangle removed. Taking the limit, this converges to the Sierpinski triangle.



Proof of Theorem 9.3. Observe that if $A_1, \dots, A_k, B_1, \dots, B_k \in \mathcal{H}(X)$, then

$$d_{\mathcal{H}}(A_1 \cup \dots \cup A_k, B_1 \cup \dots \cup B_k) \leq \underbrace{\max_{1 \leq i \leq k} d_{\mathcal{H}}(A_i, B_i)}_r.$$

If $a \in A_i$ or $b \in B_i$, then $d(a, B_i) \leq r$ or $d(b, A_i) \leq r$. Hence if $a \in A_1 \cup \dots \cup A_k$, then

$$d(a, B_1 \cup \dots \cup B_k) \leq r.$$

Also if $b \in B_1 \cup \dots \cup B_k$, then

$$d(b, A_1 \cup \dots \cup A_k) \leq r.$$

Claim. $T : \mathcal{H}(X) \rightarrow \mathcal{H}(X)$ is a contraction.

Proof of Claim. If $A, B \in \mathcal{H}(X)$, then

$$\begin{aligned} d_{\mathcal{H}}(\overline{T_i(A)}, \overline{T_i(B)}) &= \max \left\{ \sup_{a \in A} d(T_i(a), \overline{T_i(B)}), \sup_{b \in B} d(T_i(b), \overline{T_i(A)}) \right\} \\ &\leq c_i \max \left\{ \sup_{a \in A} d(a, B), \sup_{b \in B} d(b, A) \right\} \\ &= c_i d_{\mathcal{H}}(A, B). \end{aligned}$$

Also,

$$\begin{aligned} d_{\mathcal{H}}(\overline{T_1(A)} \cup \dots \cup \overline{T_k(A)}, \overline{T_1(B)} \cup \dots \cup \overline{T_k(B)}) &\leq \max_{1 \leq i \leq k} c_i d_{\mathcal{H}}(A, B) \\ &\leq \underbrace{\left(\max_{1 \leq i \leq k} c_i \right)}_{< 1} d_{\mathcal{H}}(A, B). \end{aligned}$$

Thus, T is a contraction. □

By Contraction Mapping Principle, $\exists! A_{\infty} \in \mathcal{H}(X)$ s.t. $T(A_{\infty}) = A_{\infty}$. □

END OF PMATH 351!